

MAGNETOM

Startup System

System Start-up

Symphony, A Tim System

Document Version

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1.1 Prerequisites

- Room lighting must be functioning in every room of the MR site.
- The system components have been positioned and installed according to the installation instructions.
- The RF integrity of the RF room was measured after installing the RF room and its components.



Only trained personnel, who have successfully completed the Avanto course, are allowed to perform the start-up. The warranty is invalid, if problems occur in the system due to work performed by untrained personnel or third parties.



For safety information, refer to "MR-specific safety information".

1.2 Service Tools

The Service Tools required for starting up the MR system include:

- Service tool kit (complete)
 - RF cable 2 m BNC (part-no.: 8733479, included in the kit) is required for the tune-up procedure.
- Set of standard tools
- ESD service kit
- Digital multimeter
- Protective conductor measurement device
- Helium syphon (dewar side)

1.2.1 Magnet system

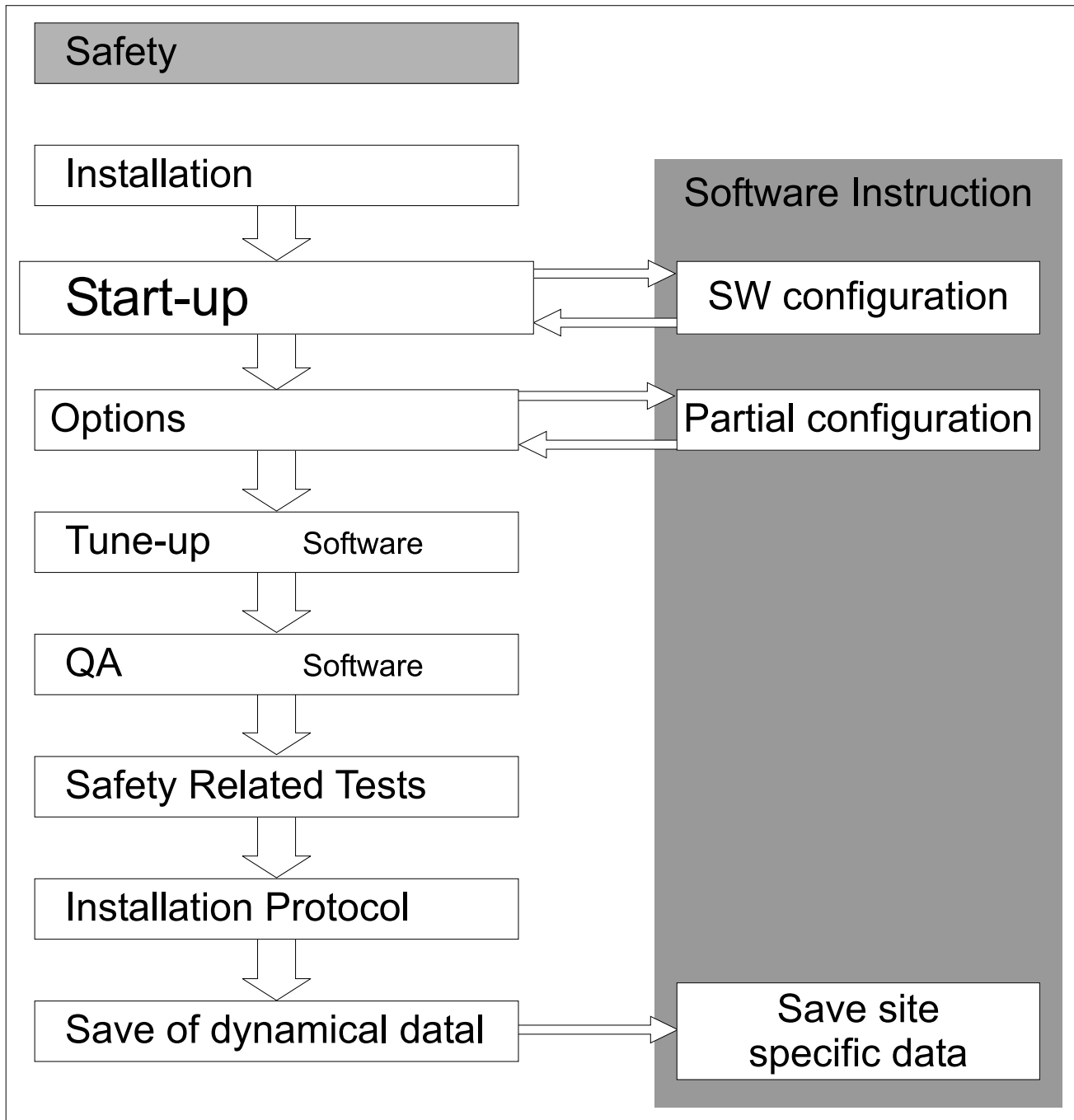
- Array shim device
- Heat gun
- Magnet power supply 3600
- 3600 Standard extension kit
- Protective clothing (face shield, gloves, safety shoes, etc.)
- Remote Display Unit for magnet helium level

1.3 Overall Flowchart



System start-up should be performed in the order of the steps described. Complete each step before you start working on the next one.

Fig. 1: Startup flowchart



1.4 Safety

For your own safety and for the safety of third parties, as well as to protect the MR system and its parts, observe the safety information in the "General Safety Notes" and "MR-Specific Safety Information" in the "Technical Documents" when performing work on the system.

Read the special safety information regarding:

- General safety for service personnel
- Attaching or checking the "Strong Magnetic Field" warning sign
- Attaching or checking the "Cardiac Pacemaker" warning sign
- Bringing ferromagnetic materials into the magnet area
- Safety information for working when the magnet is under field
- Hazards involved in cryogen filling
- Working with heavy loads
- Protective conductor test prior to switch-on and after repair work
- Reduction of the magnetic field to 20 mT when quenching the magnet

Using the service platform

- Prior to using the service platform, perform the relevant safety checks.

2.1 Checking the line voltage

- Lock all emergency shutdown buttons.
Prior to switch-on, several adaptations to the line voltage have to be made (refer to the Upgrade Instructions).
- Perform your own measurement.

The on-site line voltage is:	VAC
The on-site line frequency is:	Hz



WARNING

High Voltage

Adhere to the warning. Otherwise death or serious injury may occur.

- » Verify that no line voltage is present at the ACC, GPA, RCA and LHe compressor.
- » Switch off the on-site power supply (ACC: F100, F102) and secure it with a lock to prevent anyone from inadvertently switching it on.

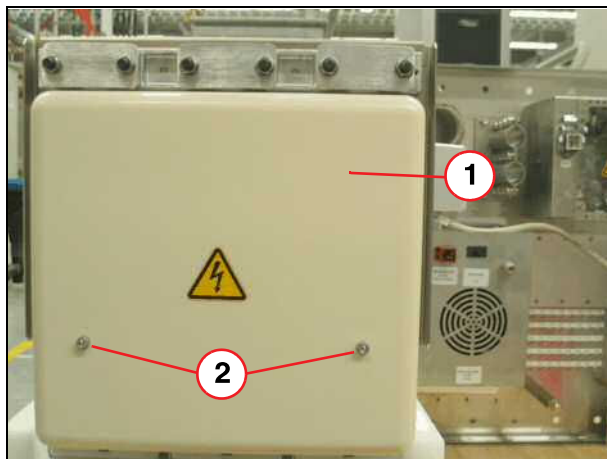


For detailed information, refer to "Upgrade Instructions, Magnet cooling".

2.2 Checking the Gradient Connections

2.2.1 Gradient cable outside HF room

Fig. 2: Protective cover, gradient filter (technical room)



- (1) Protective cover
- (2) Screws

- Removal of the protective cover (→ 1/ Fig. 2 Page 12) of the gradient filter from the filter plate.

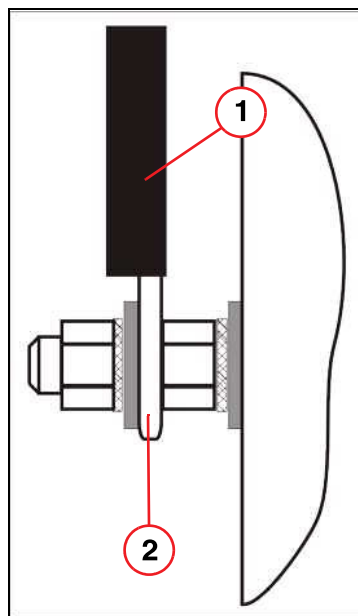


The cable lugs may point to the front or to the rear; they must not be rotated by 180°.

Fig. 3: Filter plate 1



Fig. 4: Filter plate 1/1

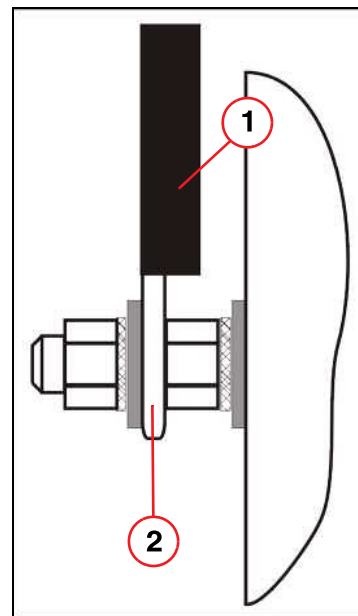


- (1) Gradient cable
- (2) Cable lug

Fig. 5: Filter plate 2

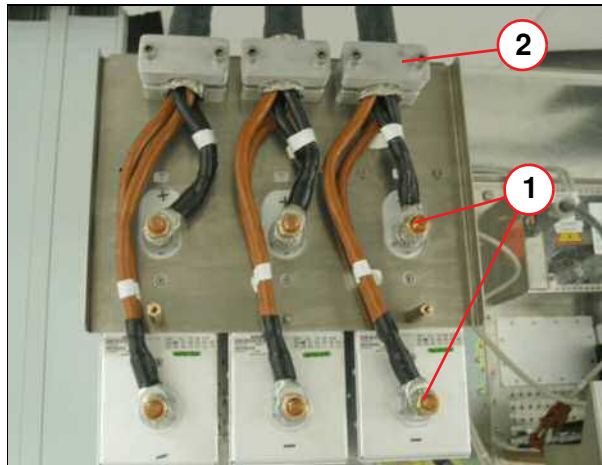


Fig. 6: Filter plate 2/1



- (1) Gradient cable
- (2) Cable lug

Fig. 7: Gradient filter gradient cable



- (1) Gradient cable
- (2) Clamp screws

- Do a polarity check of the gradient cables (X, Y, and Z) to the gradient filters.
- Check the torque of the cable connections (→ 1/Fig. 7 Page 13) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 22 ± 1 Nm
- Check whether the cable clamps (→ 2/Fig. 7 Page 13) are tightened.
- Attach the protective cover.



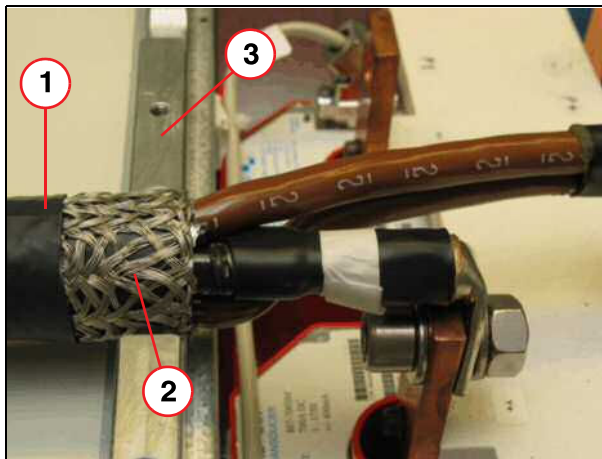
The cable clamps (→ 2/Fig. 7 Page 13) must clamp the shielding only. Ensure that the insulation is not clamped.

2.2.1.1 Check the Connection at the GPA with a 1.5m Cable Set



Ensure that the gradient cables are connected as shown in (→ Fig. 8 Page 13). The shield must be clamped between the rails.

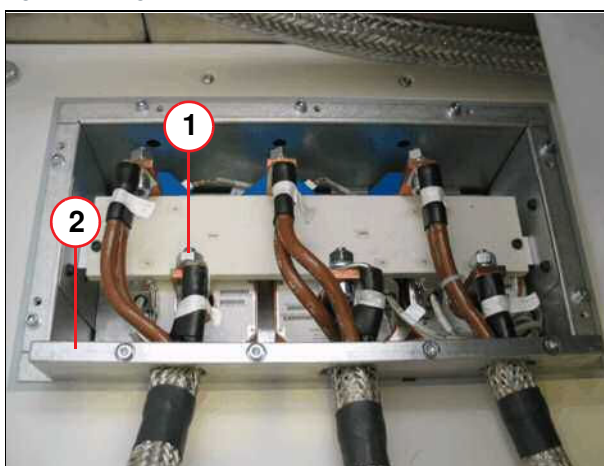
Fig. 8: GPA gradient cable



- (1) Gradient cable
- (2) Shielding
- (3) Rail

- Remove the protective cover.

Fig. 9: GPA gradient cable 1



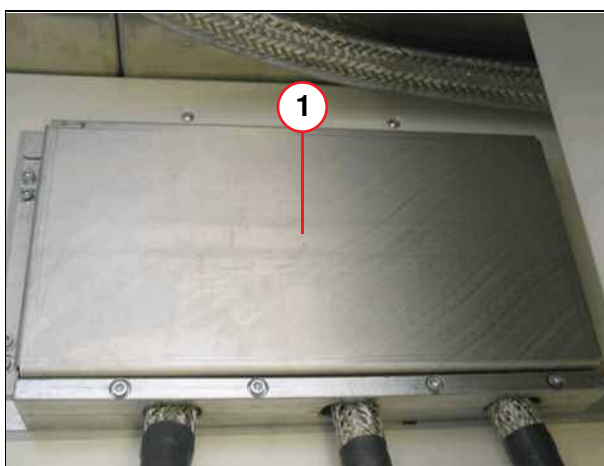
- (1) Gradient cable
(2) Rails

- Do a polarity check of the gradient cables (X, Y, and Z).
- Check the torque of the cable connections (→ 1/Fig. 9 Page 14) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 35 ± 2 Nm.
- Check whether the cable clamp rail (→ 2/Fig. 9 Page 14) is tightened.



The cable clamp rail (→ 2/Fig. 9 Page 14) must clamp the shielding only. Ensure that the insulation is not clamped.

Fig. 10: GPA gradient cover 1



- (1) Gradient cover

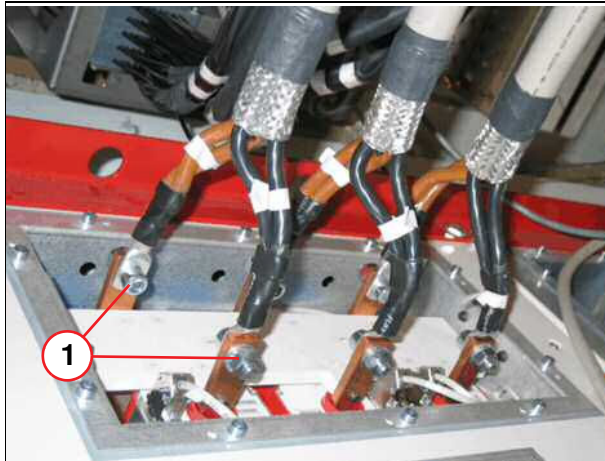
- Install the protective cover (→ 1/Fig. 10 Page 14).

2.2.1.2 Check the Connection at the GPA with a Cable Set that is Longer than 1.5m



The cable lugs may point to the front or to the rear; they must not be rotated by 180° .

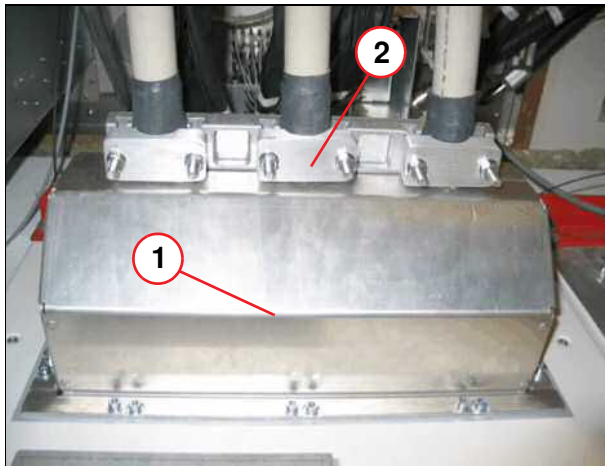
Fig. 11: GPA_gradient cable



(1) Gradient cable

- Remove the protective cover.
- Do a polarity check of the gradient cables (X, Y, and Z).
- Check the torque of the cable connections (→ 1/Fig. 11 Page 15) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 35 ± 2 Nm.

Fig. 12: GPA gradient cover



(1) Gradient cover
(2) Clamps

- Check whether the cable clamps (→ 2/Fig. 12 Page 15) are tightened.
- Attach the protective cover (→ 1/Fig. 12 Page 15).



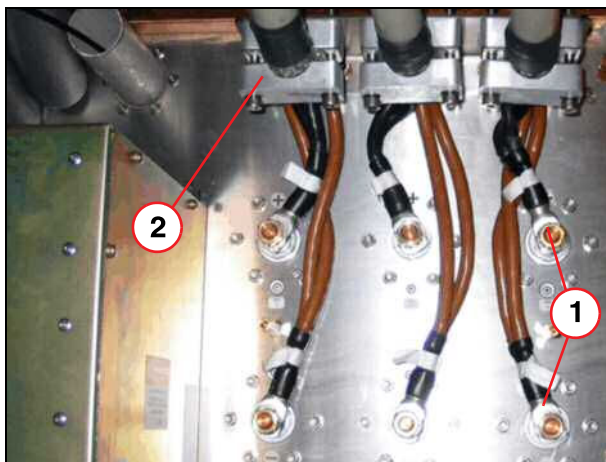
The cable clamps (→ 2/Fig. 12 Page 15) must clamp the shielding only. Ensure that the insulation is not clamped.

2.2.2 Gradient cable outside HF room



The cable lugs may point to the front or to the rear; they must not be rotated by 180° .

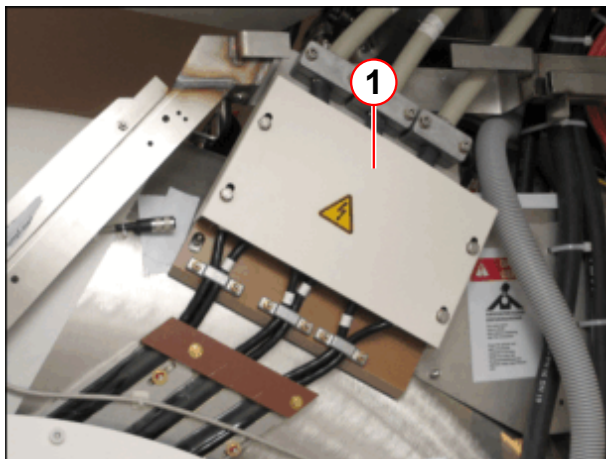
Fig. 13: Gradient filter, gradient cable UR



- (1) Gradient cable
- (2) Clamps

- Remove the protective cover.
- Do a polarity check of the gradient cables (X, Y, and Z).
- Check the torque of the cable connections (→ 1/Fig. 13 Page 16) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 22 ± 1 Nm.
- Check whether the cable clamps (→ 2/Fig. 13 Page 16) are tightened.
- Install the protective cover.

Fig. 14: Gradient cable connection plate at the magnet



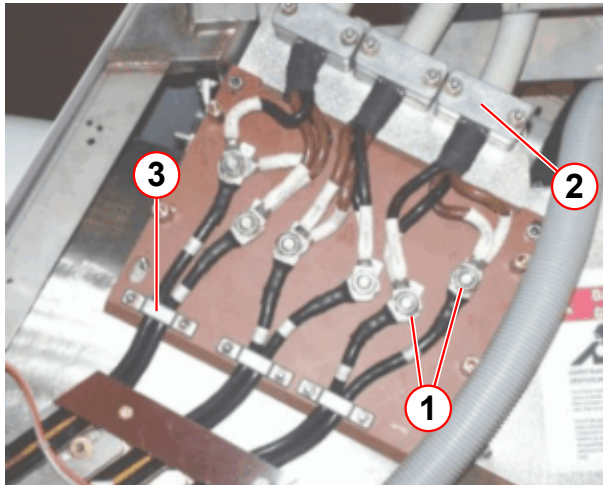
- (1) Protective cover

- Remove the protective cover (→ 1/Fig. 14 Page 16) at the magnet.



The gradient cables must neither lie across or be fixed to the basic frame of the cover.

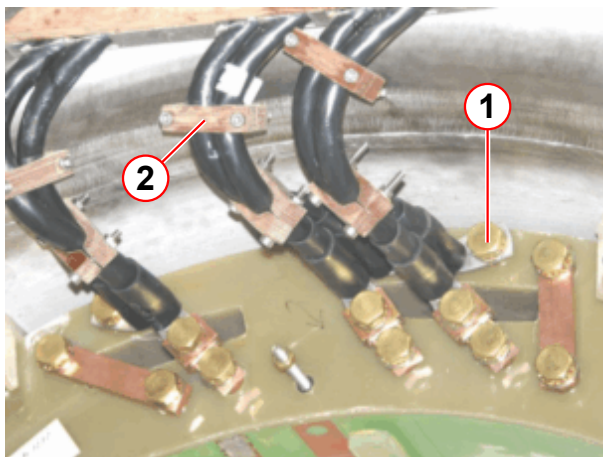
Fig. 15: Gradient connection plate



- (1) Cable connections (6x)
- (2) Upper cable clamps
- (3) Lower cable clamps

- Do a polarity check of the gradient cables (X, Y, and Z).
- Check the torque of the cable connections (→ 1/Fig. 15 Page 17) (X+ / X-, Y+ / Y-, Z+ / Z-).
 - » Apply a torque of 35 ± 2 Nm.
- Check whether the cable clamps (→ 2/Fig. 15 Page 17) and (→ 3/Fig. 15 Page 17) are tightened.
- Install the protective cover.

Fig. 16: Cables routed correctly



- (1) Cable connection (bolts SW 17)
- (2) Cable clamps

- Remove the protective cover from the gradient coil
- Visually check the correct configuration of the gradient coil connection according (→ Fig. 16 Page 17).
- Check whether all of the 6 cable clamps (→ 2/Fig. 16 Page 17) are fixed and tightened.
- Check the torque of the cable connections (→ 1/Fig. 16 Page 17).
 - » There are 13 bolts to be checked.
 - » Apply a torque of 22 ± 1 Nm.
- Install the protective cover.



The cables or cable lugs must not make contact with the fixing material at any point.



Check the cable insulation for damage in order to prevent a short. Ensure that the cables do not make contact with conductive material at any point.

2.3 Emergency shutdown buttons

For safety reasons, all emergency shutdown buttons must be tested prior to starting up the MR system.



If an UPS system is installed, the UPS system and the MR system must be shut off by pushing either one of the EPO buttons.



Refer to local codes/guidelines for the use of EPO buttons.

EPO	Function	
	o.k.	not o.k.
1		
2		
3		
4		
5		

2.4 Checking the water connection at the ACC



Check for the correct water connection on top of the ACC and on top of the RCA.
Ensure that the connection to “WATER IN” and “WATER OUT” is correct.
Incorrect water flow to the RFPA will damage the RFPA.

2.5 Switching on the system

! CAUTION

It may be unintentionally possible that cable W1139 is not installed but rather that cable W145 is directly connected to the TAS_3TM X8 at the filter plate in the magnet room.

If this is not done irreversible damage to the RF components/electronics at the magnet will occur.

- » Connect cable W1139 to TAS_3TM X8
- » Connect cable W145 to cable W1139

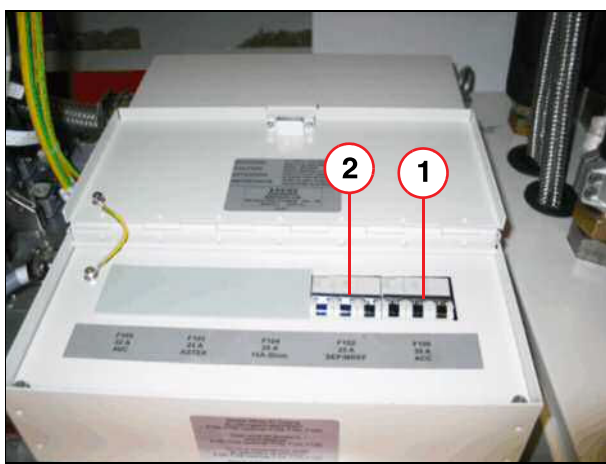
2.5.1 Circuit breakers

2.5.1.1 Circuit breakers ACC

Switch on the following circuit breakers in the ACC (top):

- F100 50A ACC
- F102 25A RCA/MREF

Fig. 17: ACC_F100_F102



- (1) F100
- (2) F102

Switch on the circuit breaker F110 in the breaker box of the ACC

- F110 GPA

Switch on the following circuit breakers in the ACC:

- F11 RFPA 1
- F29 fan box / CCS fans
- F22 RF Room
- F24 RFIS

- F21 E-Shim
- F23 AMC/RFSU
- F1 UPS
- F2 MSUP-PS
- F30 Imagecalc
- F31 MRC
- F3 36V
- F4 24V
- F5 RCA control
- F6 GPA control

Fig. 18: ACC Circuit Breaker



2.5.1.2 Circuit breakers GPA

- Switch on the circuit breakers F110 in the breaker box.

2.5.1.3 Circuit breakers MREF (helium compressor)

Switch on the following circuit breakers in the MREF:

- Circuit breaker F3 in the RCA.

2.5.2 Filling the secondary water circuit

Approximately 50 to **70 liters of distilled water** are required to fill the secondary water circuit. An additional 6 to 10 liters of distilled water are required to fill the magnet power supply.

- With the hoses at maximum hose length of 4 x 15 m between the ACC and RCA, approximately 13 more liters of water are needed to fill the circuit than at the minimum hose length of 4 x 2.5m (you need approximately 0.285 liters of water per meter of 3/4" hose used).



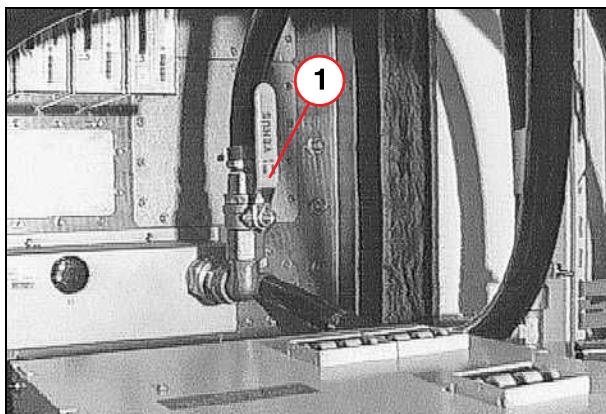
Do not use deionized water; use distilled water only.

- Remove the cover of the "cooler".
- Open the following valves in the secondary water circuit:
 - At the filter plate (quick disconnect (→ Fig. 19 Page 22)
 - Return for gradient coil and GPS at the RCA (refer to (→ Fig. 20 Page 22).
- Close the water flow valves (→ Fig. 29 Page 29) and then open them a maximum of two to three turns (at 50 Hz line frequency) or open them a maximum of one turn (at 60 Hz).



If the water flow valves are open too far, the overpressure monitor will switch off the pump shortly after it started.

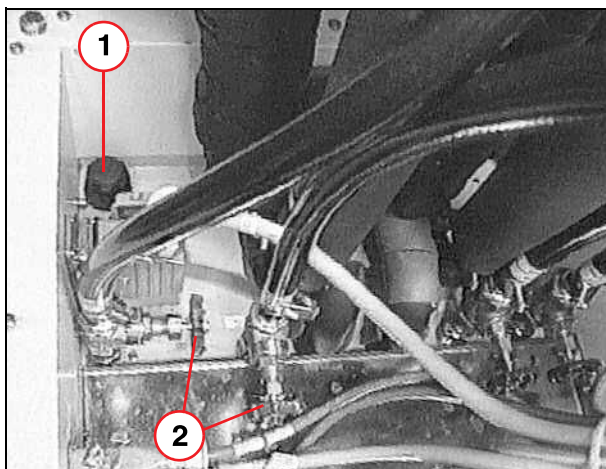
Fig. 19: Water valve opened at filter plate



(1) Ball valve

- Open the valve cap of the water tank and insert the funnel extension.

Fig. 20: Open the valve cover of the water tank and insert the extension hose of the funnel



(1) Water tank sealing cap
(2) Return valve open max.

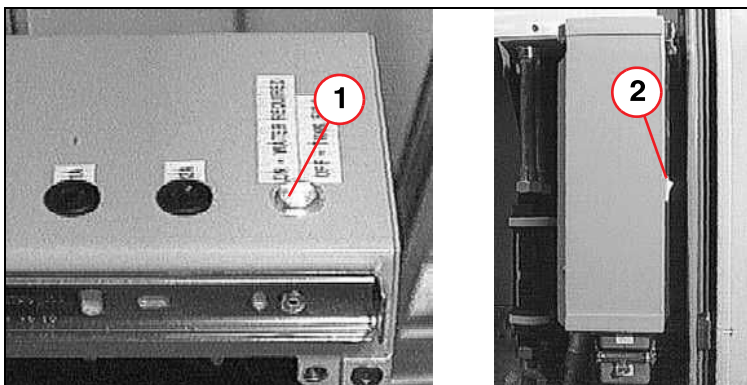
- Switch on fuse F1 "Cooler 1".
- Fill the water tank to its maximum level with distilled water (monitor the yellow lamp - refer to (→ Fig. 22 Page 23)).

Fig. 21: Filling procedure at RCA cabinet (front)



- Set the pump switch to "MAN".
 - The pump gradually starts and fills the water circuits of the gradient coil and the electronics cabinets. When the minimum acceptable water level is reached, the pump switches off automatically. No air is pumped into the circuits. The yellow lamp is illuminated.

Fig. 22: Water indicator - yellow lamp LP1



- (1) LP1
(2) Auto / Man switch

- Refill the water tank, until the yellow lamp goes out.
 - At that point, the pump will gradually begin to run again and refill the water circuit.
- The cycle begins again; refill the tank with distilled water until the yellow lamp goes out.



Refill the water only when lamp LP1 lights up. Refill only enough water until lamp LP1 goes out. A residual air pocket must remain in the tank.

- If lamp LP1 remains off when the pump is running, the refill procedure is ended. Close the cap of the water tank.

2.5.3 Adding Citric Acid to the Secondary Water Circuit

After filling has been completed, citric acid must be added to the water circuit (citric acid powder is part of the delivery volume).

The purpose of adding citric acid is to prevent germ build-up in the water circuit.

! CAUTION
Adding citric acid The citric acid can cause irritation to skin and eyes » Avoid contact with eyes and skin. Wear goggles and the gloves provided when handling citric acid. Follow the safety instructions in the enclosed "Handling Instructions".

Fig. 23: Citric acid set



- (1) Goggles
- (2) Citric acid powder, 8 grams
- (3) Gloves
- (4) Warning label
- (5) Handling instructions

- Dissolve 8 grams of citric acid in 250 milliliters of distilled water.

Fig. 24: Dissolving the citric acid



- Add the citric acid solution to the cooling circuit.

Fig. 25: Adding the citric acid solution



- Affix the warning label provided to the cooler.

Fig. 26: Warning label



(1) Position for warning label



CAUTION

Adding citric acid

The citric acid can cause irritation to skin and eyes

» Wash hands thoroughly after handling.



The coolant turns turquoise after adding citric acid.



Cooling water containing citric acid that was collected while draining the water circuit should be refilled into the secondary cooling circuit.



Follow the country-specific guidelines when disposing of cooling water containing citric acid.



The acidity of the citric acid is neutralized after it has been dissolved in water for several days.

2.5.4 Checking and setting the water flow rates

The flow rates of the primary and the secondary water circuit can be checked using the service software ("SeSo"). All following steps should be executed "on-line" at the MRC. On-line Help explains the individual steps.

- Access the service menu in the SeSo (Service Software) as follows at the MRC:
- Enter your password and select menu item **"Magnet & Cooling"**.
- Select **"Cooling Status"** in the "Action" menu (takes up to 20 sec in time).
- The current flow rates are listed in the field **"Actual Cooler Values"** under **"Water-flow"**

2.5.5 Adjusting the primary water flow rate



Check the primary water flow rate preventing temperature stability problems.

- Check and adjust the primary water flow rate
 - » The flow rate should be in the range of 45 l/min to 55 l/min
- Check the primary water flow rate via Magnet & Cooling status

Fig. 27: Magnet & Cooling status

Magnet & Cooling			
Event Log Reports Home Help			
Action			
Cooling Status			
Cabinet		ACC	25.90°C
		GPA	24.53°C
Actual Cooler1 Values			
Water Level		NORMAL (Normal, Low)	
Water Flow (only Cooler1)		prim	49 l/min
		Gradient Coil	28 l/min
		ACC/GPA Suite	38 l/min
Water Temp.		primary	10.7 °C
		secondary	19.9 °C

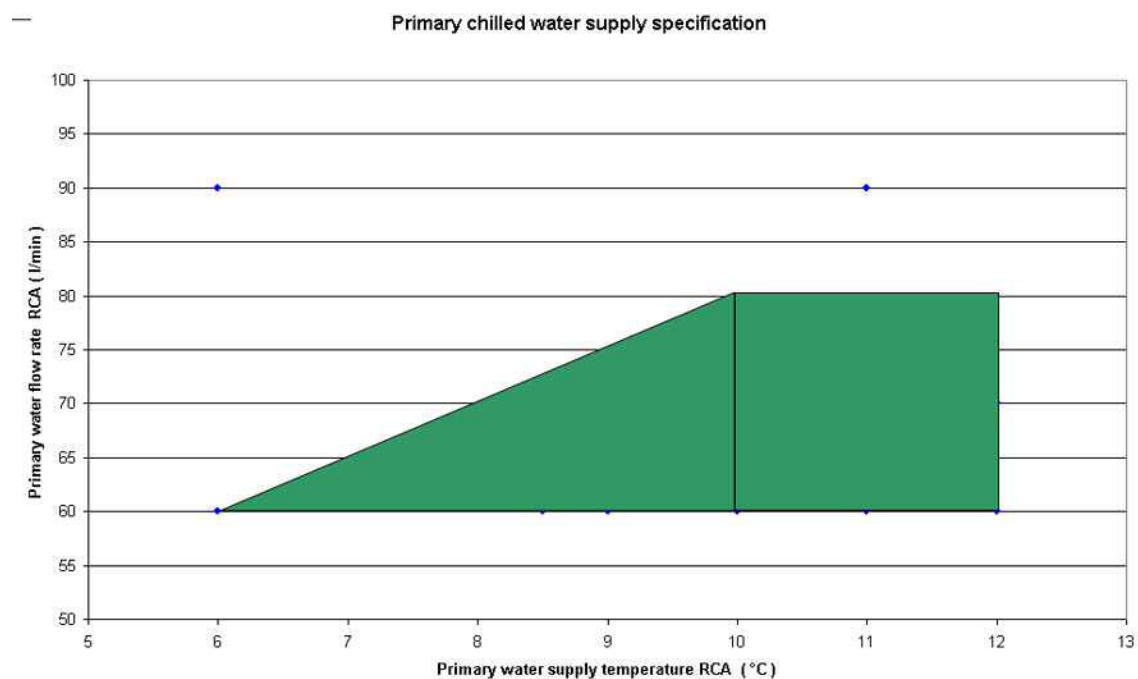


The status menu of "Magnet & Cooling" shows only the primary water flow rate for the cooler. The flow rate for the helium compressor (approx. 10 l/min) and for the water-cooled cabinet cooling system (approx. 10l/min) is not shown.

If the flow rate is out of specification:

- » Check the position of the valves of the primary water circuit on top of the RCA.
 - » Check the flow rate of the hospital central cooling or the chiller.
- Please refer to the digram for the water volume and water temperature required on-site (→ Fig. 28 Page 28)

Fig. 28: On-site chilled water supply



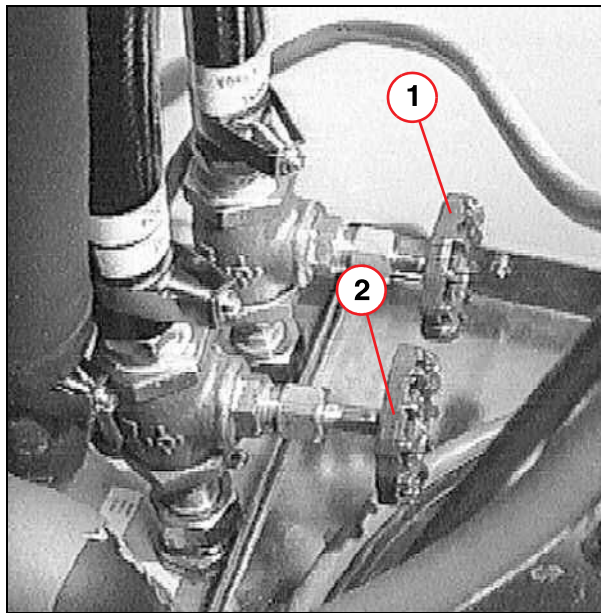
2.5.6 Adjusting the secondary water flow rate

- Adjust the flow valves (refer to (→ Fig. 29 Page 29)), make sure that the flow rates are within specification, refer to (→ Tab. 1 Page 29).



Adjust the flow from the closed position of the valve in small steps to the required value. If the water flow valves are open too far, even if the flow rates are within spec, the over-pressure monitor will switch the pump off shortly after it started. It takes some time for the software to indicate the change in flow.

Fig. 29: Slowly adjust the water flow valves for flow control



- (1) CCA/GPA
(2) Bypass valve

Tab. 1 Flow rates and warning levels for MAGNETOM Symphony, A Tim System

Symphony, A Tim System	Quantum	
	Gradient coil	ACC/GPA Suite
Flow rates	29 l/min	38 l/min
Warning level	24 l/min	33 l/min
Primary water flow rate without helium compressor and CCS	45 l/min - 55 l/min ¹	
Warning level	25 l/min	

1. Value is shown in the magnet & cooling platform

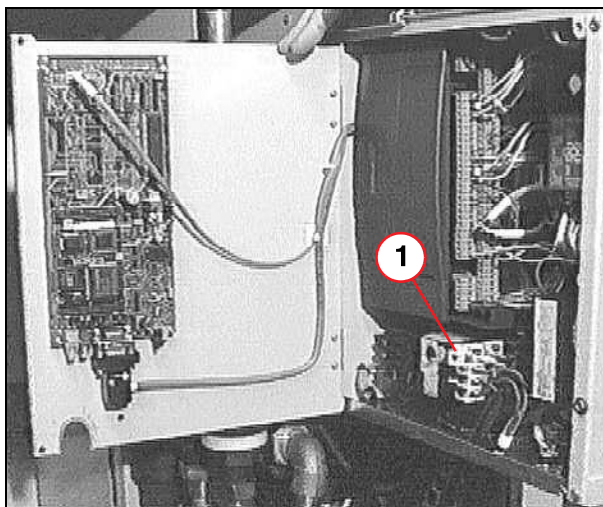
i

If the indicated flow rates cannot be reached (e.g. only 2/3 of the indicated flow), the pump rotation is not correct.

In this case, switch two of the connection wires (50, 51, 52) at the motor protection circuit in the pump control (refer to the circuit diagram). Then perform the flow adjustment again.

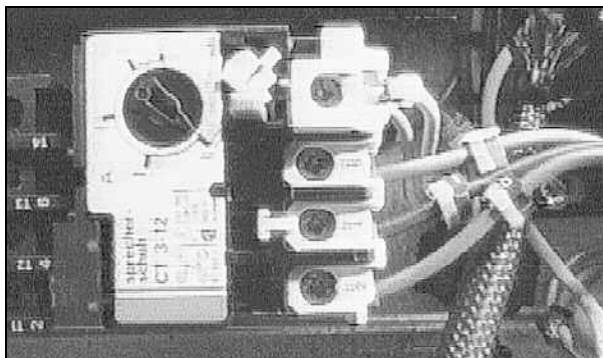
- Recheck the secondary hose connections for leaks.

Fig. 30: Electronics box



(1) F1 overload

Fig. 31: F1 overload motor protection - change the pump rotation by switching the wires



- Adjust the pump switch to "Auto" (→ Fig. 22 Page 23).
- Switch on the entire MR system at the alarm box with the "System ON" switch.

2.5.7 Setting the temperature set point

- Select "Initialization" in the "Action" menu.
- Check the standard temperature value under "cooler" in the "Temperature set point" field. The value must be 20°C to avoid water condensation.
- Save if settings have been changed.

2.5.8 Checks after 12, 24 and 48 hours

- Check the level in the water tank. If the yellow lamp LP1 (→ Fig. 22 Page 23) glows, refill with distilled water until the lamp goes out.
- Check the "flow rates" again in the service software.

2.5.9 Setting the screen reset time

- Select "Initialization" in the "Action" menu.
- Enter the time for reset under "Time Setting" in the "Magnet" field.
 - The reset must occur outside the normal time frame for patient examinations. Select e.g., '2:00 a.m.'
- Save the settings.

2.5.10 The alarm box

Fig. 32: Alarm box



The alarm box indicates the system status via LEDs:

- LINE POWER
- HELIUM LEVEL
- MAG STOP
- PRESSURE (not used with Symphony, A Tim System)
- COMPRESSOR
- BATTERY LOW
- EIS
- COMMS FAULT

The alarm box contains the following switch functions:

Previous status	Push button... to change status to...		
	Host Standby	System ON	System OFF
MRC OFF	MRC ON	MRC ON	No action
System OFF	System remains OFF	System ON	

MRC ON System OFF	No action	MRC remains ON System ON	MRC OFF System OFF
MRC ON System ON	MRC remains ON System OFF	No action	MRC OFF System OFF

Figure showing alarm box

2.6 Warning signs



Prior starting up the magnet system make sure that the warning signs are posted!

The warning signs are available in different languages. Pick the correct language. The following pictures are examples.

WARNING

Strong magnetic field.

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.
- » The sign has to be posted in a way that it is visible even when the door to the controlled area is opened!

Fig. 33: Warning sign



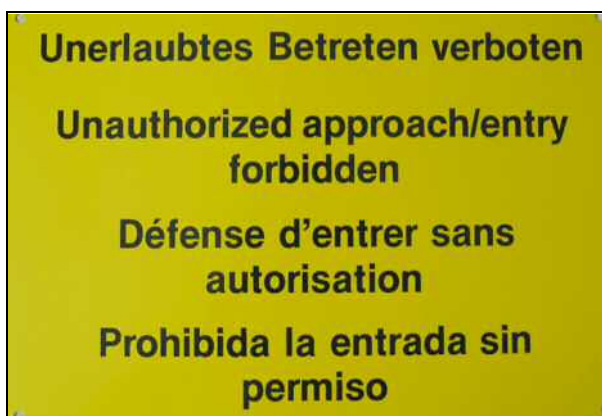
Fig. 34: Warning sign "Pacemaker"



Fig. 35: Warning sign "Filling-up"



Fig. 36: Warning sign "Unauthorized"



2.7 Hearing protection sign

The hearing protection sign is packed together with the warning signs.



The hearing protection sign must be posted at the main entrance door of the examination room. Attach the sign at eye level.

Fig. 37: Hearing protection sign

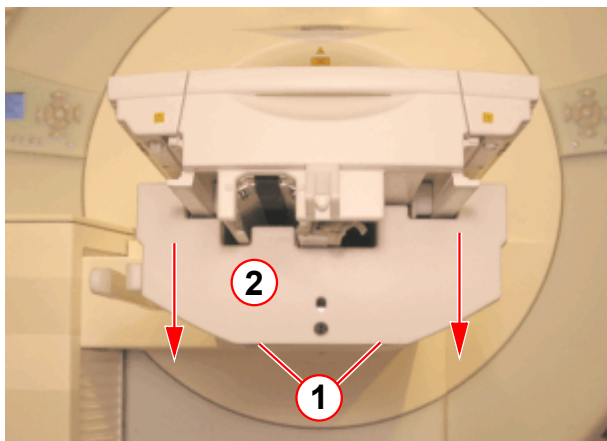


2.8 Partial drive unit (optional)

2.8.1 Check the toothed belt tension for the partial body drive unit

- Move the table top in **Home Position**

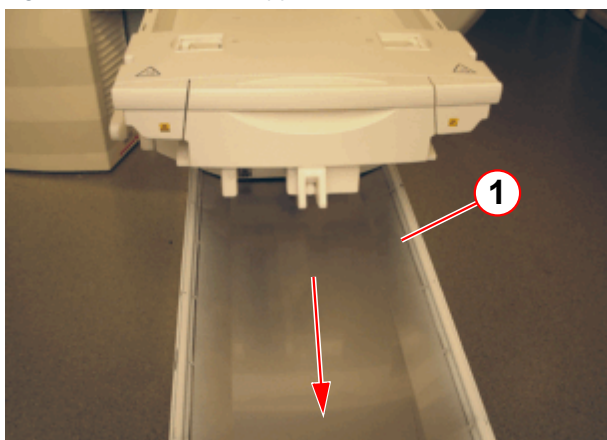
Fig. 38: Patient tabletop support frame foot end cover



- (1) 4 mm allen screws
- (2) Foot end cover

- Remove the 2 screws on the underside of the foot-end cover(1).
- Remove the foot-end cover (2) by sliding down.

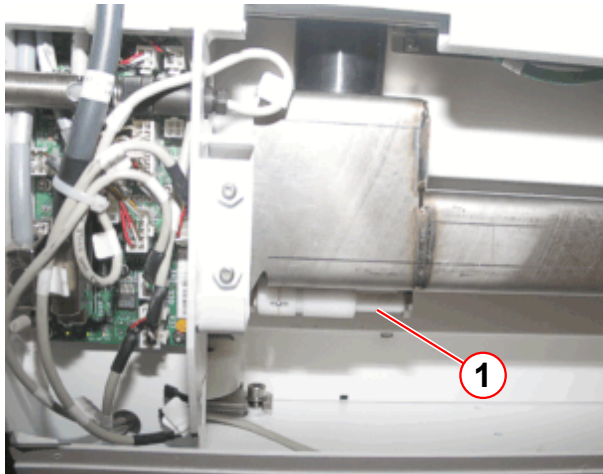
Fig. 39: Patient table support frame bottom cover



- (1) Bottom cover

- Remove the bottom support frame cover (1) by sliding out.

Fig. 40: Support frame; bottom side

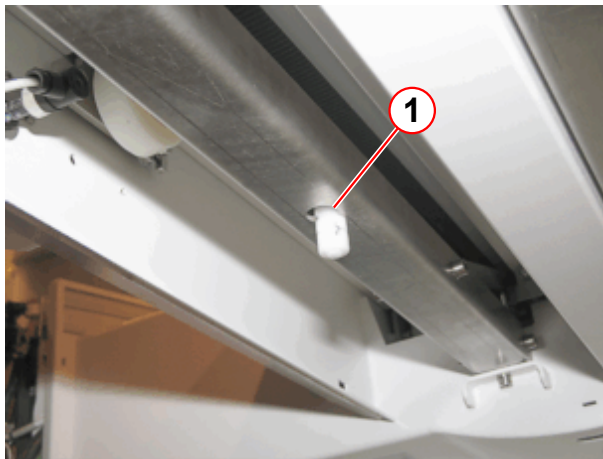


(1) Storage position of the service tool - toothed belt tension

The toothed belt tension service tool is located under the table top in the support frame.

- Remove the toothed belt tension service tool.

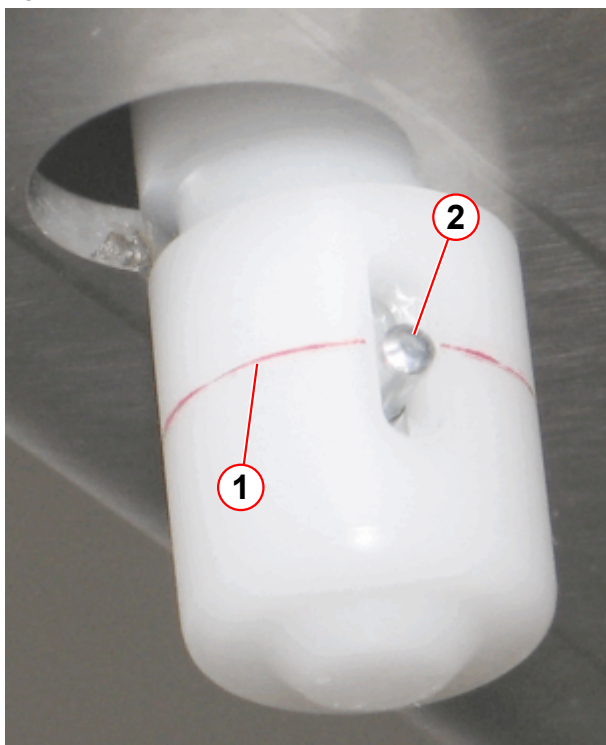
Fig. 41: Support frame bottom side



(1) Service tool - toothed belt tension in use

- Insert the toothed belt tension service tool (1) in the hole on the support frame bottom side, refer to (→ Fig. 42 Page 38).

Fig. 42: Service tool - toothed belt tension



- (1) Indication pin
- (2) Ring marking

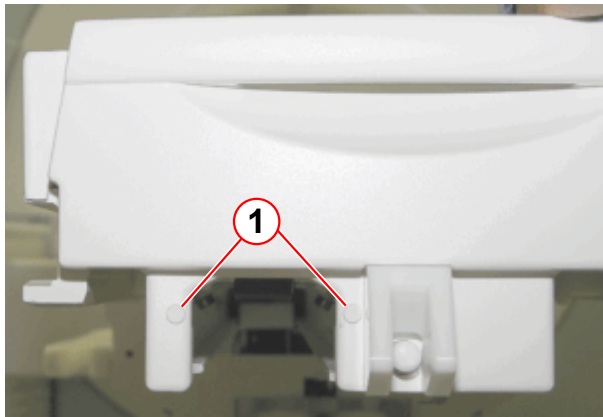
The indication pin shows the belt tension.

- » The indication pin must be aligned with the marking ring.
- » If the indication pin does not align with the ring marking, adjustment is necessary; refer to (→ Adjustment of the toothed belt tension / Page 39).

- Remove the toothed belt tension service tool and return it to its storage position under the table top in the support frame, refer to (→ Fig. 40 Page 37).
- Reinstall the bottom support frame cover, refer to (→ Fig. 39 Page 36).
- Reinstall the foot-end cover, refer to (→ Fig. 38 Page 36).
- Mark OK in the protocol.

2.8.2 Adjustment of the toothed belt tension

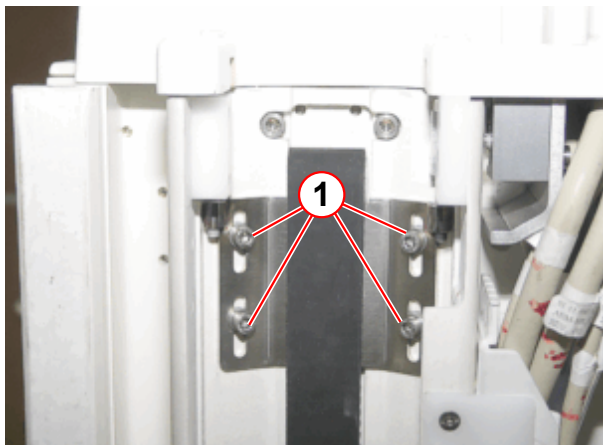
Fig. 43: Partial body patient table foot end side



(1) Blanking plug

- Remove the blanking plugs.

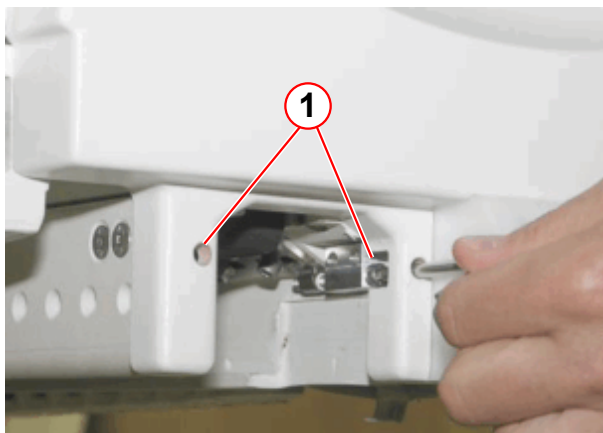
Fig. 44: Tension unit for the toothed belt



(1) Fixing screws (4 mm Allen screw)

- Loosen the fixing screws, do not remove.

Fig. 45: Toothed belt tension adjustment



(1) Adjustment screws for the belt tension

- Adjust the toothed belt tension with a 4 mm Allen key.
- Turn both adjustment screws (cw) equally until the indication pin is aligned with the marking ring, refer to (→ Fig. 42 Page 38).
- Tighten the fixing screws, refer to (→ Fig. 44 Page 39).

- Check the toothed belt tension, refer to (→ Fig. 42 Page 38).

- Remove the toothed belt tension service tool and return it to its storage position under the table top in the support frame, refer to (→ Fig. 40 Page 37).
- Insert the blanking plugs, refer to (→ Fig. 43 Page 39).
- Reinstall the bottom support frame cover, refer to (→ Fig. 39 Page 36).
- Reinstall the foot-end cover, refer to (→ Fig. 38 Page 36).

2.9 T-class (optional)

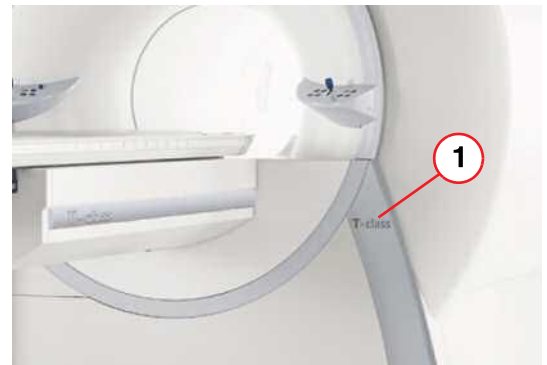
If the MR system includes the **T-class** and **removable table** options, the T-class accent strips must be attached to the left and right side of the patient table.

Fig. 46: T-class (example Avanto)



(1) T-class

Fig. 47: T-class (example Symphony, A Tim System)



(1) T-class



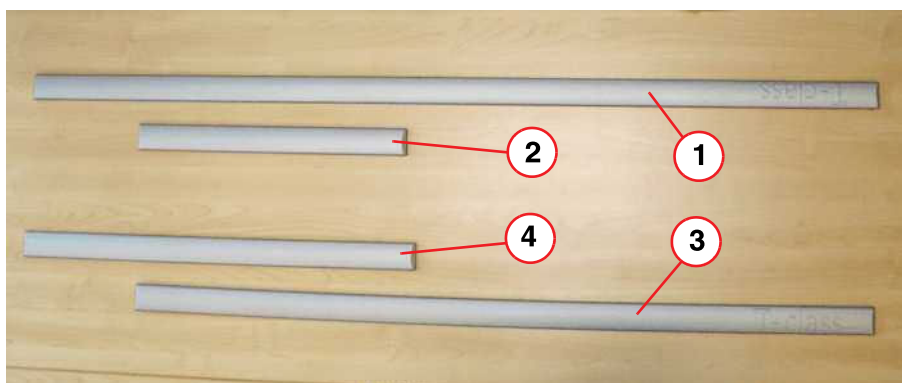
Determine the side to be used for docking the trolley together with the customer.



If the optional breast biopsy device is available, the accent strips are not used.

The T-class accent strips are stored at the system by the service provider (e.g. on the patient table).

Fig. 48: Accent strips for the removable table



- (1) Accent strip for the right side of the patient table with docking at the left (length: 1275 mm)
- (2) Accent strip for the left side of the patient table with docking at the left (length: 415 mm)
- (3) Accent strip for the left side of the patient table with docking at the right (length: 1075 mm)
- (4) Accent strip for the right side of the patient table with docking at the right (length: 613 mm)

- Remove the caps at the left and right side of the patient table (support frame) to prepare for gluing the accent strips.
- Clean the surface of the patient table to prepare for gluing the accent strips.

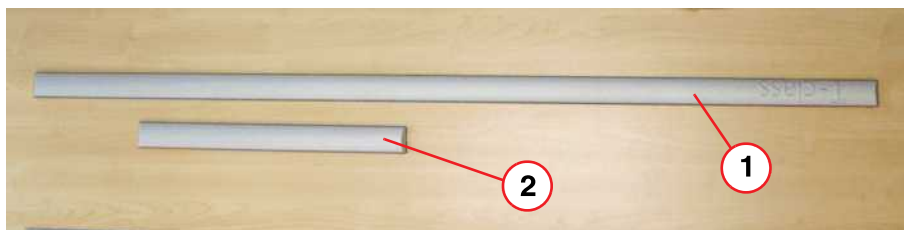
- Move the tabletop into the magnet.

2.9.1 Removable table

There are two different types of accent strips for the removable table, depending on the docking position of the trolley (left or right).

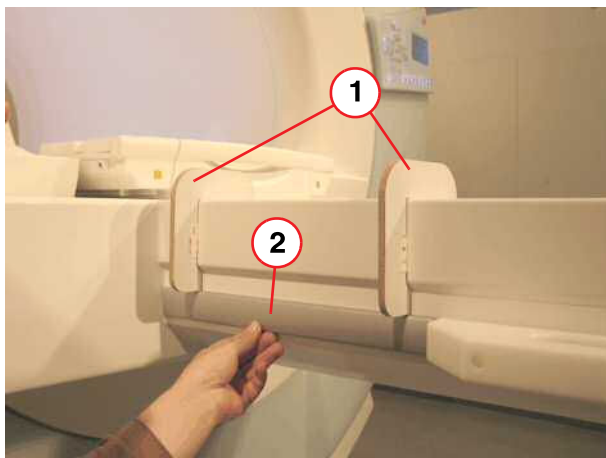
2.9.1.1 Docking position at the left side of the patient table

Fig. 49: Accent strips for the removable table with docking at the left side



- (1) Accent strip for the right side of the patient table with docking at the left side (length: 1275 mm)
- (2) Accent strip for the left side of the patient table with docking at the left (length: 415 mm)

Fig. 50: Affix the T-class accent strips (docking on left side)

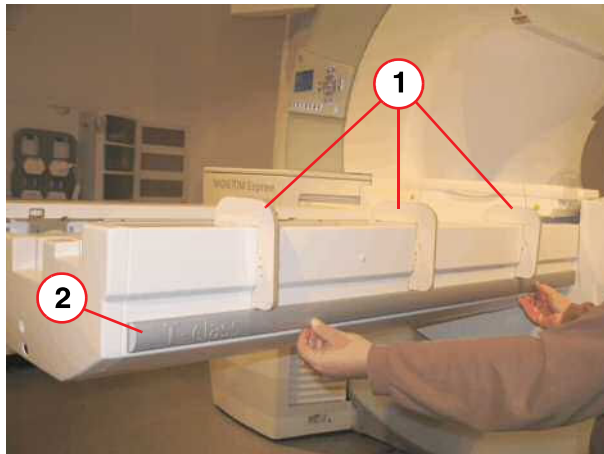


- (1) Adjustment tool
- (2) T-class accent strip (length: 415 mm)

Left side of the patient table:

- Affix the adjustment tools (1) at the patient table (support frame).
- Remove the protective foil from the T-class accent strips (length: 415 mm).
- Align the T-class accent strips (2) with the adjustment tools (1) and carefully affix them to the patient table.
- Remove the adjustment tools.

Fig. 51: Affix the T-class accent strips (right side)



- (1) Adjustment tool
(2) T-class accent strip (length: 1275 mm)

Right side of the patient table:

- Affix the adjustment tools (1) at the patient table (support frame).
- Remove the protective foil from the T-class accent strips (length: 1275 mm).
- Align the T-class accent strips (2) with the adjustment tools (1) and carefully affix them to the patient table.
- Remove the adjustment tools.

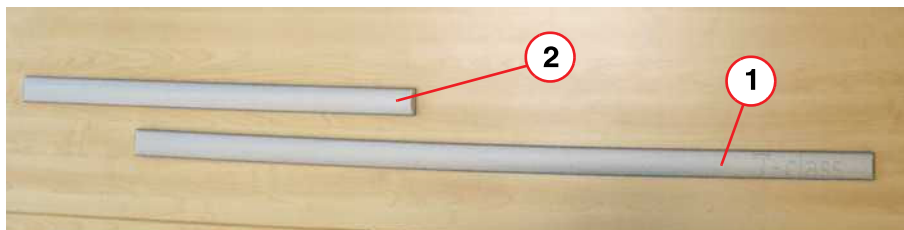
- If the mechanical end stop was removed, install it again and deactivate the emergency release of the patient table.



The accent strips for the docking position at the right side of the patient table are not required.

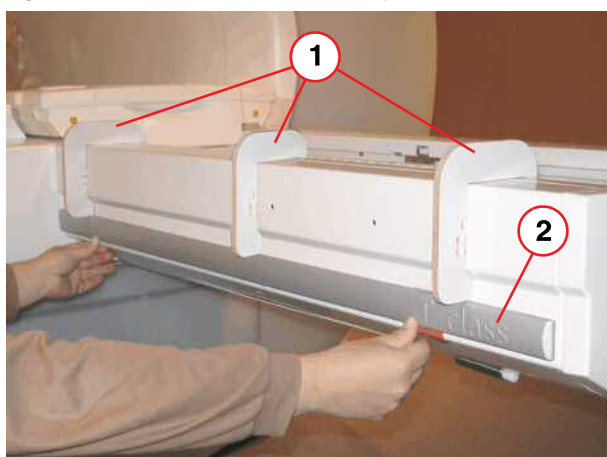
2.9.1.2 Docking position at the right side of the patient table

Fig. 52: Accent strips for the removable table with docking at the right side



- (1) Accent strip for the left side of the patient table with docking at the right (length: 1075 mm)
(2) Accent strip for the right side of the patient table with docking at the right (length: 613 mm)

Fig. 53: Affix the T-class accent strips (left side)

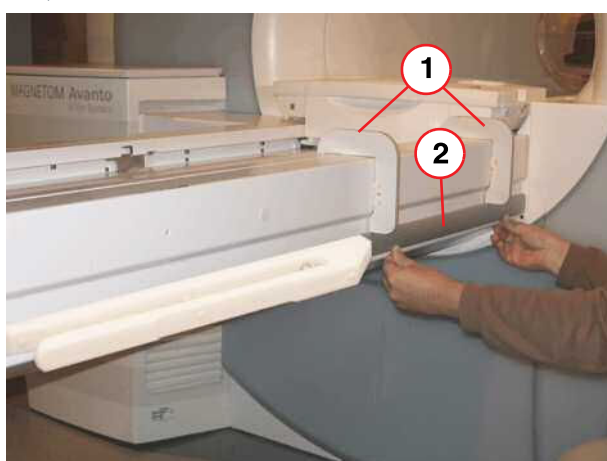


- (1) Adjustment tool
(2) T-class accent strip (length: 1075 mm)

Left side of the patient table:

- Affix the adjustment tools (1) at the patient table (support frame).
- Remove the protective foil from the T-class accent strips (length: 1075 mm).
- Align the T-class accent strips (2) with the adjustment tools (1) and carefully affix them to the patient table.
- Remove the adjustment tools.

Fig. 54: Affix the T-class accent strips (docking on right side)



- (1) Adjustment tool
(2) T-class accent strip (length: 613 mm)

Right side of the patient table:

- Affix the adjustment tools (1) at the patient table (support frame).
- Remove the protective foil from the T-class accent strips (length: 1275 mm).
- Align the T-class accent strips (2) with the adjustment tools (1) and carefully affix them to the patient table.
- Remove the adjustment tools.

- If the mechanical end stop was removed, install it again and deactivate the emergency release of the patient table.



The accent strips for the docking position at the left side of the patient table are not required.

3.1 General

Performing the software configuration of the system (Options/Service/Local Service/Configuration). Most of the software configuration steps performed are completely automatic. Follow the instructions shown on-screen. The pop-up windows will prompt you to perform the actions required.

- Check the MeasSettings in the Service Software/Configuration/Menu.

3.1.1 Remote Diagnostics



It is important to install and configure the Rdiag to get support from the HSC in case you need it during installation.

For detailed information regarding rdiag configuration, refer to chapter 7 in the "Installation Software."

3.1.1.1 Introduction

Remote Diagnostics allows you to establish a connection from a remote location in order to provide remote service to the system, on-line technical assistance, automated reporting of system quality assurance data (QDB, EDB), magnet monitoring data, and system utilization data (customer option).

Only router connections are supported for new installations equipped with the syngo MR system.

3.1.1.2 Supported connections

- ISDN router (router provided by SIEMENS)
- Customer ISDN router
- Direct connection via Internet Security EndPoint (VPN tunneling) Authentication: SHA1 or MD5 supported encryption: DES or 3DES
- Router with analog modem interface via modem (router provided by SIEMENS)
- Customer router with analog modem interface via modem

3.1.1.3 Responsibility

The project manager is responsible for planning a remote connection in cooperation with the customer.

3.1.1.4 Support

Only Internet Explorer 6.0 has been tested and released as a service browser for remote service involving syngo MR systems. The "SIEMENS Remote Services TM" will support router connections only for NEWLY delivered systems. See <http://cs.med.siemens.de/> "For Service", "Remote Services" for the most recent information regarding the router configuration. The project manager is responsible for planning a router connection to the Siemens Remote Diagnostic server. In case you have questions regarding your preconfigured router from Siemens, please contact **srs_final@med.siemens.de** or your responsible remote diagnostic technician.

3.2 TestTools

Mandatory testing of the TestTools is required at the end of the installation.

Instead of running all tests in one sitting, it is recommended to perform them individually over a period of time (e.g., during wait times, lunch breaks, etc.).

To get an idea of how much time is needed for the tests, please refer to the following table. Please plan to perform the tests accordingly:

Test time in hh:mm:ss ¹	functional	communication/status	interactive	complete
AMC	00:25:50	00:01:19	00:05:00	00:31:09
RF	00:54:21	00:00:45	03:00:08	03:55:14
RCCS-int			02:30:00 ²	
GPA	00:24:54	00:00:20		00:25:14
IMAGER	00:05:50			00:05:50 ³
HOST	00:00:05		00:10:00	00:10:05
All comm. tests		00:02:19		

- 1.
2. If the system is equipped with eight receive channels, the RCCS-int test requires approx. 45 minutes.
3. If the system is equipped with eight receive channels, approximately four minutes are required for the IMAGER test.

Normally it is not necessary to perform the cable tests in the RF-section. They should be used to isolate defective components during troubleshooting only.

The following tests can be performed prior to Tune-Up/QA as they do not require a tuned system:

- AMC complete
- Imager complete
- Host complete
- Communication tests complete



Before handing over system, please ensure that all tests of the TestTools have been performed and are within specification.



The RCCS-int test requires no interaction, only a pop-up has to be confirmed at the start of the test (not at the start of the test session!).

4.1 Magnet & Cooling

4.1.1 Checking the correct MSUP ID plug

- Install the MSUP display unit
- Check the message on the display unit.

Fig. 55: MSUP Display Unit



- Mk I Cold head
 - » 0701 OR70-2

Fig. 56: MSUP Display Unit



- Mk II Cold head
 - » 0070 OR70-2

4.1.2 Procedure of calibrating the helium level probe

- Open Magnet & Cooling, Initialization platform.
- Enter the resistance of the helium level probes.
 - » "0%" values are in the "Installation data sheet" from SMT delivered with the magnet.
 - » Default value for "100%": 5.0 ohm

Fig. 57: Initialization

Siemens Med Service Software - Microsoft Internet Explorer provided by Siemens AG Medical Soluti...

Magnet & Cooling | Event Log | Reports | Home | Help

Action
Initialization

Magnet Helium Level Probe 1 min(0%) 105.2 Ohm max(100%) 5.0 Ohm
Probe 2 min(0%) 105.2 Ohm max(100%) 5.0 Ohm

Time Setting Daily at 02:00 o'clock

Magnet Current 583.50 A

Cooler1 Temperature Set Point 20 °C

Ready | Save Settings | Default | Upload | Download

- Save the settings

4.1.3 Checking the parameters

- Select the menu item "Magnet & Cooling".
- Select "Initialization" in the "Action" menu

Fig. 58: Initialization

Siemens Med Service Software - Microsoft Internet Explorer provided by Siemens AG Medical Soluti...

Magnet & Cooling | Event Log | Reports | Home | Help

Action
Initialization

Magnet Helium Level Probe 1 min(0%) 105.2 Ohm max(100%) 5.0 Ohm
Probe 2 min(0%) 105.2 Ohm max(100%) 5.0 Ohm

Time Setting Daily at 02:00 o'clock

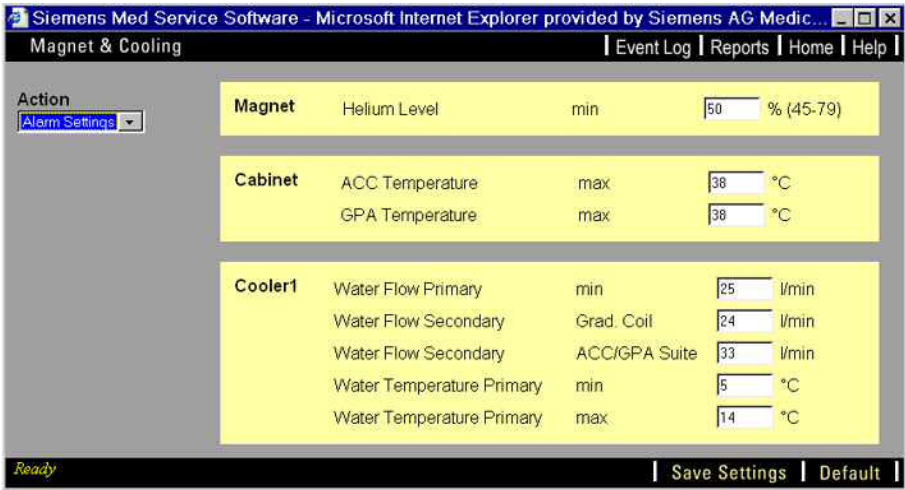
Magnet Current 583.50 A

Cooler1 Temperature Set Point 20 °C

Ready | Save Settings | Default | Upload | Download

- Check and compare the magnet-specific values with the values in the SMT document.
- Check "Time Setting"; the default setting is 02:00.
- Check the set point of the water temperature; the default setting is 20 °C.
- Select "Alarm Setting" in the "Action" menu.

Fig. 59: Alarm Settings

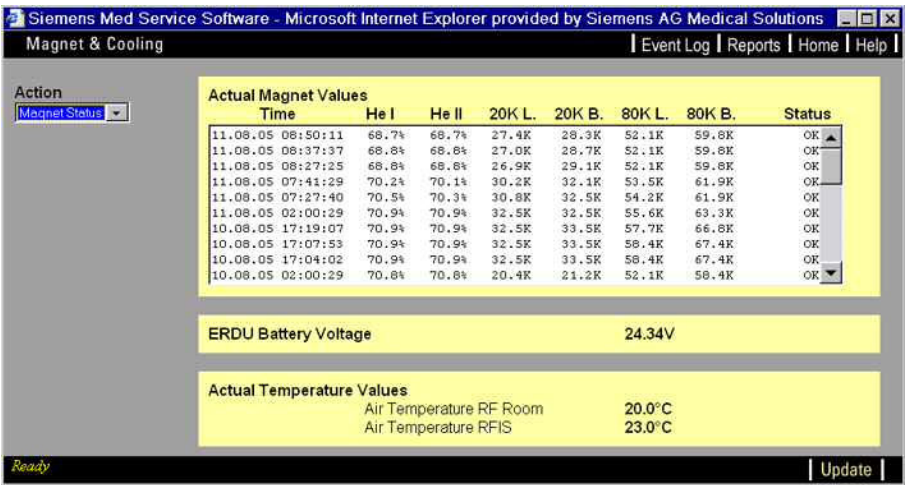


- Check the warning value of the helium level. If necessary, change it.

4.1.4 Checking the Magnet & Cooling values

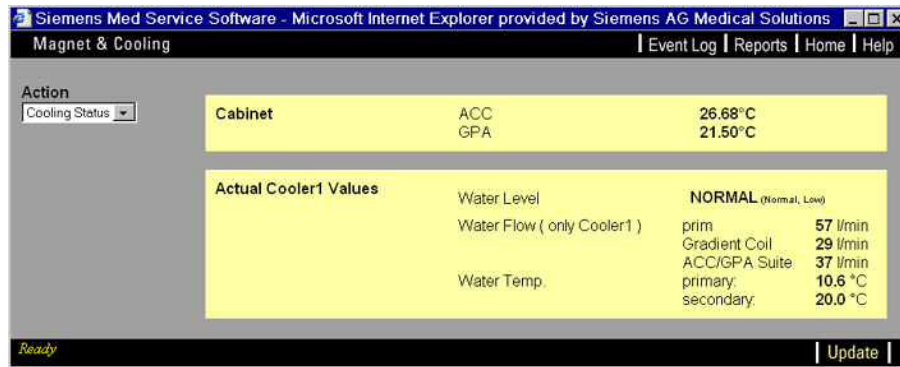
- Select "Magnet Status" in the "Action" menu
By selecting "Magnet Status", a helium measurement will be performed.

Fig. 60: Magnet Status



- Select "Cooling Status" in the "Action" menu

Fig. 61: Cooling Status



4.2 Serial number set-up



Necessary only for a Symphony, A Tim System upgrade!

Serial numbers of the magnet and the MSUP have to be set up in the MSUP.

Using a Laptop or the HyperTerminal of the MR system to do the set-up.

- Connect the HOST via the RS232 cable to the MSUP X5. RS232 cable of the MPS can be used.
- Use the HyperTerminal with the following settings:

Baud rate	9600
Data bits	8
Parity	No
Stop bit	1
Flow control	No
Terminal emulation	VT100

To enter the magnet and the MSUP serial / revision number via the HyperTerminal, the following commands should be performed:

- Type **"Run"** or **"R"** to enter the process display

Fig. 62: HyperTerminal

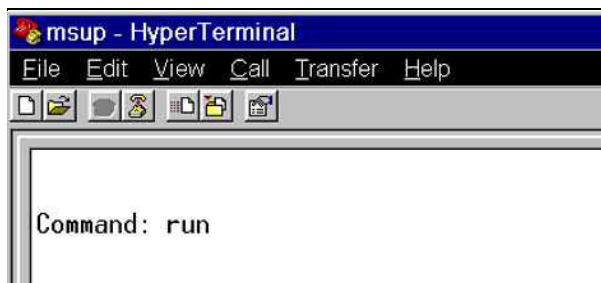
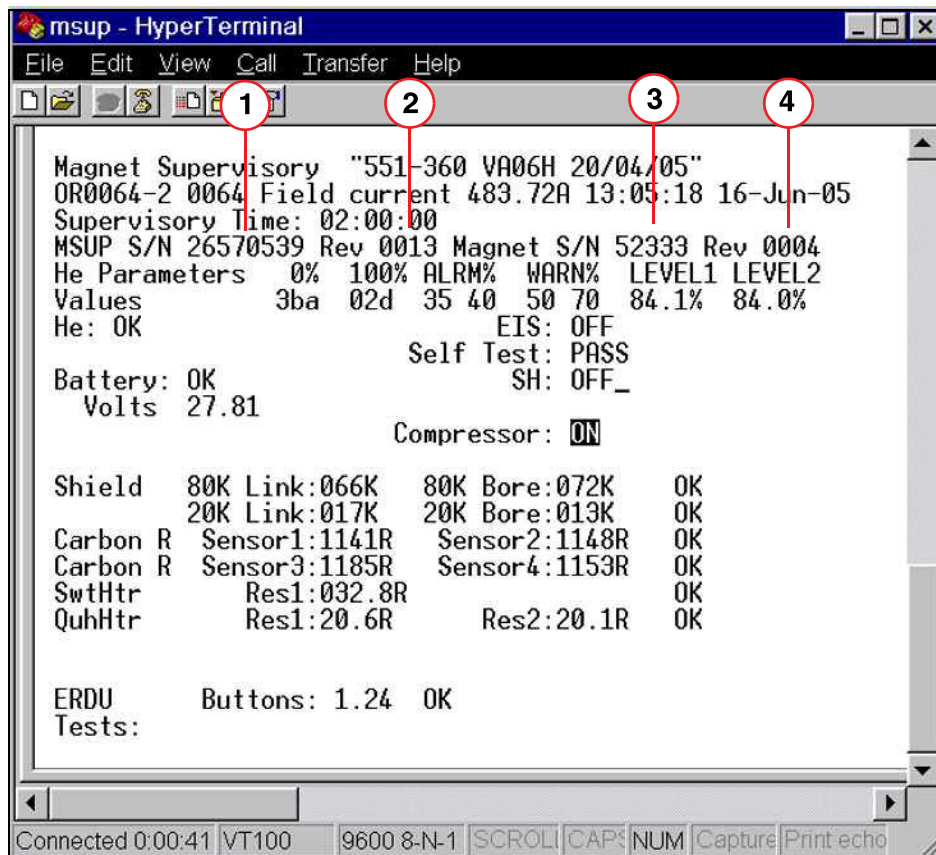


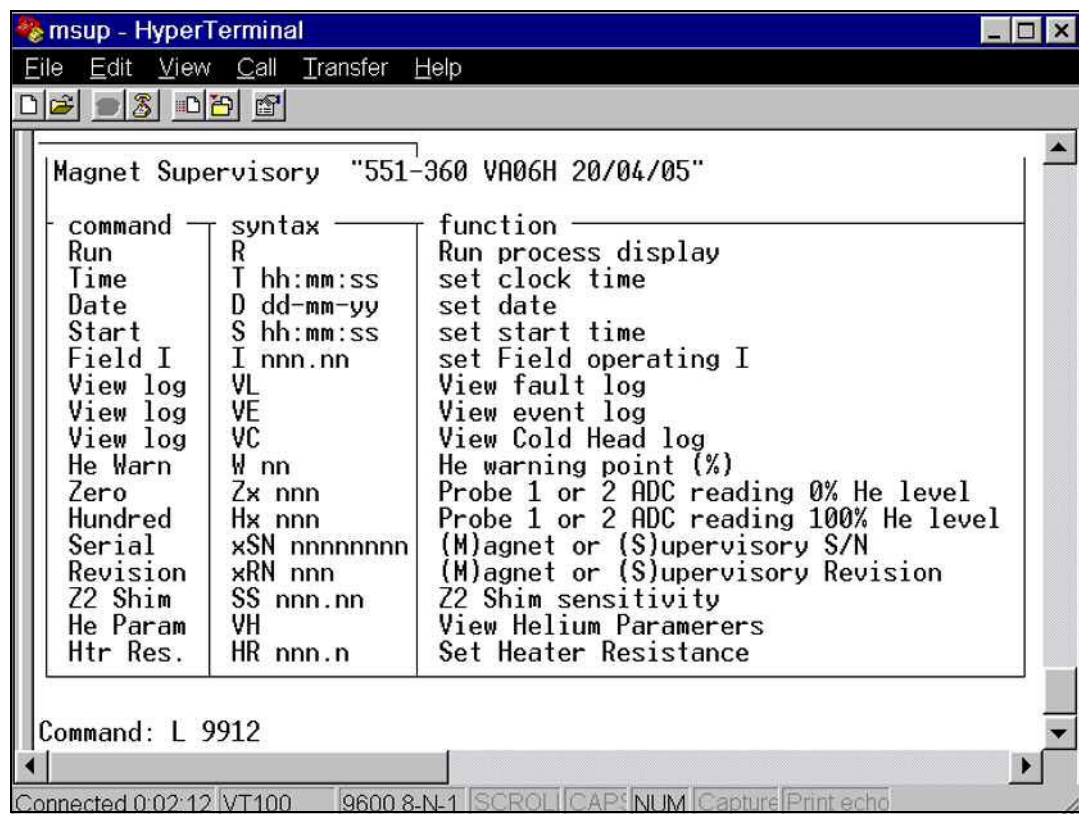
Fig. 63: Magnet Supervisory Run Process Display



- (1) MSUP Serial Number
- (2) MSUP Revision Number
- (3) Magnet Serial Number
- (4) Magnet Revision Number

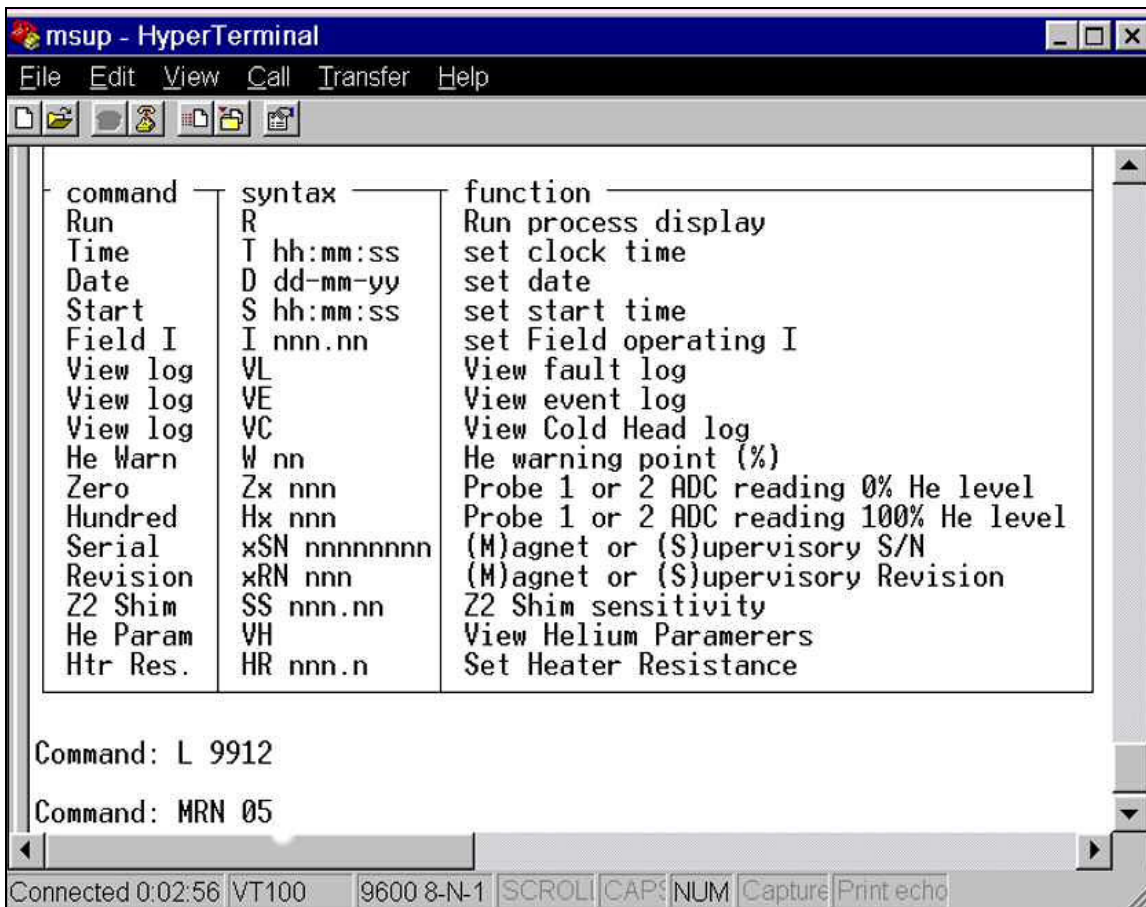
- Switch back to the command display by pressing "ESC"

Fig. 64: Magnet Supervisory Command Menu



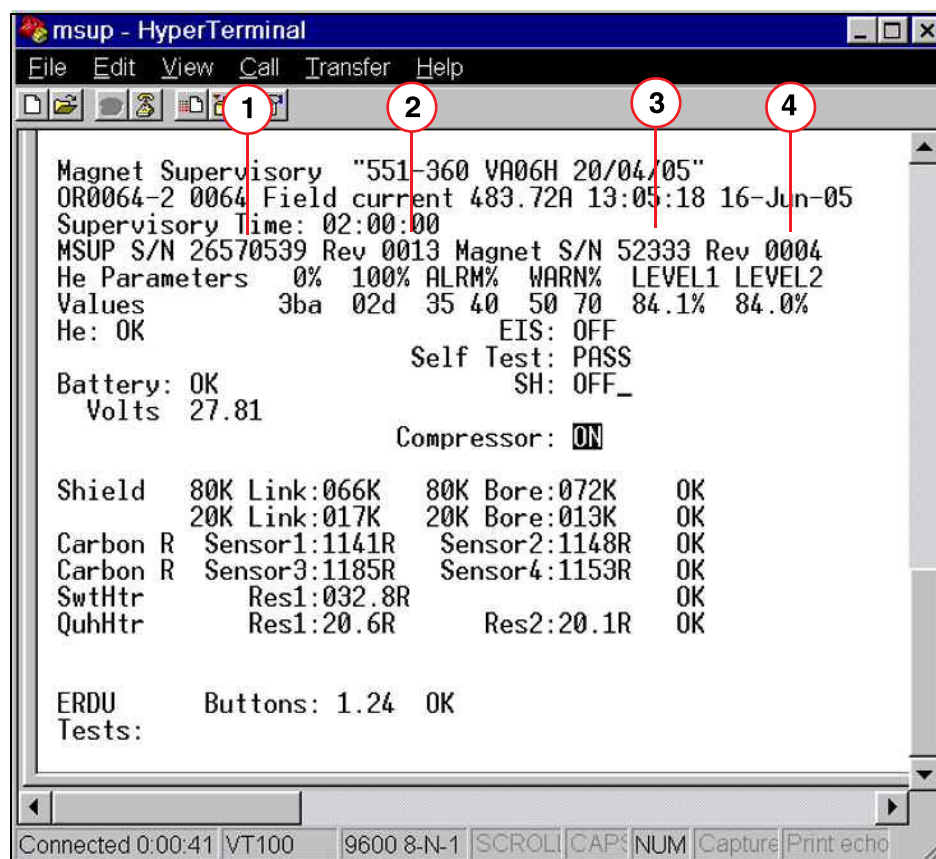
- Enter "L 9912" to unlock the access for changing the serial/revision numbers.

Fig. 65: Magnet Supervisory Command Menu



- Enter the magnet/MSUP serial/revision number
 - » Magnet Serial Number -> Type "**MSN xxxxx**" Return (e.g. 54321)
 - » Magnet Revision Number -> Type "**MRN xx**" Return (e.g. 13)
 - » MSUP Serial Number -> Type "**SSN xxxxx**" Return (e.g. 54321)
 - » MSUP Revision Number -> Type "**SRN xx**" Return (e.g. 14)
- Check the serial/revision number
 - » Type "**Run**" or "**R**"

Fig. 66: Magnet Supervisory Run Process Display



- (1) MSUP Serial Number
- (2) MSUP Revision Number
- (3) Magnet Serial Number
- (4) Magnet Revision Number

- Disconnect the connection.

5.1 General Overview

This procedure describes how to fill OR70 magnet systems with the standard SMT dewar syphon and with the High Efficiency Syphon (HES) or Smart syphon when performing the fill of the magnet at its full operating field.

The procedure also explains the different filling conditions for stationary systems compared to those used for mobile systems.

It is extremely important that the procedures are read carefully paying particular attention to the 'Notices and Warnings' as well as all the health and safety requirements when working with cryogenic liquids and magnet systems that are at field.

The "Filling the Magnet" procedure is divided into sections following the logical process used.

The type of LHe syphon fitted to the system will need to be established so that the correct syphon fill procedure is used.

The main process steps are:

- Safety & auxiliary tooling
- System, syphon & dewar preparation
- Precooling the dewar syphon
- Helium transfer procedure with HES syphon
- Helium transfer procedure with Smart syphon
- Stopping the transfer & disconnecting the syphons
- Changing the LHe supply dewars
- After the transfer & leak check

NOTICE

Establish type of syphon fitted and read appropriate procedures.

- » The filling procedures for the HES and SMART syphon types is completely different!!! Please read, be familiar with and carefully follow the appropriate procedure for your specific syphon in order to prevent an unintentional magnet quench.



WARNING

Procedures and warnings not followed.

Personal injury or damage to equipment or the magnet system may occur.

- » Read all the safety instructions, important notes and auxiliary remarks. Tools described. They are valid for all system and syphon types.

5.2 Safety

This section on safety is valid for the different filling procedures and syphon types used.

- For your safety, the safety of third parties, and to protect the MR system and its parts from damage, adhere to the safety information contained in:
 - General safety notes (→ General Safety Notes / TD00-000.860.01)
 - MR-specific safety information (→ MR-specific safety information / MR-000.860.01)
- Prior to beginning work, read the special information in the manual regarding:
 - general safety for service personnel,
 - restrictions regarding ferro-magnetic materials,
 - safety when working with the magnet at field,
 - working with heavy loads,
 - the information on cryogen handling.
 - the information on handling pressurized gas cylinders.



WARNING

Protective clothing, gloves and eye protection must be worn during the following procedure.

Personal injury may occur.

- » Only trained personnel should perform the following.



WARNING

Check ferrous content of equipment (including LHe dewars).

Personal injury may occur or the equipment may be damaged.

- » Only non-magnetic equipment may be taken into the magnet room.



WARNING

Do not vent Helium gas into confined areas.

Possible asphyxiation of personnel.

- » Follow correct HES procedures to minimize gas venting.
- » Depressurize all dewars outside the imaging suite or by connecting them to the system's vent line.

**WARNING**

Restrict access.

Danger of asphyxiation, cold burns. Ferrous materials entering imaging suite.

- » Unauthorized personnel must be excluded from the imaging suite.

**WARNING**

Use appropriate warning signs and fence in area to limit access.

- » "Warning: Authorized personnel only"
- » "Warning: Filling area for liquid helium"
- » "Warning: Danger of cold burns and asphyxiation"
- » "Warning: Strong magnetic fields"

**WARNING**

Dewar valve positions.

If the access or vent valves on the dewar are left open, air will freeze in the neck of the dewar with the strong possibility of causing an explosion.

- » See LHe supplier's manual for safe operation.

5.3 Auxiliary tooling

- Sufficient helium supplied in the largest possible supply dewars for the site.

NOTICE

Transfer efficiency and prevention of air ingress.

- » **Larger dewars will increase efficiency and minimize the risk of air ingress into the syphons.**

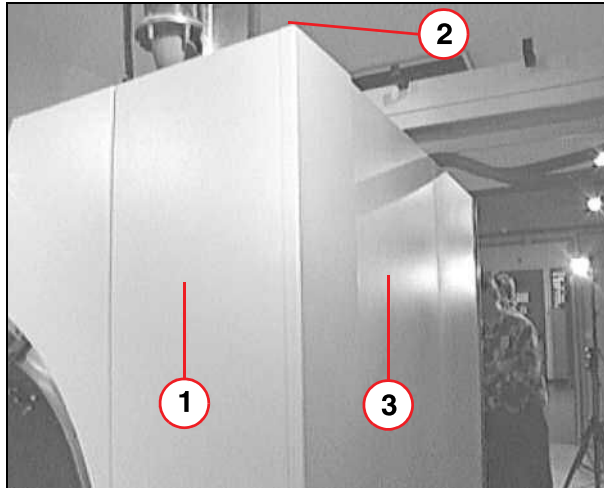
- Helium gas cylinder (at least 99.996%, equals 4.6 quality in some regions).
- Pressure regulator 200 bar/ 0 - 1.5 bar
- Helium syphon (dewar end).
- Gas tubing for pressurizing the dewar.
- Syphon/dewar adapter for the LHe dewar (syphon diameter = 16 mm).
- Replacement bursting disk and poppet seal "O" ring for quench valve (if needed).
- Heat gun and paper towels.
- Safety stepladder.
- Protective clothing: gloves, safety glasses, face shield, lab coat, safety shoes and personal oxygen monitor.
- Non-magnetic flashlight (beam type) with ECL inspection sight glass.
- Leak detection fluid or suitable electronic leak detector.
- Read descriptions and operating manuals for the cryogen dewars and gas transfer equipment used.
- Appropriate warning signs.

5.4 LHe filling of stationary and mobile systems

5.4.1 System preparation

Fig. 67: Cover - back corner

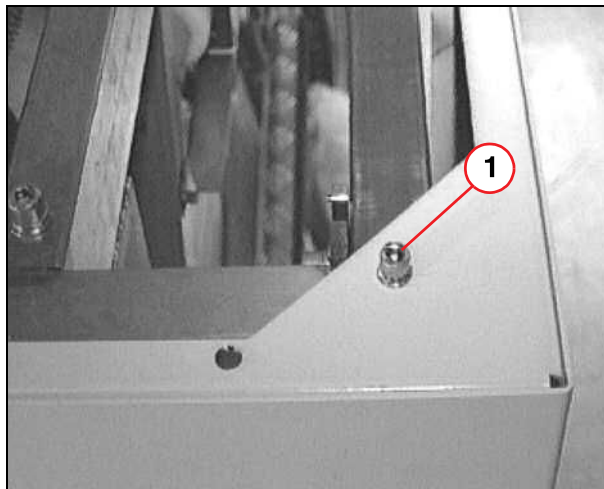
Removing the cover



- (1) Cover - back corner
- (2) 6 mm Allen screw
- (3) Side panels

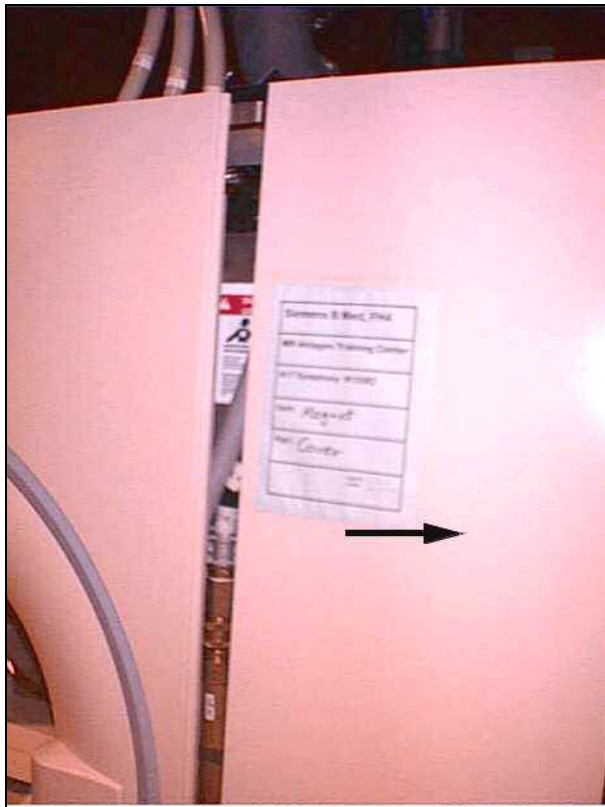
Fig. 68: Screw - cover for back corner

- Remove the 6 mm Allen screw on top of the cover for the back corner.



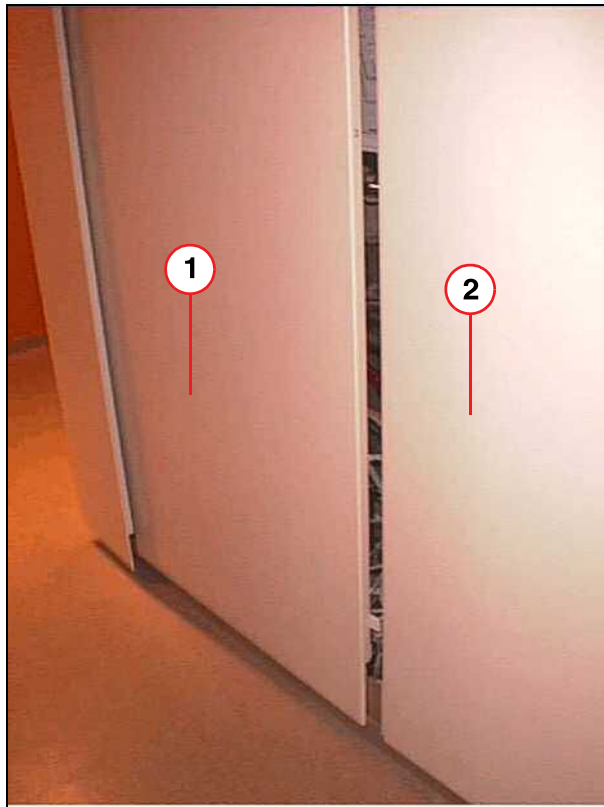
- (1) 6 mm Allen screw

Fig. 69: Cover - back corner



- Move the cover for the back corner to the right by approximately 5 cm (2 inches)

Fig. 70: Side panels



(1) Side panel A
(2) Side panel B

- Move the (→ 1/Fig. 70 Page 63) to the left
Move it behind the cover of the back corner
- Move the (→ 2/Fig. 70 Page 63) a little to left until you have access to both sides of the side panel B
Lift the side panel with both hands

- The magnet system syphons are designed to be permanently fitted to the system and should remain in position at all times. If the syphon needs to be replaced or if the system needs to be filled from empty or at nitrogen temperature, contact the SMT Service Department.
- Static and Mobile systems have different vent valve ratings. Before commencing LHe filling, check the system type and operating pressure expected.



Mobile and stationary systems are fitted with different magnet venting assemblies.

STATIONARY SYSTEMS

OR70's are fitted with a vent assembly with two 1/3rd psi valves in series and with a bypass valve in parallel. When gas freely vents through the valves, a pressure of 0.6 psi [0.041 bar] will be seen on the system pressure gauge.

MOBILE SYSTEMS

OR70's are fitted with a 1/3rd psi valve and 16 psia absolute valve which will give a higher differential pressure as high as 4.2 psi [0.28 bar] on the system gauge at high altitudes. Standard atmospheric pressure will see a system gauge reading of about 1.6 psi [0.11 bar].

Refer to (→ Fig. 87 Page 88) for pressure readings at different altitudes.)

5.4.2 Checking the Magnet Pressure

- Check to ensure that the system pressure gauge is reading a POSITIVE pressure. If not, stop the transfer. It is essential for the system to be at a positive pressure to ensure that there is no possible air ingress or ice within the turret. Both will cause inefficient LHE transfer.

NOTICE

A positive system pressure is required for correctly cooling it the HES syphon prior to connecting it with the dewar syphon.

- » Follow procedure "connecting to the HES syphon".

- If the system is at atmospheric (zero) pressure, there is a leak. This will result in air ingress which will cause icing within the system.
- Before continuing the filling procedure, the system must be checked for leaks.
- Check the turret, venting, and system valves.
- Check the syphon end cap.
- Check the quench valve burst disk and "O" ring.
- See section: (→ Magnet System Leak Check / Page 85).
- If no leaks can be found, contact Uptime Service Center.

NOTICE

Ice will prevent efficient transfer.

- » Using an ECL sight glass, check for large amounts of ice when looking down the current lead (ECL) port. If ice is visible, contact the Uptime Service Center.



Pressure can be raised in the system by operating the EIS switch heater when the magnet is at field.

- Check that the connections to the system's helium exhaust port are intact, and that the helium exhaust is venting freely.

5.4.3 Current lead (ECL)

- Prior to transferring helium, the demountable current lead (ECL) must be removed from the system if the magnet is at field.

WARNING

Remove the ECL before starting LHe filling.

- » If the demountable current lead is left in the system when starting to transfer helium, the risk of a magnet quench increases considerably.
- » Remove the ECL and replace it with the baffle assembly.

5.4.4 LHe Dewar Preparation

Certain operations need to be done without delay to ensure efficient helium transfer.

- Ensure that the helium dewar is provided with open and clear access to the magnet
- Similarly, ensure that the syphon is provided with open and clear access (in particular height restrictions)
- If the helium storage dewar is highly pressurized, release the pressure slowly to minimize helium loss.

WARNING

Venting of helium gas in a confined area.

Possible asphyxiation.

- » The dewar must be depressurized outside the imaging suite or by connecting it to the magnet's vent line.

- Check the quantity by weighing, or by other means, as available.
- Fit the syphon dewar (neck) adapter to the top of the dewar.
- Check to ensure that the pressure gauge on the LHe dewar is accurate and displays zero bar after being depressurized. (or connect an electronic pressure meter if available).
- Connect the pressurizing gas supply hose to the dewar after purging it of air, ensuring that the hose is long enough to allow the dewar to be moved near the magnet but with the gas cylinder remaining outside the imaging suite.

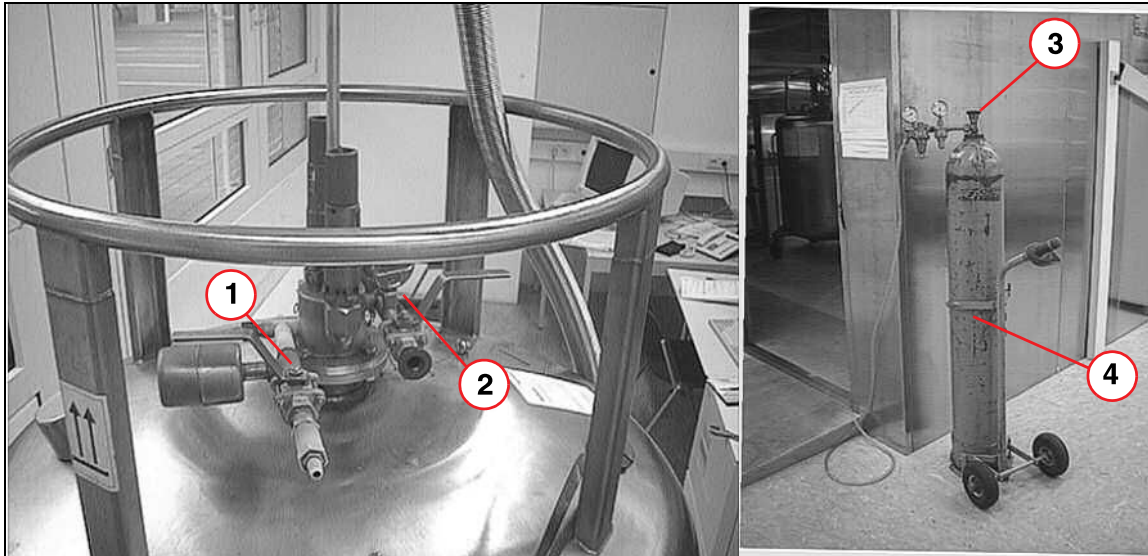
CAUTION

Ice will prevent efficient transfer.

Impure helium will cause build up of sludge and ice.

- » Use only pure [at least 99.996%] helium to pressurize the dewar.

Fig. 71: He dewar and He gas cylinder



- (1) Pressure relief valve
- (2) Vent valve
- (3) Pressure regulator
- (4) He gas cylinder

5.4.5 Preparing the dewar syphon

- Fit extensions as necessary to suit the size and type of LHe dewar being used, including the sludge inhibitor, to the dewar syphon.
- Check the physical condition of the syphon. The flow valve opens freely and ensure that the bayonet leg, threads and seals of the coupling are serviceable and not damaged.

! CAUTION

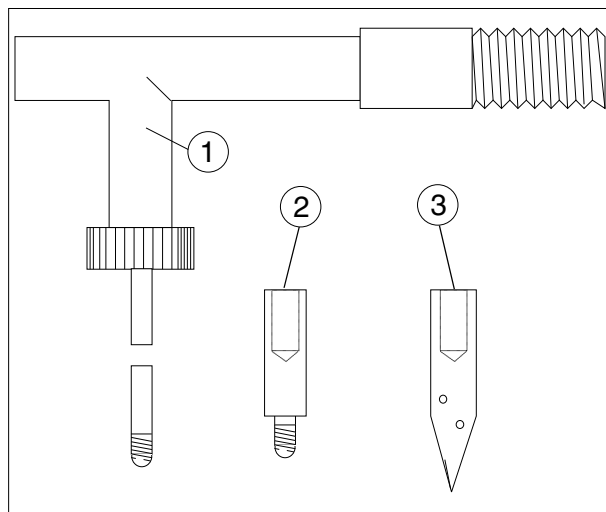
When using the LHe syphon without the extension and sludge inhibitor (→ 3/ Fig. 72 Page 67).

Check to ensure that the syphon end is at least one inch above the bottom of the dewar.

» This will prevent ice and sludge from blocking the syphon.

Fig. 72: Components parts of syphon

(→ Fig. 72 Page 67) shows component parts of syphon.

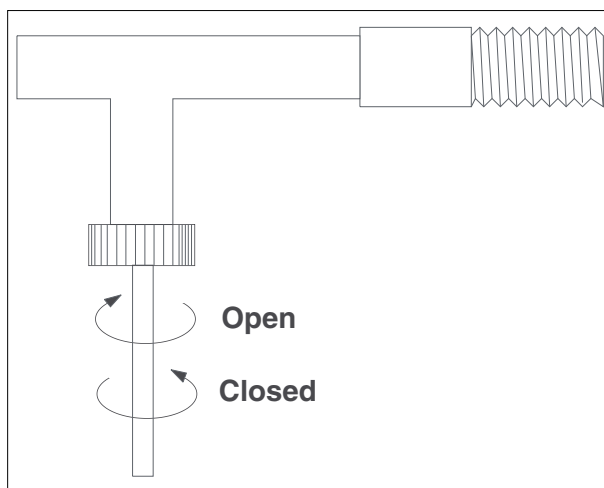


- (1) Dewar syphon
- (2) Syphon extension leg.
- (3) Sludge inhibitor

5.4.6 Precooling the dewar syphon

- Close the vent and low-pressure relief valves on the dewar.
- Open the dewar syphon flow valve fully (approximately 4 turns).
(→ Fig. 73 Page 67).

Fig. 73:



NOTICE

About four turns clockwise will open the valve fully. [viewed from the top]

- » The flow valve is fully open when the two guide pins at the top of the knurled brass nut are flush with the top of the stainless steel body.

- Ensure the sludge inhibitor is fitted to the dewar syphon down leg.

- Insert the syphon down leg into the dewar adapter and tighten the adaptor "O" ring clamp so as to make a gas seal through which the syphon can slide. The seal does not have to be completely leak-tight at this stage.
- Open the dewar access valve, and slide the syphon slowly into the dewar. Gas will begin to flow through the syphon.
- Push the syphon fully down into the dewar until the sludge inhibitor touches the bottom.
- Now tighten the syphon neck adapter "O" ring clamp.
- Open the gas supply on the regulator from the gas cylinder
- Allow the syphon to cool until a cold "cascading white plume" appears (refer to (→ Fig. 74 Page 68). Do not allow high pressure to build in the dewar, only enough to cool the syphon.
- Regulate the dewar pressure by opening the dewar pressure release valve if required.
- Set the gas supply pressure to 4 psi [0.28 bar].
- When the syphon is cold, close the flow valve and move the dewar close to the magnet so that the syphons can be coupled without excess bending.

Fig. 74: "Plume" during syphon cool down



NOTICE

Do not bend the syphons beyond the permissible ratio.

- » Tight bends in the syphons reduce the efficiency of helium transfer.

NOTICE

Do not open the vacuum valve of the dewar syphon near the bayonet.

- » This valve is the syphon vacuum valve. Use it only when re-pumping the syphon. Contact the Uptime Service Center if it is damaged.

5.5 Helium Transfer with High Efficiency Syphon (HES)

You can easily recognize whether you have a High Efficiency or a Smart Syphon fitted. (→ Fig. 75 Page 70) shows the configuration with a High Efficiency Syphon (beginning August 2002, a new revised HES was fitted to the systems).

The HES syphon will have a second vent tube fitted to the syphon bracket for venting away gas during syphon cool-down.

The magnet system is designed to have the HES syphon permanently fitted. If the magnet needs to be filled from empty (Nitrogen temperature), or the syphon needs to be removed for any reason, contact the Uptime Service Center.



CAUTION

Damaged HES syphon

Damage to ring nut, syphon leg or end cap could cause air ingress.

» Contact the Uptime Service Center for advice.



CAUTION

Do not connect the two syphons immediately.

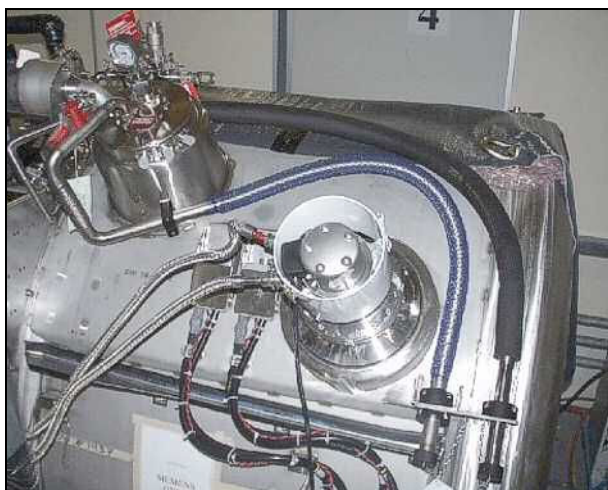
The HES syphon must be fully precooled before connecting the dewar syphon.

» Warm gas pushed into the system will cause the magnet to quench.

5.5.1 Important Notes

- Monitor the filling procedure continuously by checking:
 - the transfer pressure and fill level
 - the service turret and venting (check gas flow through valves and venting).
 - that all connections are properly sealed and leak tight.

Fig. 75: New High Efficiency Syphon on Magnet



5.5.2 Cooling the HES System Syphon

- ONLY when the system is pressurized to its operating pressure can you begin the cool-down of the HES system syphon (gauge needle in the green). System pressure will vary depending on atmospheric pressure. However, the minimum for any system must be 0.5 psi [0.34 bar].
- Remove the magnet syphon and the vent tube end caps.
- Fix the short "U" tube in place (refer to (→ Fig. 76 Page 71) so that the initial gas is vented into the exhaust pipe. When the U-tube starts getting frosty, remove it to get a better visual control of the gas flow.
- Allow the syphon to vent until it is cold (2-4 minutes, looks like "pluming", see (→ Fig. 77 Page 71).

NOTICE

The magnet should not be completely depressurized!

- » The system pressure is required to precool the HES syphon.

Fig. 76: "U" tube fitted for venting down



- Open the dewar syphon flow valve again FULLY and cool down until the 'long white cascading plume' can be seen.

NOTICE

OPEN the system bypass valve and leave it open for the entire refill!

- » This procedure is now common for mobile and stationary systems.
- » Refer to (→ Fig. 84 Page 83) and (→ Fig. 85 Page 83).

Fig. 77: "U" tube removed from the syphon end and cap fitted to the vent-down tube



5.5.3 Connecting to the HES syphon

! WARNING

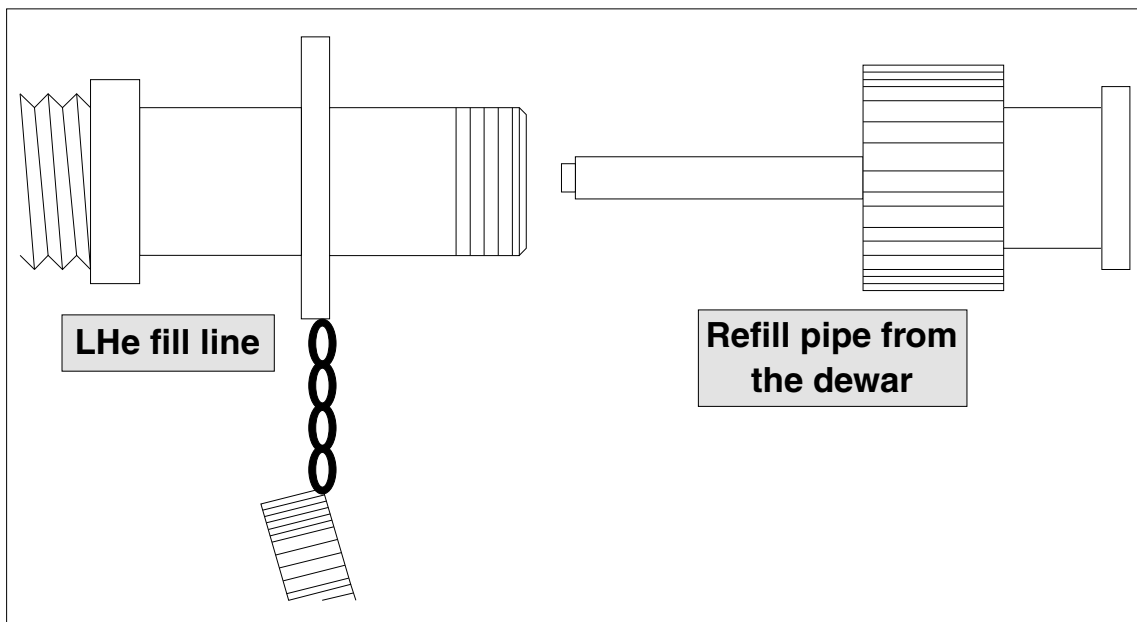
If either syphon is not fully cold and visibly plumbing before connecting together, the magnet will quench when the transfer is started.

The magnet will quench when the transfer is started.

» Connect syphons with a **HIGH FLOW** of liquid from the dewar syphon.

- With system and dewar syphons fully cold, couple them quickly together and fasten the securing nut.
- If this joint is well made, ice will not form at this point. If after a few minutes, gas is weeping from the coupling, warm it with a heat gun and re-tighten.

Fig. 78: Overview: Bayonet coupling

**! WARNING**

Transfer pressures lower than 2 psi [0.14 bar].

May cause the magnet to quench.

» Dewar pressure must be maintained at 4psi [0.28 bar] on initial connection.

**WARNING**

Do not start the transfer if either syphon feels cold on the outside, or becomes covered with condensation or frost during initial cool-down.

Again stop the transfer if either syphon begins to frost on the outside during transfer.

- » The syphon must be replaced or re-pumped. There is a danger of quenching the magnet if a soft syphon is used.
- » Contact the Uptime Service Center for advice.

NOTICE

Vibration of the pressure gauge needle of the magnet can occur.

- » This will stop when the system pressure has settled.

NOTICE

Both excessive and not enough pressure may cause poor efficiency.

- » Maintain a dewar pressure of about 4 psi [0.28 bar] for STATIONARY systems (with bypass valve open).
- » Maintain a dewar pressure of about 4 psi [0.28 bar] for MOBILE systems (with bypass valve open).
- » If available, an electronic pressure meter fitted to the dewar will provide more accurate pressure monitoring.

- Start a helium level measurement intermittently. Record the initial level. An increase should be seen within 15 minutes. If not, stop the transfer and investigate the reason for poor transfer efficiency.
- After connecting the syphons, remove the 'U' tube from the vent down line as soon as possible and cap the end of the vent tube. Put 'U' tube in a safe place.
- During the transfer, continue to monitor the helium level and the dewar pressure.
- Maintain a constant 4 psi transfer pressure. At this pressure, the flow rate will be about 10 liters/min. A 250 liter dewar will therefore empty in about 25 minutes.
- Never leave the transfer unattended.
- As soon as the level meter reads 100% or the dewar empties, stop the transfer immediately.

- Close the flow valve on the dewar syphon leg.

NOTICE

Helium dewar empty.

- » Good indications of the dewar becoming empty will be a sudden drop in the dewar pressure accompanied with a whistling sound from the syphon.

**WARNING**

It is important to know when the dewar is empty when using a HES syphon for transfer.

If left open, the magnet could quench.

- » The dewar flow valve must be **CLOSED** immediately.

- Continue by going to the section on (→ Stopping the LHe Transfer and disconnecting the Syphons / Page 82).

5.6 Helium Transfer with Smart Syphon

5.6.1 General Overview

Some OR70 magnet systems are equipped with a Smart syphon (→ Fig. 79 Page 75). The Smart syphon has an additional valve mechanism fitted internally that will open only when cooled to LHe temperature. The initial warm gas is diverted directly from the syphon into the system venting to prevent the magnet from quenching.

The magnet system is designed to have the Smart syphon permanently fitted. If the magnet needs to be filled from empty (Nitrogen temperature), or the syphon needs to be removed for any reason, contact the Uptime Service Center.



CAUTION

Damaged Smart syphon

Damage to ring nut, syphon leg or end cap could cause air ingress.

- » Contact the Uptime Service Center for advice.

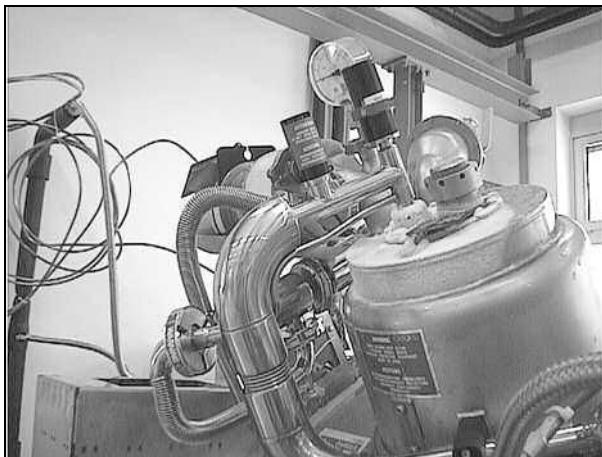
NOTICE

The Smart syphon does NOT require any initial precooling prior to connecting the dewar syphon.

- » The initial warm gas is vented away from the magnet.

- It is recommended that the precool procedure is used at all times for the dewar syphon.

Fig. 79: Smart Syphon on magnet OR70 or OR24



**WARNING**

Prevention of accidents to personnel and equipment.

Personal injury could occur as well as damage to the equipment and the magnet system.

- » Read all the health and safety instructions applicable to helium filling and cryogenic handling before beginning this procedure.
- » Refer to (→ Safety / Page 58)

- Check that the system pressure gauge is reading a positive pressure. If not, the transfer must not proceed (refer to (→ Checking the Magnet Pressure / Page 64)).
- Static and Mobile systems have different vent valve ratings. Before starting the fill procedure with LHe, check the system type and operating pressure.



Mobile and stationary systems are fitted with different magnet venting assemblies.

STATIONARY SYSTEMS

OR70's are fitted with a vent assembly with two 1/3rd psi valves in series with a bypass valve in parallel. When gas vents freely through the valves, a pressure of 0.6 psi [0.041 bar] will be seen on the system pressure gauge.

MOBILE SYSTEMS

OR70's are fitted with a 1/3rd psi valve and 16 psia absolute valve which will give a higher differential pressure, e.g. as high as 4.2 psi [0.28 bar] on the system gauge at high altitudes. Standard atmospheric pressure will see a system gauge reading of about 1.6 psi [0.11 bar].

Refer to (→ Fig. 87 Page 88) for pressure readings at different altitudes.)

- Prepare the LHe dewar as instructed in the section on (→ LHe Dewar Preparation / Page 65)
- Prepare & precool the dewar syphon as instructed in the section on (→ Preparing the dewar syphon / Page 66) and (→ Precooling the dewar syphon / Page 67).

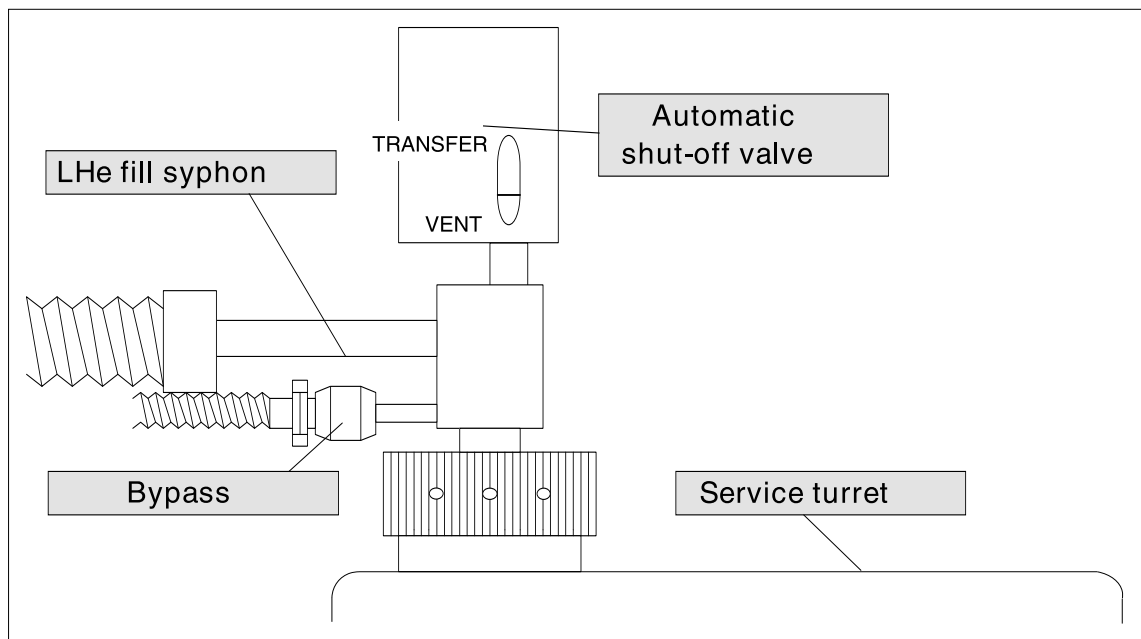
**WARNING**

Remove the ECL before starting the LHe filling procedure.

If the demountable current lead is left in the system when starting to transfer helium, the risk of a magnet quench increases considerably.

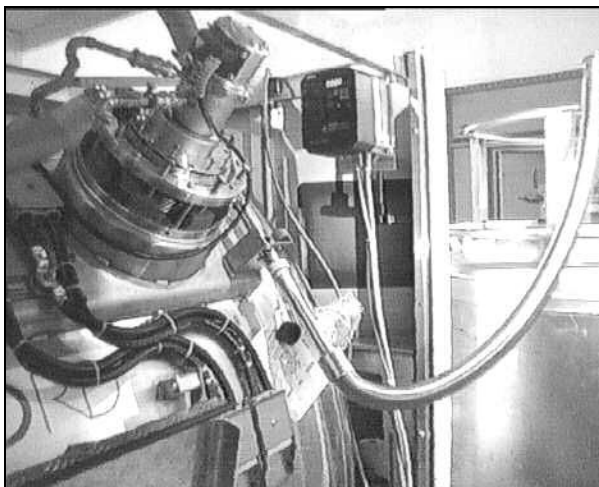
- » Remove the ECL and replace it with the baffle assembly.

Fig. 80:



5.6.2 Smart syphon LHe fill procedure

Fig. 81: LHe dewar in the magnet room



NOTICE

Do not depressurize the magnet prior to transfer.

» Fills with Smart syphons can proceed with pressure in the magnet.

5.6.3 Connecting to the magnet Smart syphon

**WARNING**

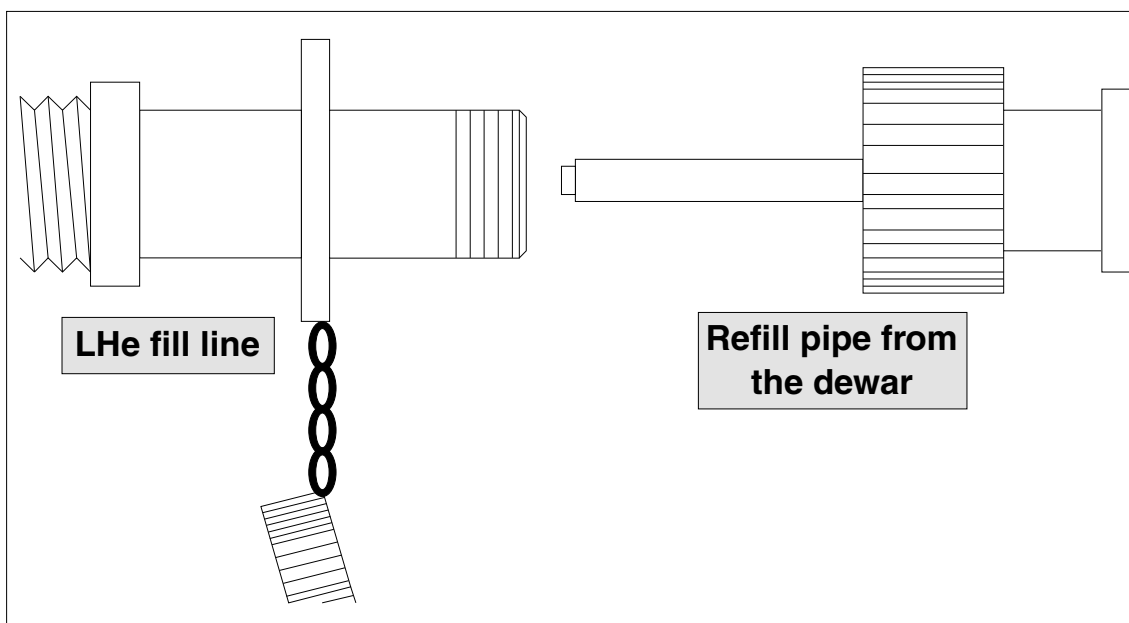
Check that the Smart syphon valve is in the "vent" position. Physical damage may have occurred to the valve assembly.

The magnet will quench if a transfer is attempted.

» Do NOT proceed, continue with troubleshooting.

- Remove the end cap from the magnet syphon (if gas vents from the smart syphon, replace the end cap and contact the Uptime Service Center).
- Check that the "O"ring in the bayonet assembly is okay and securely located
- Check that the dewar syphon bayonet is clean and dry before connecting.
- Check there is no physical damage to the components (→ Fig. 82 Page 78).
- Open the dewar syphon flow valve fully to cool down and create a flow of helium before inserting it into the magnet syphon. (refer to (→ Precooling the dewar syphon / Page 67)).

Fig. 82: Overview: Bayonet coupling



**CAUTION**

Do not start the transfer if either syphon feels cold on the outside, or becomes covered with condensation or frost during initial cool-down.

Again stop the transfer if either syphon begins to frost on the outside during transfer.

- » The syphon must be replaced or re-pumped. There is a danger of quenching the magnet if a soft syphon is used.
- » Contact the Uptime Service Center for advice.

NOTICE

This procedure is different for mobile and stationary systems.

- » For **MOBILE** systems **OPEN** the system bypass valve.
- » For **STATIONARY** systems leave the bypass valve **CLOSED**.

- With the dewar syphon cold and 'pluming' (liquid will be seen white and cascading downwards) connect the syphons together and fasten the securing nut.
- Gas will flow through the Smart syphon, venting initially from near the top of the magnet syphon into the quench tube until the syphon is cooled to near 4.2 Kelvin. At this point the Smart Syphon valve will operate to let liquid into the helium vessel.
- If this joint is well made, ice will not form at this point. If after a few minutes, gas is weeping from the coupling, warm it with a heat gun and re-tighten.

NOTICE

Syphon cooling will take 5-10 minutes.

The syphon is fitted with a visual indication "TRANSFER" of the valve position. Initially the valve may vibrate until fully open

- » Maintain the dewar pressure at 4 psi [0.28 bar].

NOTICE

Gas escaping from around the valve body.

- » Indicative of a broken seal on valve body. The transfer must be stopped and the Uptime Service Center contacted.

NOTICE

Vibration of the pressure gauge needle of the magnet can occur.

- » This will stop when system pressure has settled.

5.6.4 What to do, if the Smart Valve does not open

If the Smart Syphon valve has not moved to the transfer position, it is highly likely that the smart valve has become stuck in the closed position.

- Try warming the Smart valve body for 2-3 minutes while still maintaining dewar transfer pressure

Before moving the valve manually, check the following:

1. Syphon has been cooled for at least 10 minutes
2. The vent line from the top of the syphon has become very cold (visible fog falling, or liquid oxygen dripping - this occurs after about 10 minutes under normal circumstances).
3. The syphon on the dewar side is installed correctly (stinger fitted at the dewar end).
4. The shut-off valve at the syphon on the dewar side is completely open.
5. The syphon is covered with ice, check the vacuum of the syphon. If vacuum is bad, replace the syphon.
6. The pressure shown at the helium dewar is o.k. (4 psi equals 0.28 bar)
7. Does the helium dewar contain liquid helium.

After you checked all items, the Smart valve of the syphon can be opened manually.

- Apply even pressure to both sides of the ring, while lifting it gently to the top of the slot in the housing (about 10mm).
 - » If resistance is felt, do not continue!
- The syphon valve should remain open now, if helium flow and the valve mechanism are o.k.
- At the end of the transfer, the valve should move to the closed (vent) position.

**WARNING**

Manual operation of the lifter.

The magnet will QUENCH.

- » Do not manually move the valve unless the syphon has had adequate time to cool (5-10 minutes).

NOTICE

Too much or too little pressure may result in inefficiency.

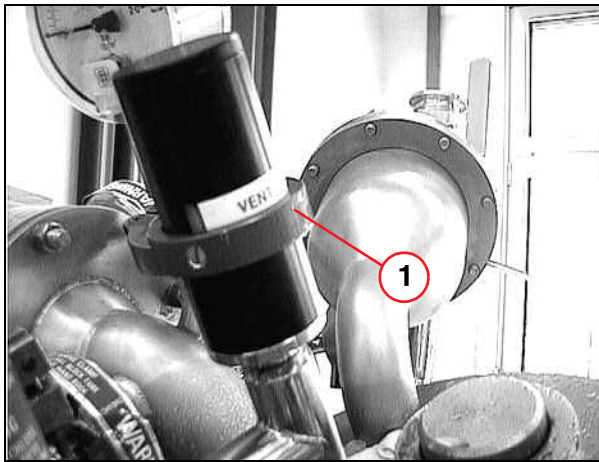
Maintain a pressure of about 4 psi [0.28 bar] for STATIONARY systems.

- » Maintain a pressure of about 4 psi [0.28 bar] for MOBILE systems. (with by-pass open).
- » If available, an electronic pressure meter fitted to the dewar will provide more accurate pressure monitoring.

- » If the syphon still does not open, remove the syphon, refer to (→ Flushing procedure for the Smart Syphon / Page 86).

- During the transfer, continue to monitor the Helium level and the dewar pressure.

Fig. 83: Smart Syphon



(1) Lifting Ring

Maintain a constant 4 psi [0.28 bar] transfer pressure. The flow rate at this pressure will be about 10 liters/min. A 250 liter dewar will therefore empty in about 25 minutes.

- Never leave the transfer unattended.
- As soon as the level meter reads 100%, or the dewar empties, stop the transfer immediately.
- Close the flow valve on the dewar syphon leg.
- Wait until the Smart syphon valve has returned to the "vent" position (1 to 2 minutes).

NOTICE

Manual operation of lifter.

- » If the Smart Syphon valve has not moved to the transfer position, it is highly likely that the Smart valve has become stuck in the closed position. In this case, gently warm the ring on the top of the magnet syphon which can then be used to close the smart valve.

5.6.5 Stopping the LHe Transfer and disconnecting the Syphons

- Close the flow valve on the dewar syphon leg.
- Close the gas supply to the dewar on the cylinder regulator.
- Slowly open the depressurization valve on the dewar.
- If the dewar is empty, refer to (→ Changing supply dewars / Page 83).
- Warm the syphon joint to room temperature using a heat gun.
- Stand clear of the syphon joint, and disconnect the syphons. Take care. As soon as the syphons are disconnected, cold gas will exhaust. Wipe away any moisture from the end of the magnet syphon with a clean, lint-free cloth or paper, and cap the end of the syphon.

NOTICE

The syphon end cap must be refitted and leak checked immediately after disconnecting the two syphons.

If it is left off, loose, or the "O" ring/seal is missing, air will enter the syphon with the possibility of causing air ingress.

- » Check that the end cap is fully tightened by applying gentle heat to the area. DO NOT apply heat continuously to the plastic cap.
- » The end cap must be leak checked.

- Check the dewar to ensure that it is fully depressurized and then remove the dewar syphon.
- Immediately close the access valve on the top of the dewar; then close the top vent valve and the dewar depressurization valve.
- Open the valve leading to the low-pressure relief valve.



WARNING

Dewar valve positions.

If the access or vent valves on the dewar are left open, air will freeze in the neck of the dewar with the strong possibility of causing an explosion.

- » See LHe supplier's manual for safe operation.

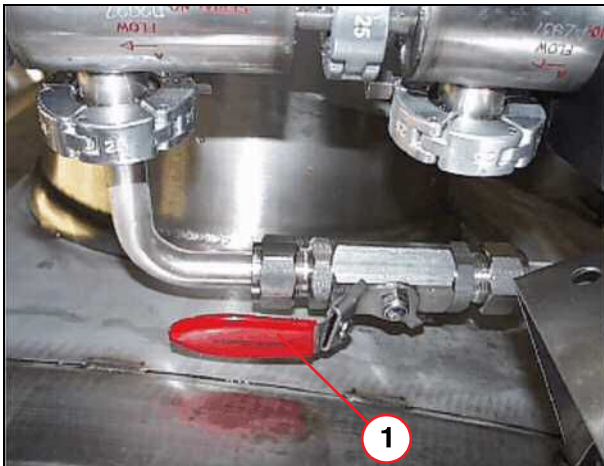


WARNING

Danger of air ingress.

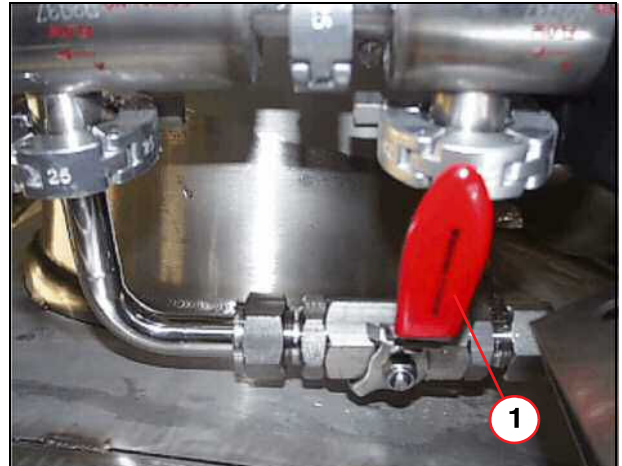
- » CLOSE THE SYSTEM BYPASS VALVE.
- » This must be warmed before closing to ensure excessive force is not used (see (→ Fig. 84 Page 83) and (→ Fig. 85 Page 83), open and closed bypass valve positions).

Fig. 84: Depressurization valve - open position



(1) Bypass valve open

Fig. 85: Depressurization valve - closed position



(1) Bypass valve closed

- Warm the dewar syphon and store it in a safe place.
- Changing to a new full LHe - dewar: Refer to (→ Changing supply dewars / Page 83).

5.6.6 Changing supply dewars

- If the supply dewar empties before the magnet is filled, it will be necessary to replace the supply dewar with a full one. Care will be necessary during this operation to ensure that efficiency is maintained and the dewar syphon is not iced.
- When the dewar empties (whistling of the syphon or dewar pressure drop noted), stop the transfer immediately by closing the dewar syphon flow valve.
- Turn off the gas supply and vent the storage dewar down to atmospheric (zero) pressure. Close the dewar valve after venting down and open the low-pressure relief valve.
- Warm the syphon joint to room temperature.
- Stand clear of the syphon joint and disconnect the two syphons. Take care. As soon as the syphons are disconnected, cold gas will exhaust from the cryostat syphon. Wipe away any moisture from the end of the cryostat syphon with a clean, lint-free cloth or paper, and cap the syphon end.
- Ensure that the full dewar is at hand and is depressurized and ready to receive the syphon (vent valve and low-pressure relief valve closed).
- Some pressure may have built up in the old dewar. Open the vent valve and ensure that the dewar is depressurized. Close the vent valve.
- Remove the syphon from the empty dewar and immediately put it into the full dewar. If necessary, transfer the dewar adapter with the syphon. If a different adapter is used, the gas hose must have previously been changed to the full dewar. As the sy-

phon is entered into the dewar, wipe off the frost from the surface of the down leg. Allow the syphon to cool, as previously, until a cold "plume" appears.

NOTICE

Avoid air ingress to syphon and dewar.

- » **Quickly exchange the syphon. The dewar syphon may be pushed down into the dewar more quickly than when it was fully hot, as less liquid will be boiled off.**

- After removing the syphon from the empty dewar, close the dewar access valve as soon as possible and open the valve leading to the low-pressure relief valve.
- When the dewar syphon is cooled, close the flow valve and move the dewar near the magnet so that the dewar syphon will reach the cryostat syphon without excessive bending.
- Continue the fill procedure by following either (→ Connecting to the HES syphon / Page 72) or (→ Connecting to the magnet Smart syphon / Page 78).

5.6.7 After the transfer

For correct and safe operation of the magnet after a helium transfer, ensure the system is fully leak checked so that the system will maintain a safe working pressure.

NOTICE

Ramping the magnet.

- » **After a helium fill, it is advisable to wait one hour before ramping the magnet.**

- Use a heat gun to warm up the turret, quench valve, and vent valves. Use paper towels to remove excess water.
- To help pressurize the magnet system, the EIS switch heater circuit may be used.
- Use a suitable leak check solution or electronic detector to check turret, venting, valves and the syphon end cap for leaks.
- Before leaving the site, the system **MUST** be checked to ensure that it has pressurized back to its standard operating pressure.



Before leaving the site, the system **MUST** be checked to ensure that it has pressurized back to its standard operating pressure.

5.7 Magnet System Leak Check

The magnet system must be checked for leakage after work has been performed during a service visit or a helium fill visit, if the system turret and venting have been subjected to extreme temperatures.

The magnet system must remain pressurized at all times. Before leaving an installed system site, a full leakage check must be performed and the system observed as it is turning back to its normal operating pressure.



WARNING

There must always be a positive pressure in the system.

Internal "icing" of the magnet may result causing the magnet to quench and the helium vessel to rupture.

- » **Follow the leakage check procedures during all service operations.**

If the system pressure remains at zero but there is no apparent leak, check the quench valve integrity. Refer to the (→ Magnet / M6-015.840.17).

- For stationary systems, the normal operating pressure is 0.6 psi [40 mbar] at sea level.
- For mobile systems, the normal operating pressure is 1.6psi [1.1 bar] at sea level.

NOTICE

The system operating pressure for mobile and stationary systems will vary with changes in local atmospheric pressures.

- » **Mobile systems taken to high altitudes (above 6000 feet) will see operating pressures up to 4.5 psi [300mbar].**

5.7.1 Procedure

An electronic helium leak detector should be used to search for leaks around all joints on the service turret. (Follow manufacturer's instructions for correct use of the electronic detector.)

- The leak rate using an electronic device should not exceed 1×10^{-4} mbar-l/sec.



WARNING

Observe caution when using an electronic leak detector with the magnet at field.

Accidents to personnel and equipment damage will occur.

- » **Check for magnetic parts before use.**

- If electronic equipment is not available, a suitable leak detection fluid must be used.
- Leak check fluid must be applied liberally by either brushing or squirting it onto the joints and seals. Leaks will be detected by observing bubbles where the fluid has been applied.

- The turret, quench valve and venting must be warmed to room temperature and dried before commencing the leak check. Remove condensation and water with paper towels.

The specific areas to leak check are:

- Clamps and seals on vent lines
- ECL and syphon seals
- System pressure gauge
- Turret top plate
- Syphon end cap
- The turret electrical connections
- Flexible tubing

Where leaks are detected you should:

- Tighten or replace clamps, if damaged.
- Check that the correct type of seals are fitted.
- Check seal's faces for damage. Replace, if damaged.
- Tighten syphon end cap and the central retaining screw for the chain.
- Tighten Syphon / ECL seals. If leak persists, replace the copper seals.
- Check quench valve burst disk and "O" ring. Replace, if necessary.
- Check that bolted flanges are correctly torqued.
- Seals covered in vacuum grease should be cleaned and inspected. Refit the seal or "O" ring using NO vacuum grease.

If problems persist with leaks to the turret or venting, contact your local Uptime Service Center.



Do not use any form of vacuum grease on O rings or seals.

It is recommended that these be wiped clean and checked for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.



Replace used high load seals.

The high load seals (copper (Cu) or Heli-flex types) must be replaced with a new one each time the joint is broken.

5.7.2 Flushing procedure for the Smart Syphon

The syphon may be partially blocked with ice. To clear any ice, the Smart syphon must be removed from the system.

- Depressurize the system to almost zero pressure by opening the system bypass valve.
- Using the syphon ring nut spanner, release the syphon nut and remove the syphon.
- Replace the syphon baffle plug into the magnet to prevent air ingress into the system.
- Close the system bypass valve.

- Position the Smart syphon and dewar syphon on the floor and connect them together.
- Connect a gas supply of pure helium (99.999%) gas to the end of the dewar syphon.
- If still frosty, gently warm up the Smart syphon system leg and valve assembly.
- Purge the two syphons with helium gas for 10 minutes at a pressure of 5 psi.



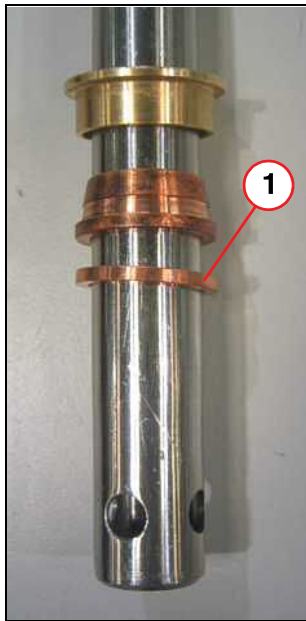
The helium gas will vent through the Smart syphon vent valve.

- After 10 minutes, stop the gas purge and disconnect the gas line from the dewar syphon.
- Disconnect the two syphons and refit the syphon end cap.
- Remove the syphon baffle plug from the system and re-fit the Smart syphon.



A new copper seal will be required for the syphon before refitting (→ Fig. 86 Page 87). Only the flat copper ring must be replaced, the other parts of the sealing are compressed and should not be removed!

Fig. 86:



(1) Copper seal

Copper seal to be replaced after removal of syphon (before re-inserting it)

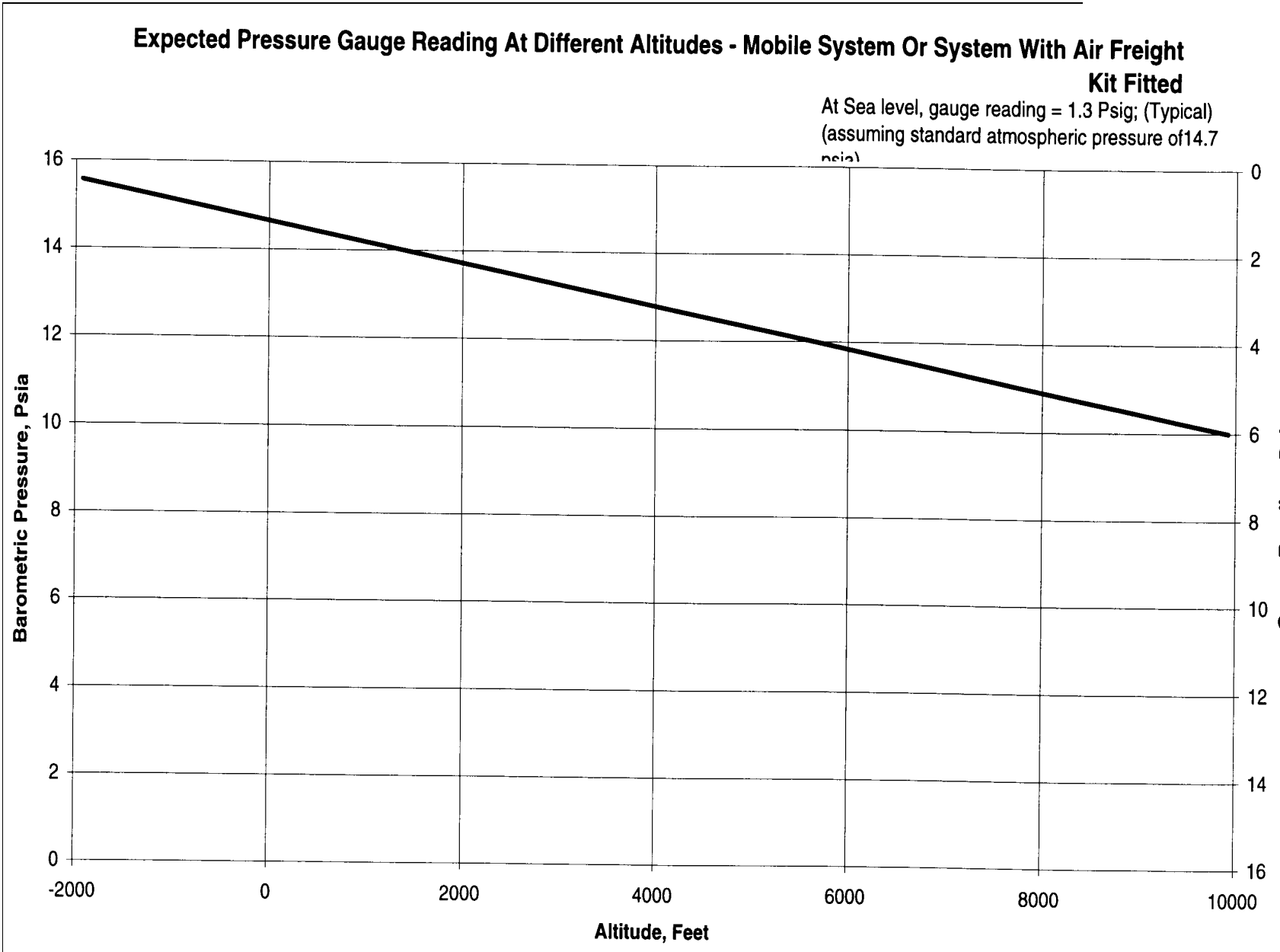
- Tighten syphon ring nut by hand and complete using the ring nut spanner.



The system must be allowed to repressurize and the syphon seal nut leak checked using a suitable leak check solution.

- If the Smart syphon still fails to operate, a replacement is recommended.

Fig. 87:



6.1 Turret Leakage Check

6.1.1 General

On non-recondensing systems, leaks can lead to air ingress and blockage of the neck.

Leakproof is defined as: The total integrated loss across all leaks must be less than 1×10^{-3} mbar-l/s. Individual leaks must be of the order 10^{-4} mbar-l/s, or less.

A total leakage rate of 7×10^{-3} mbar-l/s corresponds to a liquid helium loss of ~180 liters/year.

6.1.2 Safety



CAUTION

Pressurized helium vessel

Cold helium gas

Failure to observe the following safety information may result in physical injury.

- » Do not disconnect any of the fittings without first depressurizing the helium vessel.

6.1.3 Equipment Required

- Handheld Leakage Detector, material number 99 00 978

Fig. 88: Leak detector



- Calibrated leakage

- MSUP display unit

6.1.4 Prerequisites for Checking Leakages

- Helium vessel at normal operating pressure i.e. 0.6 psig to 1.3 psig by inspecting the gauge for non-recondensing systems at sea level.
- Ensure that all joints to be checked are at room temperature and dry.
- Ensure that the atmospheric environment is not saturated with helium gas, i.e. do not test for leakages shortly after a helium fill, magnet ramp up, or depressurization.



Determine the background level in the magnet room. The background must be at least one order of magnitude lower than the smallest leak expected, i.e. for leaks of 10^{-4} mbar-l/s, required background in the order of 10^{-5} mbar-l/s.

6.1.5 Procedure

- Switch on the leakage detector and allow it to warm up.
- Check the calibration using the calibrated leakage before and after use.



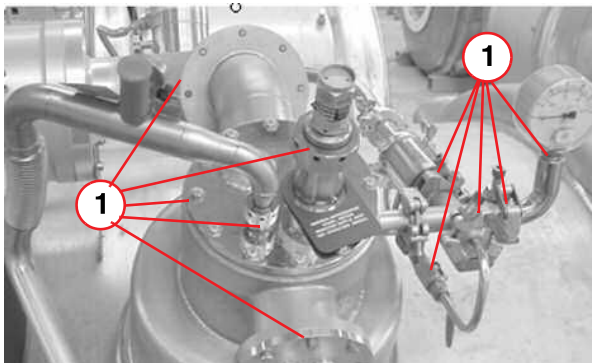
Before you check the leakage, ensure that the instrument is working correctly by testing it with calibrated leakage.

- Move the sample tube slowly, or hold it directly at the joint for a few seconds to give the detector time to respond.
- Hold the instrument at the suspected point until the reading has stabilized.
- Repeat it at the suspected point and check to see if the leakage is indicated again.

6.1.5.1 External leakage check

- (→ Fig. 89 Page 90) indicates seals on the high-pressure side of the turret that require leakage testing. This includes the turret top plate and the syphon port.

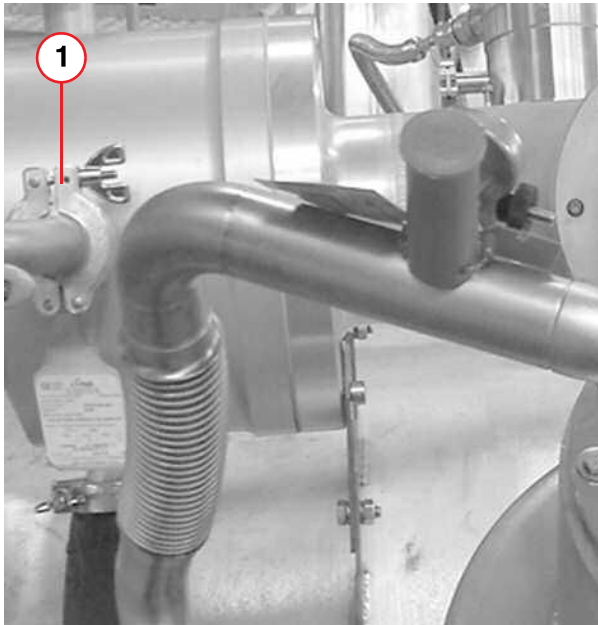
Fig. 89: Leakage checkpoints on high pressure side



(1) Leakage checkpoints

6.1.5.2 Internal leakage check

Fig. 90: Leakage test points



(1) Clamp



CAUTION

Risk of asphyxiation and hypothermia.

Failure to observe the following safety warning may result in dizziness or loss of consciousness for personnel.

In case of a quench, the RF-room will be filled with cold helium gas.

- » Perform an internal leak check at zero field.
- » Unauthorized personnel have to be excluded from the MR imaging suite.
- » Protective clothing, gloves, and eye protection have to be worn.

- Remove clamp from the quench valve and leak check the quench valve "O" ring and burst disk.
- Refit venting system on completion

6.1.6 Final Work Steps

- At the end of leakage testing, ensure once more that the instrument is working correctly by testing it with the calibrated leakage.
- Ensure that all parts of the venting system have been installed again.



Leakage detection fluid may be used on a few easily accessible points for leakage detection when the magnet is at field. Suitable points are the syphon port and the turret top plate seal.

6.2 Leak Repairing on Turret

6.2.1 General

The following points must be observed at all times:

- Joints on the "High" pressure side of the venting system will generally be fitted with metal seals, the only exception being the silicone seal in the quench valve. DO NOT use rubber seals as a substitute.
- DO NOT re-use metal seals, always replace with a new seal.
- Before disassembly OF a joint, ensure that replacement seal is available.
- DO NOT use vacuum grease.
- After repair, leak test the entire venting system, not just the joints that have been repaired.

6.2.2 Pipe Couplings

For leaks found at pipe couplings, first try tightening the clamp or the securing screws. Beware of overtightening. If tightening does not solve the leak, the coupling will have to be dismantled.



CAUTION

Hazardous conditions exist by allowing air into the system if the joint is on the high-pressure side.

- » As soon as the coupling is undone, plug the open tube with a wad of paper to prevent air ingress.
- » Do not have two joints open at the same time.
- » Do not undo joints on the high-pressure side until the system is fully de-pressurized.



Do not leave the system open for longer than is necessary to do repairs. Ensure that replacement seals are at hand.

Having dismantled the coupling, ensure that the sealing faces are clean and dry plus the sealing faces should not be scratched.

6.2.3 Syphon Port

The fixed syphon is sealed with a set of olives and a copper ring. After first tightening, the olives cannot be easily removed from the syphon down leg.

Remove the syphon and plug the port. Ensure that the square section copper ring has been removed from the port. Clean the lower face of the olives and ensure that the surface is clean and undamaged. Also verify that the olives are intact.

Fit a new, clean, undamaged copper seal to the syphon leg. Ensure that the port surface is clean, and re-fit the syphon to the same height as previously. Tighten the ring nut firmly.

6.2.4 Quench Valve

If the quench valve is leaking, refer to the "Replacement of parts."

6.2.5 Bypass valve

If the bypass valve body or the compression seals is leaking at its ends and the seal leaks cannot be closed by careful tightening, contact the USC.

7.1 Testing the ERDU circuit



Before initially ramping up the magnet, the following ERDU test has to be performed until the new ERDU test load is available.

If the ERDU test load is available perform the test referring to Safety related tests.



CAUTION

If the magnet is at field.

Magnet will quench!

- » **Perform the ERDU test (following description) before the magnet is ramped to field.**

Procedure:

1. Remove the ERDU button 1 from the system: Unplug "ERDU Button 1" cable at the flying lead on X6 at the MSUP. Do not unplug X6 at the MSUP.
 - » The alarm box will show a "MAG STOP" alarm and the buzzer will sound.
2. Reconnect the "ERDU Button 1" cable at the flying lead on X6 at the MSUP.
3. Press Acknowledge on the alarm box to silence the buzzer and clear the "MAG STOP" LED.
4. Write down the magnet pressure displayed on the pressure gauge.
 - » Pressure should be in the yellow or green section.
Depressurize if magnet pressure is above yellow or green section.
5. Press the ERDU button on the alarm box.
 - » The alarm box will signal with a Mag Stop alarm (red) LED and the buzzer will sound.
6. Monitor the magnet pressure and ensure that it increases at a steady rate.



At approx. 1/3 psia, the pressure will not raise any further due to the 1/3 psi valve opening. Cold helium gas will pass through the venting system, this is evident from the frost build-up.

It is not necessary to wait for this condition before releasing the ERDU button.

7. Once you are sure that the ERDU works, release the ERDU button and press the "Acknowledge" button.
 - » This clears the MAG STOP alarm LED on the alarm box and silences the buzzer.
 - » Helium pressure will start to reduce due to the cooling effect of the cold head.
8. Repeat for the other ERDU buttons in the system.



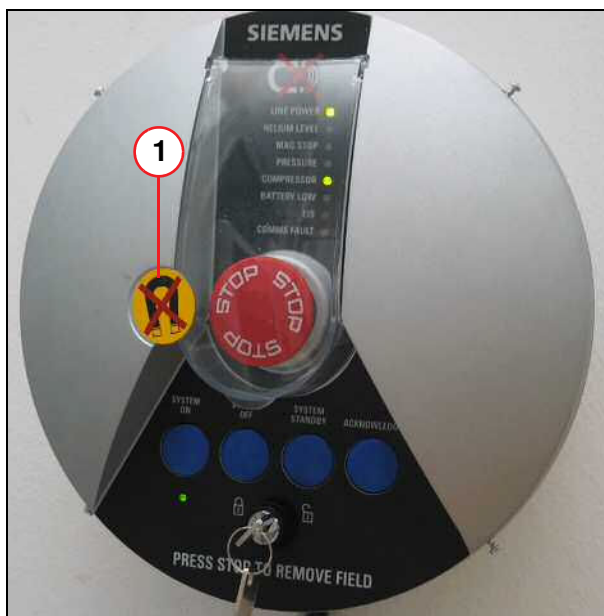
It may be necessary to wait for pressure to reduce to normal levels before testing each button.

9. Document the ERDU test in the safety-related test protocol.

7.1.1 Affix the tamper-proof seal

After the ERDU test is completed, the tamper-proof seal has to be affixed to the alarm box and ERDU buttons.

Fig. 91: Tamper-proof seal location



(1) Tamper-proof seal



The tamper-proof seal is delivered with the system.

7.2 Co-siting Situation

Co-siting means installing two MR systems with identical field strengths in close proximity.

Details are written in the planning guide.

NOTICE

Co-sited systems have to be ramped to different field strengths.

» The nominal frequencies should differ by 100 kHz.

7.2.1 Example

	1 Tesla systems	1.5 Tesla systems
System 1	40,53 MHz	63,68 MHz
System 2	40,43 MHz	63,58 MHz
Lower limit for re-ramping	40,35 MHz	63,50 MHz



Shimming is only optimized with both magnets ramped up during the shimming procedure.

7.3 Safety

- Please **read the following information carefully** to avoid personal injury and/or damage to the magnet.
 - Only Siemens CSEs trained in ramping magnets are allowed to perform this work.



Read the entire document including the flowchart before you begin any of the procedures.



Adhere to the general safety information and the special safety information included in this chapter when performing the steps indicated.



Ensure the Quench Line installation is complete before performing the first ramping procedure.

The Quench Line Acceptance Protocol must be checked before the first ramp.

Check that the Quench Line Acceptance Protocol is signed off by the Quench Line Manufacturer.



WARNING

Magnet safety checks

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Before the magnet is ramped up initially, perform "Testing the ERDU circuit."
- » The procedure described in the safety related test can be used if the ERDU test load is available.



WARNING

Strong magnetic field

Failure to observe the following safety warning may result in physical injury and equipment damage.

- » Before ramping up the magnet, ensure that no ferrous tooling or equipment are present in the imaging suite.
- » Ensure that the warning signs identifying the controlled access area (0.5 mT zone) have been posted at the entrance to the RF room as well as all other areas leading to the 0.5 mT zone.
- » The sign has to be posted in a way that it is visible even when the door to the controlled area is opened.

**CAUTION**

Risk of asphyxiation and hypothermia, if helium gas is vented into the magnet room.

Failure to observe the following safety warning may result in dizziness or loss of consciousness for personnel.

- » The quench valve must be connected to the quench pipe.

7.3.1 Always

- Always restrict access to the magnet room when ramping the magnet. Use flashing lights and signs.
- Always inspect all equipment and tools for defects prior to use.
- Always inspect connections of ramp leads at the magnet and at the filter plate.
- Clean the area of the magnet room, dispose off empty cartons or other packing materials
- Always inspect magnet room for the presence of non identified objects. Before ramping, the nature of those objects must be clarified.
- Prepare a minimum of 4 pictures of the room **prior to first ramping**.

These pictures must be colored and should have 2 to 4 megapixel resolution. They must show the complete room, all 4 corners and walls. In case there are un-removable cupboards installed, their doors must be open to prove that they are empty. Later those pictures are to be attached to the installation report.

7.3.2 Never

- Never use defective or damaged equipment.
- Never allow unauthorized people to stand too close to the magnet during ramping.
- Never block emergency exits.
- Never smoke or use open flames in the examination room.
- Never ramp a magnet when you are unsure about magnetic parts inside the magnet room. Those parts could be hidden in e.g. supposed empty boxes or in closed cupboards or sliding of a desk.

7.3.3 Protective clothing

The following protective clothing must be available on-site for all technical personnel.

- Face mask
- Protective gloves (cryogenics) and leather work gloves
- Lab coat or overalls (cotton or linen)

- It is a requirement that a personal oxygen monitor is worn in the imaging suite.

7.4 Ramping the magnet

CAUTION

Initial ramping with helium lower 70 %.

Risk of quench

- » When initially ramping the magnet during the installation of the system, the LHe level **MUST** be at least 70%.
- » Later it is possible to ramp with 50%.



Make sure that no water enters the electronic components (RFIS, RFAS).

Use a towel or a cloth to catch the water.

Cover the components with a towel or cloth.

7.4.1 Required Tools

- Magnet Power Supply (p.n.: 74 61 960)

Fig. 92: MPSU



(1) MPSU LID

- 3600 Standard Extension Kit (p.n.: 8396827)



The 3600 Standard Extension Kit consists of the Accessory kit and the Standard current lead kit.

Fig. 93: 3600 standard extension kit



- (1) Accessory kit
- (2) Standard current lead kit

- Heat gun
- Non-magnetizable tools



CAUTION

Pressurizing the system using the switch power supply and lead kit.

This method must only be used when the magnet is at zero field. Examples: before ramping or after a quench.

- » The magnet will quench if used at field (for detailed procedure refer to (→ Pressure Raising Procedure for a Magnet NOT at Field / Page 183))

7.4.2 ECL Inspection and Repair

7.4.2.1 Introduction

ECL's must be visually checked by following this maintenance procedure prior to ramping a magnet system.

Existing ECLs that do not meet the criteria of the Inspection and Repair procedure should be reported to the Uptime Service Center, who will advise on the correct action to take.

7.4.2.2 SUMMARY

A number of changes have been made to the manufacturing process of the ECL.

These changes will improve:

- Improved gas flow through the central pin connector, reducing heat during ramping.
- Ensure better contact of central pin to the magnet connector.
- Ability to change the outer contact bands.
- Lower resistance across lead when ramping.

The procedure also provides details of the key components of the ECL that must be checked prior to insertion into a magnet system. The critical checks and measurements are:

- Ensure there are Two (2) inner multi-contact bands.
- Length of inner contact conductor pin.
- Condition of Nomex conductor sleeving.
- Depth of inner conductor pin central vent hole.
- General cleanliness of the top St. collar.
- General slackness of joints and locking collars.
- Condition of inner and outer contact bands.
- Condition of fly leads.

7.4.2.3 Procedure

Fig. 94: ECL (Enhanced Current Lead)

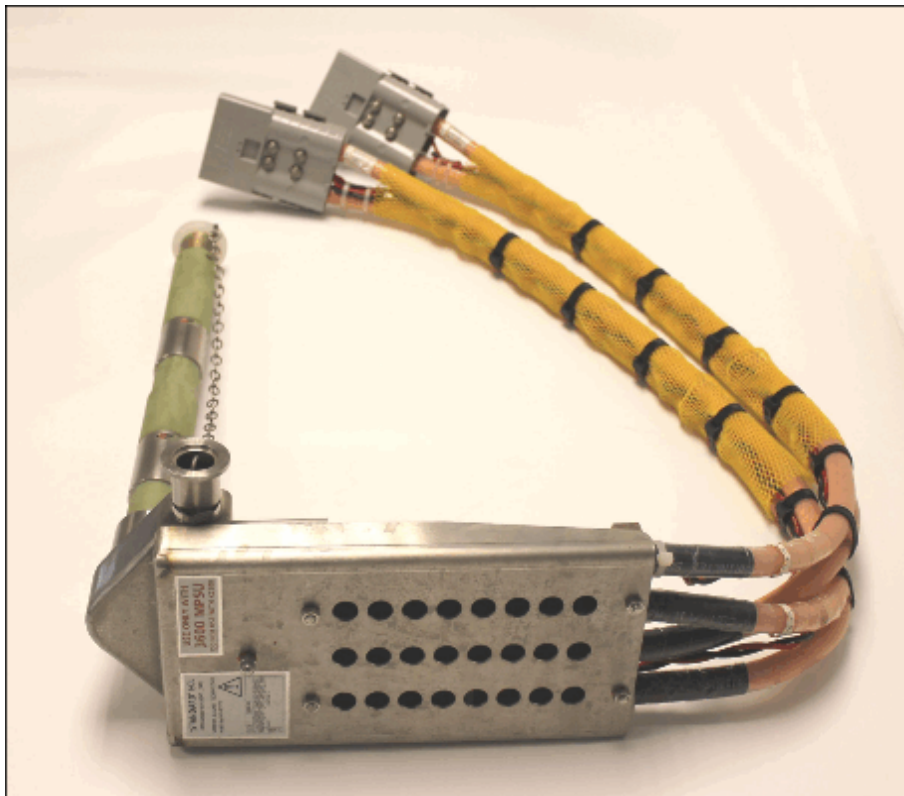
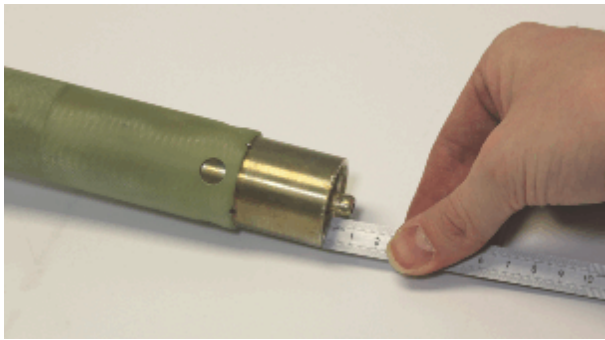


Fig. 95: Contact Bands



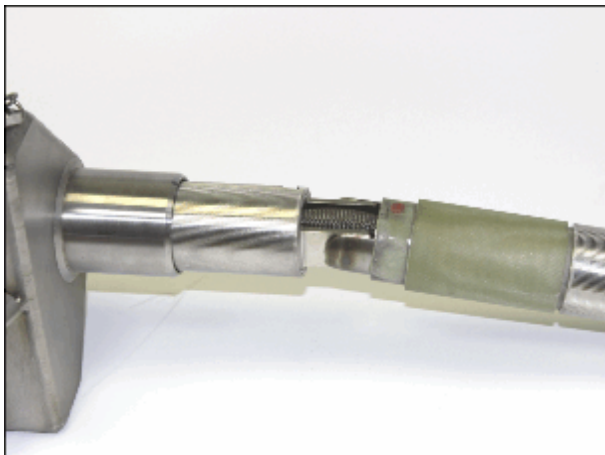
- Check the multi-contact inner conductor pin, it should be of the two-conductor band type.

Fig. 96: ECL Measurement



- Measure the distance that the inner conductor protrudes beyond the outer conductor.
- The tolerance range is 4.5mm to 8mm.
- The ECL should not be used if measurements are outside the range

Fig. 97: Sleeving inspection



- Slide the locking collar back and check that BOTH inner conductors are covered with 'Nomex' sleeving. If they are not, the ECL should not be used

Fig. 98: ECL vent hole check



- Check the depth of the central vent hole in the inner conductor using a thin rod, e.g. a 2 mm welding wire. It must be 82-84 mm (passing the pin in (→ Fig. 98 Page 105))

If a GRP pin is obstructing the vent hole, then it should be drilled through (via the vent hole) using a 4.4 mm drill.

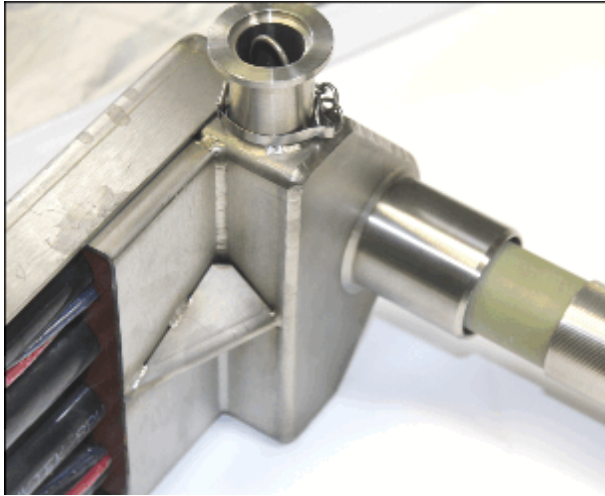


To avoid obstructing any of the coaxial connectors venting, vacuum out the swarf. Do not blow swarf up into the breather hole and housing.

7.4.2.4 General Cleaning and Examination

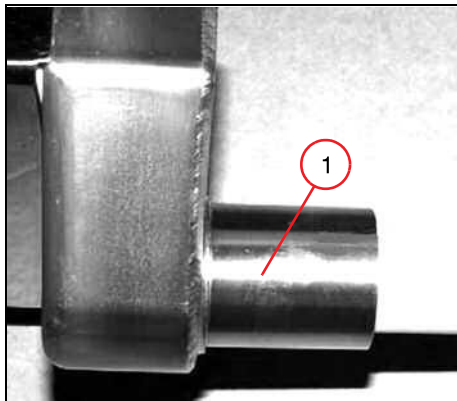
Clean off all the external and internal parts (undo sliding joints to gain access to internal parts), visible dirt and contamination with clean lint free cloth.

Fig. 99: ECL collar surface check



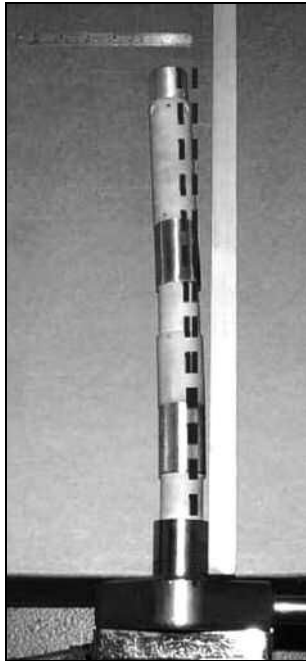
- Check for any scratches or marks on the collar. These should be 'dressed off' with fine/medium wet and dry emery paper.
- Finish off with 'Schoolteacher', (polishing cloth)

Fig. 100: ECL/DCL collar surface check



(1) Collar

Fig. 101: ECL/DCL slackness check



- Check for any excessive slackness in the joints and between the GRP and the locking collars
- Slackness is defined by the distance measured from the center of the inner conductor,s position (when the lead assembly is straight) to the furthest position that it can travel/bend away from this position.
- If there is more than 20 mm of travel, the ECL should not be used.

7.4.2.5 Contact bands

NOTICE

Bad contact bands

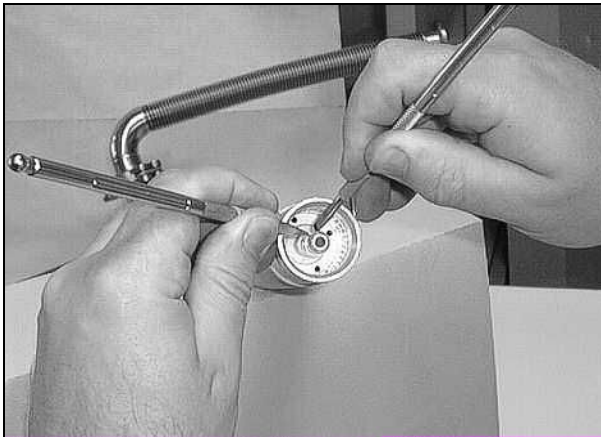
- » For the first use on a site, the contact bands of the inner and outer ECL contact have to be replaced.

Carefully remove contact bands from inner conductor and outer conductor without damaging the connector base material or silver plating.

- Place the ECL on a soft surface.

- Use two small screwdrivers to pull the spring apart at the opening. This allows the springs to be easily removed from the contact.

Fig. 102: The spring can be removed using two small screwdrivers



7.4.2.6 Contact Band Refitting

Fig. 103: Contact Bands



Great care must be exercised when fitting the contact bands; they must not be bent or distorted in any way.

Inner and outer contact bands need to be replaced regularly, and a record kept.

The outer band should be fitted first to avoid possible damage to the inner contacts.

If you notice that the band for fitting into the coaxial conductor socket groove has an insufficient spread effect, stretch the band out a little bit at the separating points before you fit it in order to reach the spread strength in the socket. By doing so it is also possible to easily correct slightly twisted bands.

Fig. 104: Outer contact bands



Fig. 105: Outer contact bands - Diagram

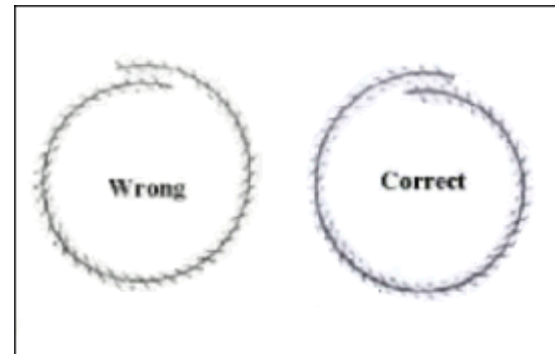


Afterward the band is being rolled by hand, so that the vertical planes are not being caught in each other. The band should only be rolled as small as necessary (diameter of the socket entry), so that it can be put into the groove with ease.

Fig. 106: Inner bands



Fig. 107: Inner contact bands - Diagram



Insert the band into the socket recess part. The inside of the socket must be clean.

Insert the band into the provided socket groove, (back-turn it inside the socket). The band should be clean and with sufficient tension to hold it against the socket wall by itself.

Fit the inner bands by sliding them over the center pin (start with the back contact). A thin screwdriver will help to locate them in their grooves.

Be careful not to scratch or bend the bands.

7.4.2.7 ECL Fly Leads

Inspect the condition of the cable insulation especially where they enter the end of the four brass tubes. Do not use if damaged.

7.4.3 Preliminary steps for ramping

7.4.3.1 List of activities which should have been completed

Completed activities	YES / NO
Magnet installation has been completed	
Cold head is in operation	
ECL inspection and repair have been performed ¹	
The LHe level must be higher than 70% for the initial installation ramp ²	
He exhaust system has been rigidly installed and is operational and leak tight	
Ventilation system has been installed and is operational	
The ERDU function has been started up successfully	
All magnetizable material must be removed from the RF-room	

1. Refer to (→ ECL Inspection and Repair / Page 102).

2. For more information, refer to (→ Footnotes / Page 176)

All of the items above must have been confirmed with **"Yes"** before continuing.

7.4.3.2 Restricting hazardous areas

 **CAUTION**

Strong magnetic field

If not observed, physical injury can occur

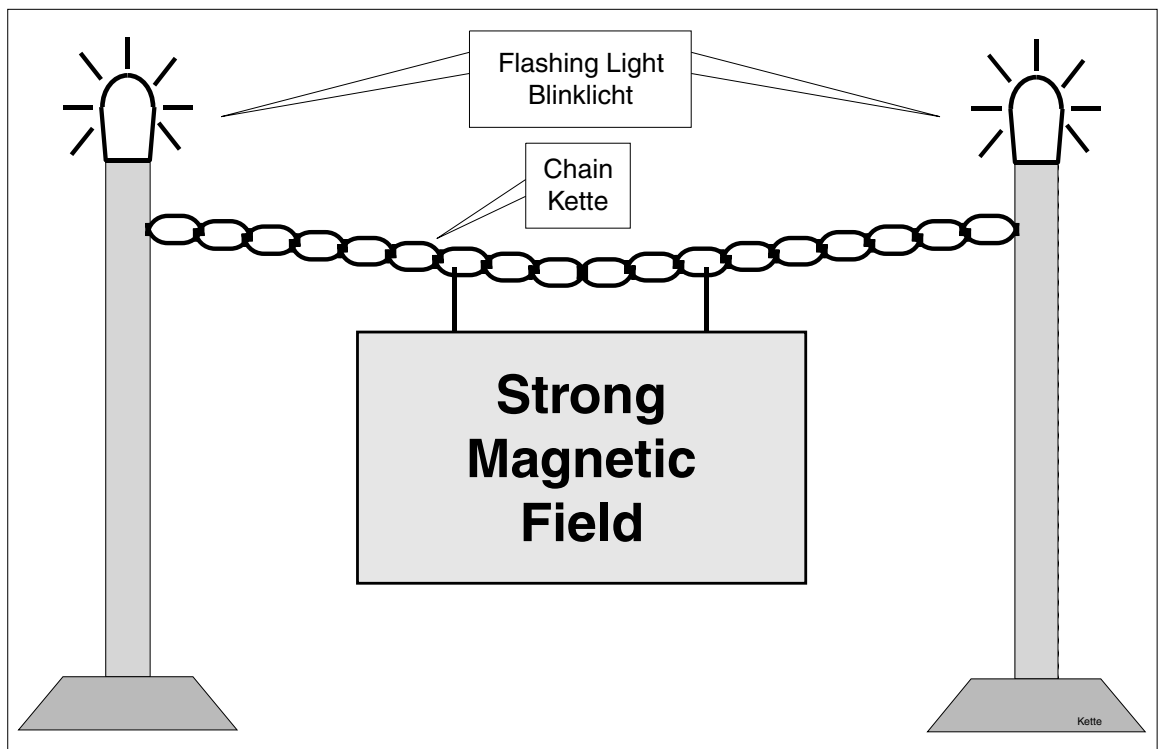
- » All magnetic field warning signs (especially "Strong Magnetic Field" and "Pace-maker-Warning") have been posted at appropriate locations.
- » Restrict access to the magnet room. Use chains and flashing lights as well as signs.

- Switch off the GPA cabinet before ramping the magnet.

K2217

F3 / F14

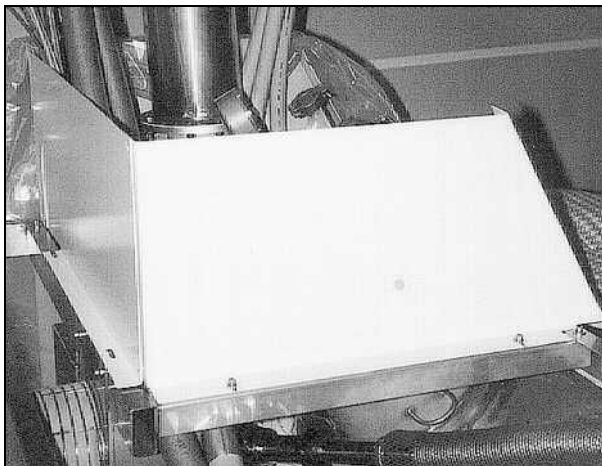
Fig. 108: Use flashing lights and chains to restrict access to the magnetic field



7.4.3.3 Removing the covers

- Remove the magnet side and back covers.
- Remove the magnet turret covers.

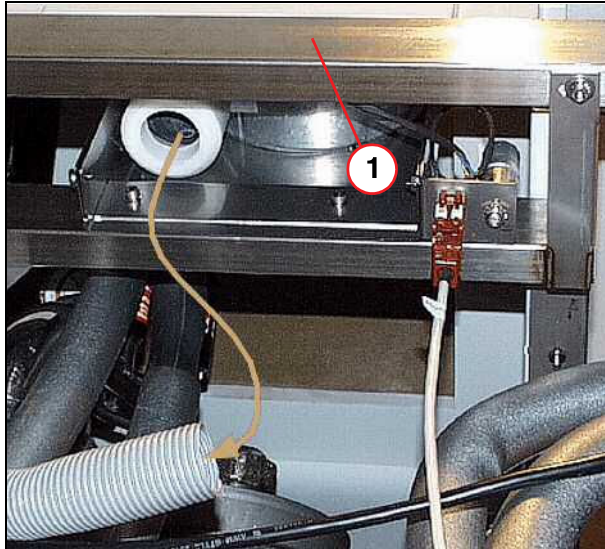
Fig. 109: Remove these covers



7.4.3.4 Tilting the patient air ventilator

- Remove the front rail (→ 1/ Fig. 110 Page 112) of the cover frame (three screws).

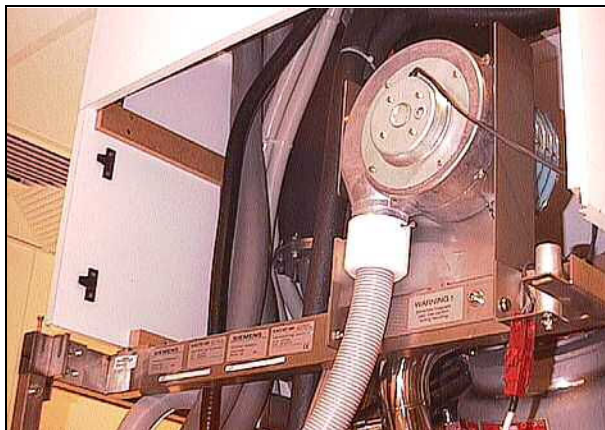
Fig. 110: Patient air hose removed from the ventilator



(1) Front rail

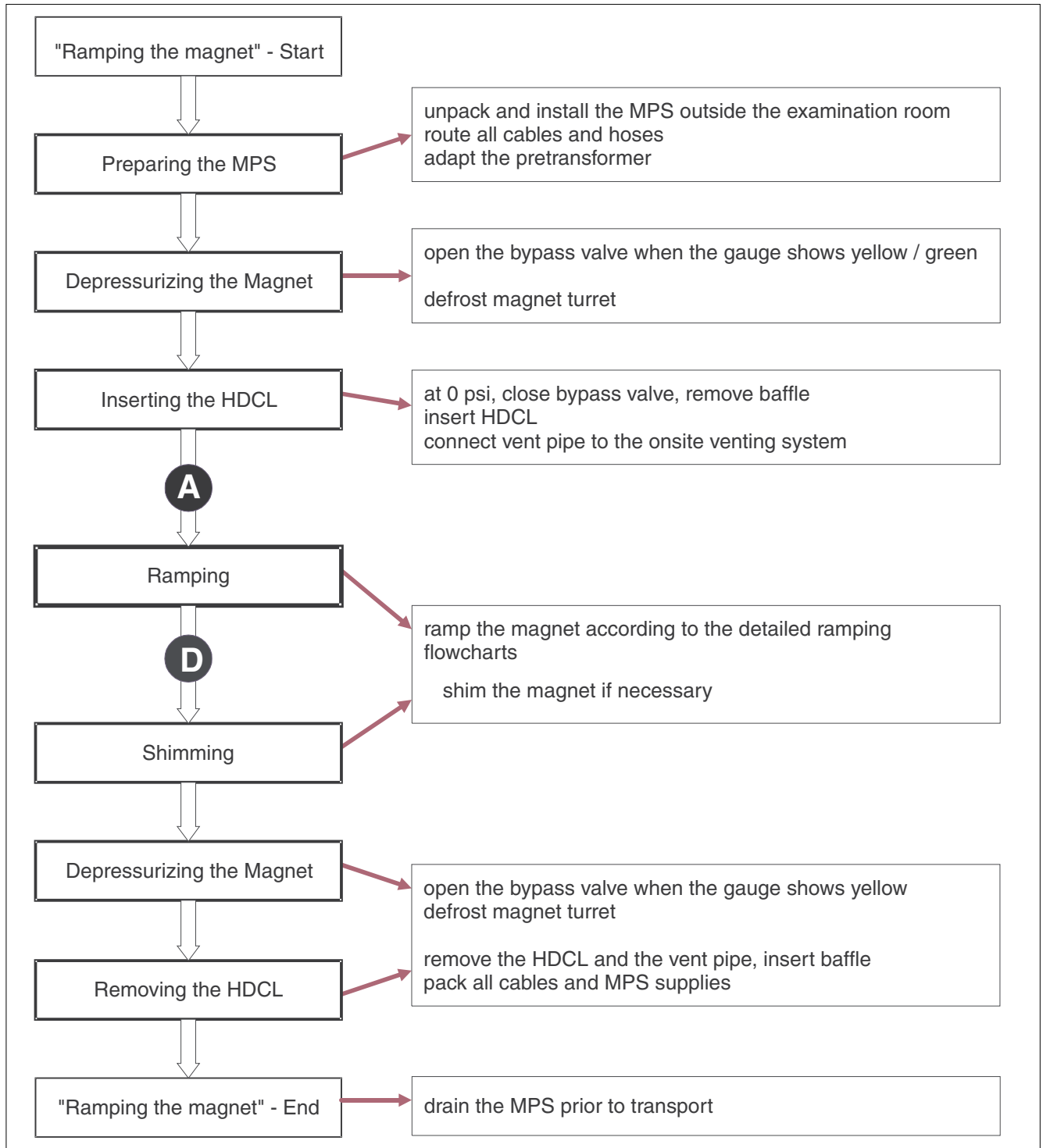
- Tilt down the ventilator by removing 3 screws (→ Fig. 111 Page 112) to access the service turret.

Fig. 111: Patient air ventilator



7.4.3.5 Complete Ramping - Flowchart

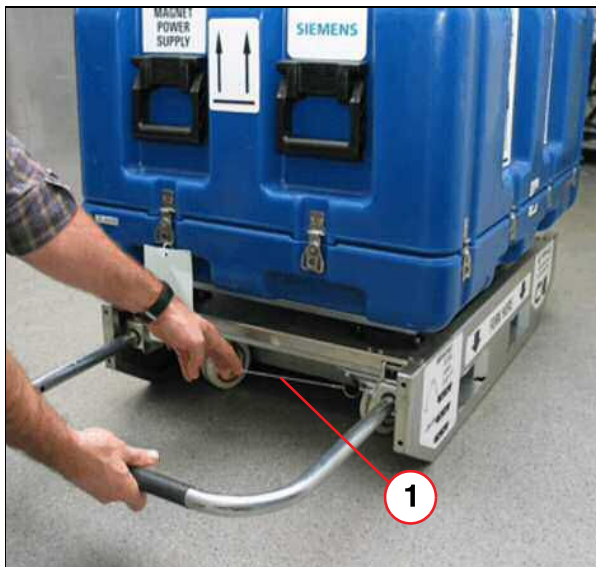
Fig. 112: General flowchart



7.4.4 Preparing the MPSU

- Position the MPSU outside the RF room door. Lower the trolley handle into the horizontal position. This will raise the wheels and allow the trolley to rest on the two skids.
- Pull the wire strap to unlock the handle. Slide the handle into the recess under the trolley.

Fig. 113: MPSU



(1) Wire locking strap

- Remove the lid of the shipping by unclipping the 12 spring clips.
- Open the hinged lid of the MPSU by pulling the two silver knobs and turning them a quarter of a turn followed by raising the lid.



CAUTION

Incorrect lock of the current lead

If not observed, magnet and MPSU can be damaged

» **Connect and secure the cables**

7.4.4.1 Installation

- Switch circuit breaker CB1 to OFF.
- Perform the cable connections as shown in the table.

Tab. 2 3600 MPSU cable connections

Cable	From	To	Part of
CAN OPEN (5m)	MPSU X7	ACC X125	Accessory kit

Cable	From	To	Part of
RS232	MPSU X6	ACC X12 (open-end cable from Ser_Distribution Port 6 (Ser_MPSU))	MPSU
MPS power lead	MPSU X1	ACC X130	MPSU

- Connect the ramp cable from the MPSU to the magnet after the transport test.

CAN OPEN lead longer than 5 m

- Connect the CAN OPEN lead.
 - » Using the CAN repeater with the PSU module if 5 m cable is too short.
Additionally using the 25 m CAN bus lead and the 25 m power cable.

Fig. 114: CAN repeater



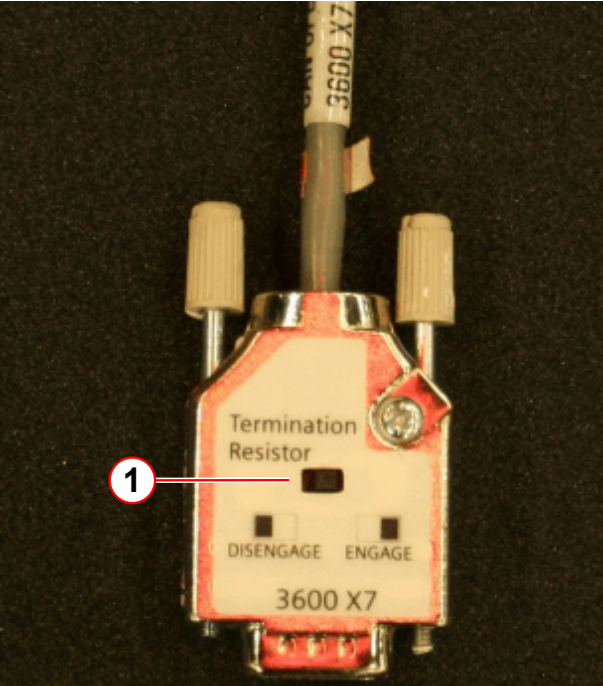
i

When using the extension kits to ramp a magnet the setting of the switch (→ 1/ Fig. 115 Page 116) on cable 617-371-25T is critical. It will ensure that communications remain good during the ramp process between the 3600 magnet power supply and the rest of the CAN bus system. If the communication does fail it could result in a magnet quench.

When cabling up the system review the status on the 3600.

- If the status is level 15 or more the switch should be in the engaged position.
- If the status is level 14 or less the switch should be in the disengaged position.

Fig. 115: Switch setting on cable 617-371-25T



(1) Switch for termination resistor

Tab. 3 CAN repeater connection

Cable	From	To	Part of
CAN OPEN (25m)	CAN repeater Can1	MPSU Can open X7	Accessory kit
CAN OPEN (5m)	Can 2	ACC X125	MPSU
Power module	CAN repeater	On-site power	Accessory kit

7.4.5 Transport Test

The transport test has to be performed with MPS revision level 17 or higher
If the revision level is lower than 17, continue with section (→ Ramp Cables / Page 126)

7.4.5.1 Introduction

During transport, it is possible that the MPSU suffered internal damage due to rough handling. As a result, the transport test is a mandatory test.

This test should be performed prior to connecting the ramp cables and using the MPSU on the magnet.

7.4.5.2 Tool Description

The transport test tool consists of 2 x 470 ohm resistors located in a grey Anderson plug. The center tap of the resistors is connected to a flying lead which is permanently bolted to the 3600 chassis. The magnet voltage sensor connectors in the Anderson shell are used and connected as shown in (→ Fig. 116 Page 117).

- Connect the transport test plug.

Tab. 4 Transport test plug

Cable	From	To	Part of
Transport test plug	MPSU X3	MPSU ground	MPSU

Fig. 116: Transport Test Plug Circuit

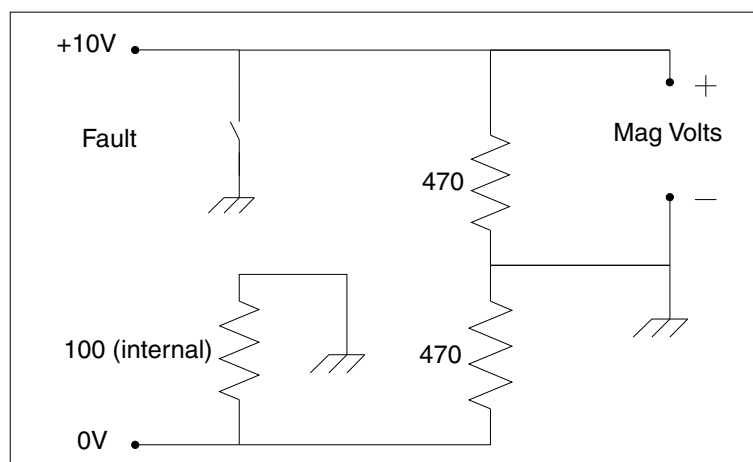


Fig. 117: Transport Test Plug - fitted correctly



Only 10 mA will flow and the fault can be detected by looking at the magnet voltage sensor input. The resulting voltages are below:

Tab. 5 Fault conditions

Condition	Magnet voltage detected
-ve bus bar fault.	+ 9.5
Test passed	+ 8.5
Test plug ground lead not connected.	+ 4.7
+ve bus bar fault.	0
Test plug not connected.	0
Rundown load diode shorted to chassis	0 - 4.5

7.4.5.3 Set up communication

Start the Hyperterminal (Password for "Advanced User": refer to speedinfo "SyngoMR password")

» Programs / Accessories / Communication

Start the Hyperterminal application and set it up as follows:

Fig. 118: Hyperterminal



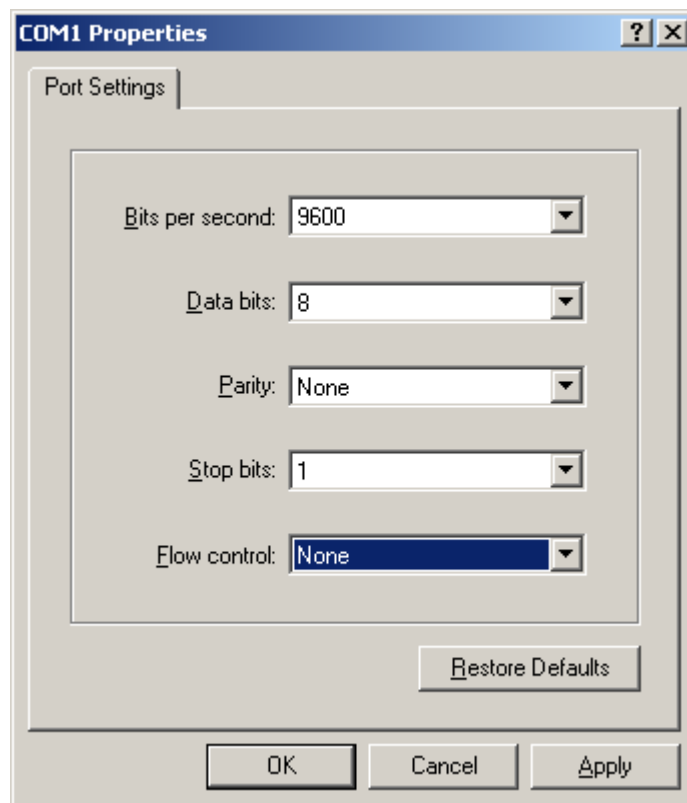
- Connect MPS using an available COM Port.

Fig. 119: Port connection



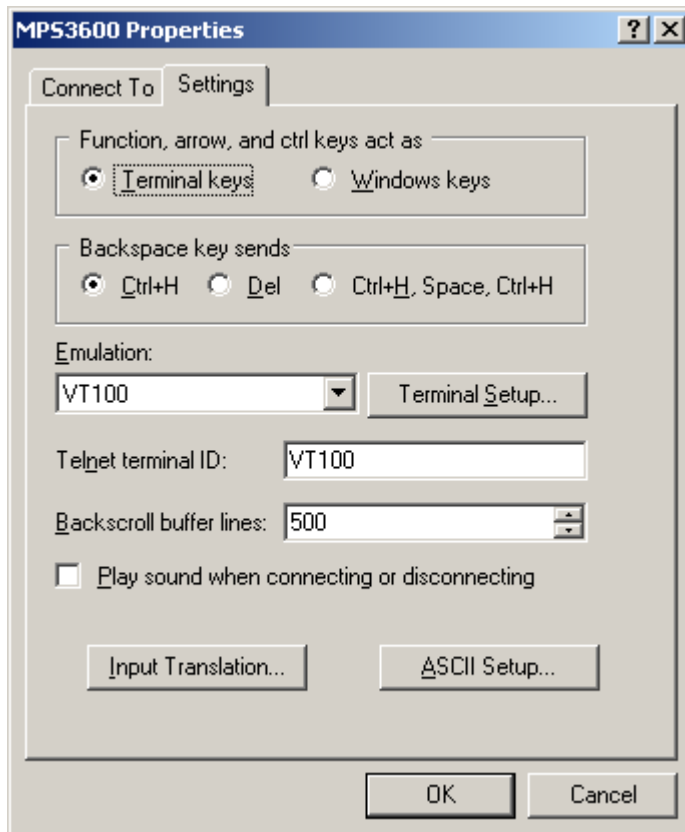
- Set the Hyperterminal protocol as follows:

Fig. 120: COM Port Properties



- Set Emulation to VT 100 in File > Properties > Settings.

Fig. 121: Terminal setting



- Save the settings on the computer using a suitable name.

7.4.5.4 Transport Test Procedure

- When the power supply arrives on site, the transport test plug should already be connected. If not, connect it to X3 as shown in (→ Fig. 117 Page 118) above. The test must be passed before the plug may be removed and the ramp cables connected.

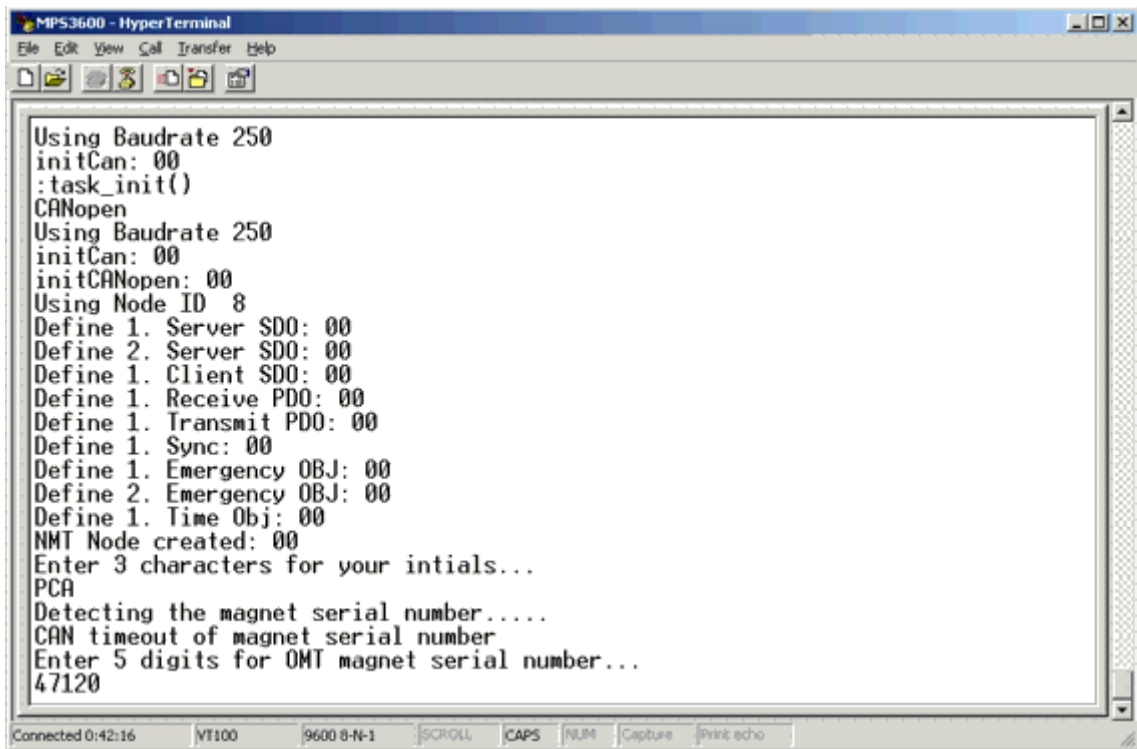


This test only needs to be completed the first time the MPSU is used on site.

- Ensure that circuit breaker F1 at the auto transformer is set to the ON position.
- Switch on circuit breaker F28 in the ACC and CB1 at MPS .
- Check that the green LED 'ON' indicator glows.

- The following screen should be displayed on the computer:

Fig. 122: System data input



```

MPS3600 - HyperTerminal
File Edit View Call Transfer Help

Using Baudrate 250
initCan: 00
:task_init()
CANopen
Using Baudrate 250
initCan: 00
initCANopen: 00
Using Node ID 8
Define 1. Server SDO: 00
Define 2. Server SDO: 00
Define 1. Client SDO: 00
Define 1. Receive PDO: 00
Define 1. Transmit PDO: 00
Define 1. Sync: 00
Define 1. Emergency OBJ: 00
Define 2. Emergency OBJ: 00
Define 1. Time Obj: 00
NMT Node created: 00
Enter 3 characters for your initials...
PCA
Detecting the magnet serial number.....
CAN timeout of magnet serial number
Enter 5 digits for OMT magnet serial number...
47120

Connected 0:42:16 VT100 9600 8-N-1 SCROLL CAPS NUM Capture Print echo

```

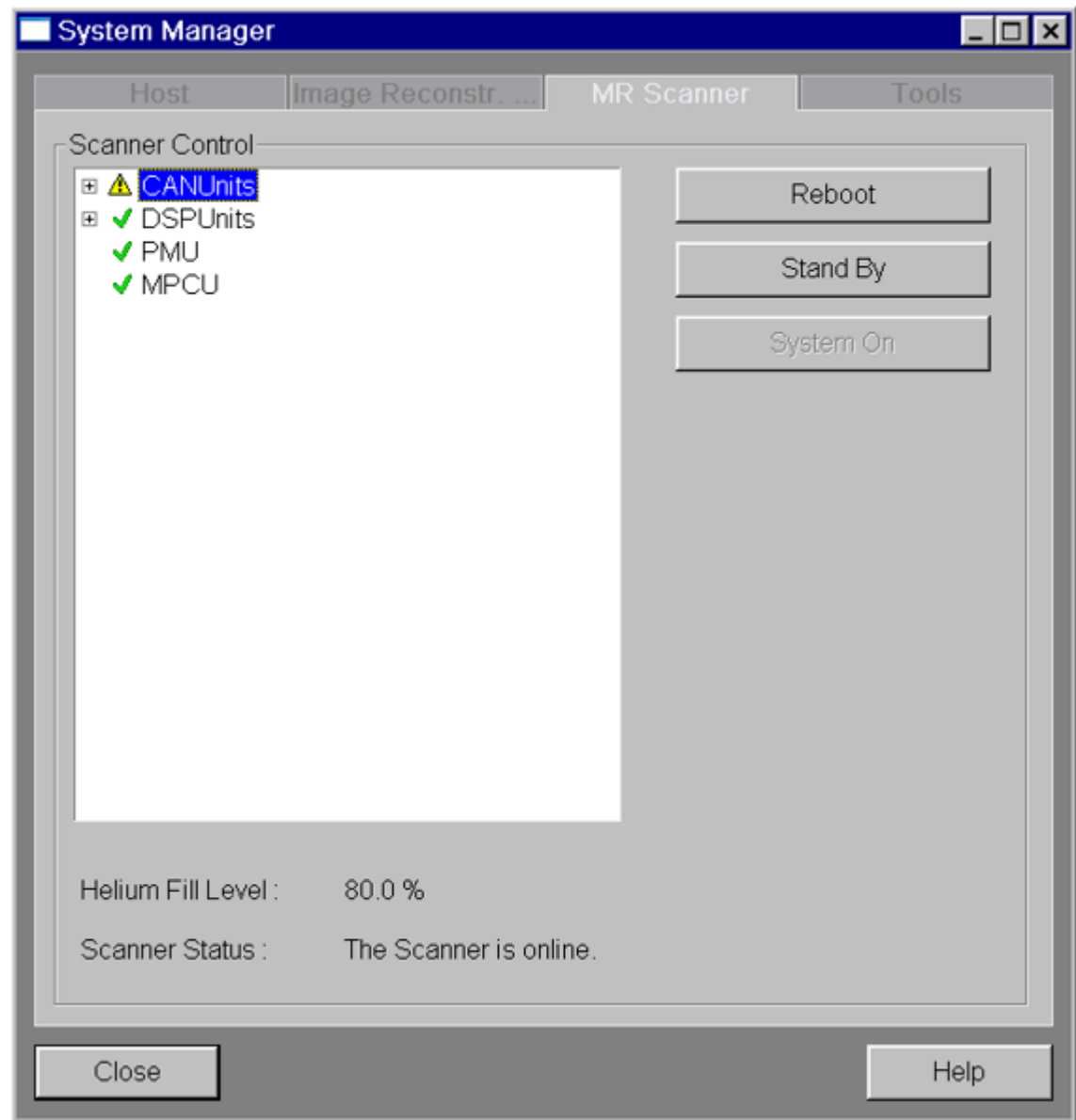


If the screen does not appear, check whether the correct COM port and settings have been selected. If they have, switch off the MPSU and then switch it back on using circuit breaker CB1.

It is recommended to save the settings in the Hyperterminal file using 'Save As' and naming it 3600.

- Reboot the MR system

Fig. 123:



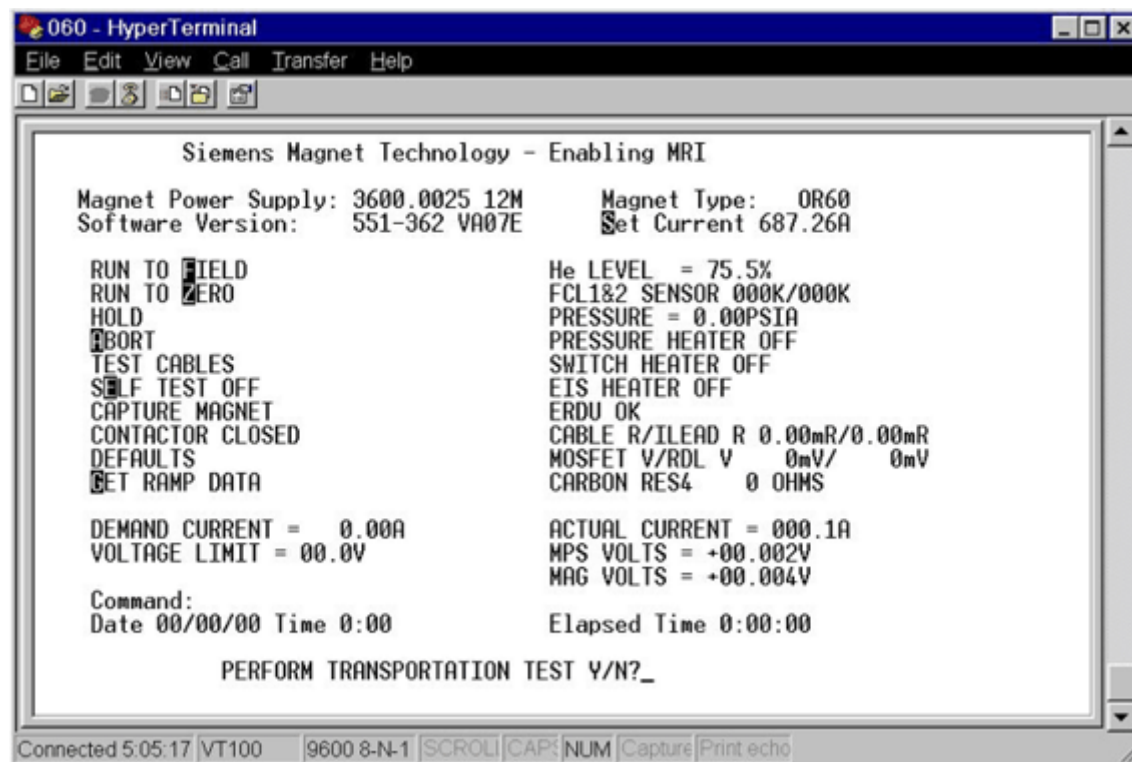
- » The reboot will update the MPS loadware.
- » The MPSRampLog.dat file will be copied to c:\medcom\log after the magnet ramp procedure is finished.



Reboot the MR system with the MPSU switched on.
 Always reboot the MR system after the MPSU is switched off and on again.
 Rename the logfile, e.g. "MPSRampLog1.dat" after the ramp is finished.
 If not renamed, the file will be overwritten by the next ramp logfile.
 Repeat booting of the MR system if the following warning message still persists: "MPS not operational".

- Enter your initials and magnet serial number when prompted.
- If the system includes a type 2657 or later Magnet Supervisory, the MPSU will now retrieve the magnet type, last set current, He level, date, and time from the Magnet Supervisory Unit. Older magnet systems that are not equipped with this type of supervisory unit will only be able to detect the Magnet ID.

Fig. 124: Screen 2 (requiring transport test)



- The screen will prompt you to 'Perform Transportation Test'. Type Y and press Enter.



If this test has already been successfully completed, type 'N' at this point to cancel the test. The software will then continue to the next part of the ramping sequence.

- Failure of the Transportation Test will be displayed briefly in the event the test detects an internal fault. This will be followed by a request to check the connections and repeat the test.

Fig. 125: Transport test failed

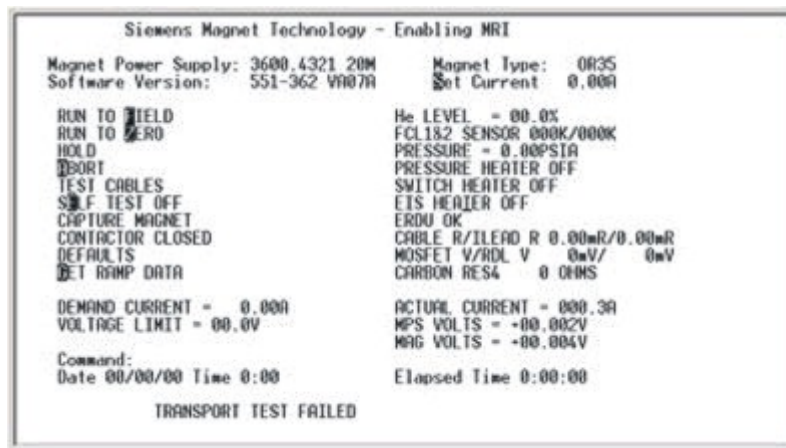
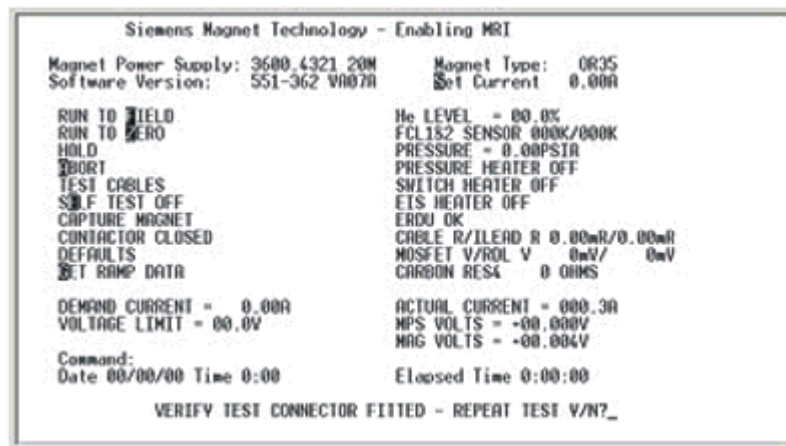
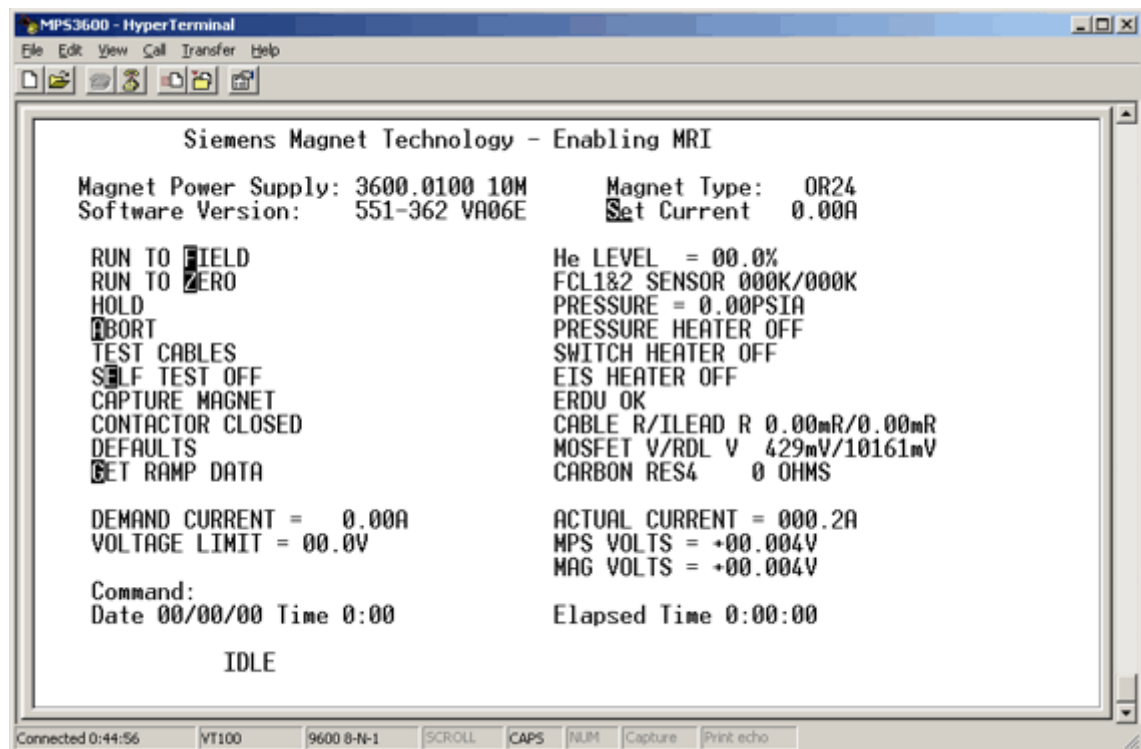


Fig. 126: Verify test connector fitted



- If the test is successful, the power supply will continue to perform an internal self test. The Self Test LED indicator will light. The self test runs up to 700 A with the output shorting contactor closed. This enables the voltage across the internal Run-Down Load and MOSFET switch to be measured and displayed on the screen.
- Upon completion of the self test, the operational screen is displayed:

Fig. 127: Operational Screen



- After 3 successive failed Transport Test attempts, further operation of the MPS is prevented. It is recommended that the MPSU be returned for repair and a replacement obtained.
- Line power to the MPS has to be switched OFF and then ON to allow for further testing attempts.
- When packing the MPSU after use, ensure that the transport test plug is refitted and ready for the next site.

7.4.6 Ramp Cables

If the Transport Test has been successfully completed or is not necessary, the following connections must be made/changed prior to ramping the magnet.



There is no need to switch off the MPSU at this point.

- Remove the transport test plug.
- Connect the two magnet ramp cables to Anderson Sockets X3 and X4 on the top of the MPSU, refer to (→ Fig. 128 Page 127).

- Route the ramp cable from the MPSU to the magnet.
Connect the ramp cables to the lower connectors at the magnet.
Secure the connectors.

**CAUTION****Incorrect locking of the ramp cables**

If these instructions are not observed, the magnet and MPSU can be damaged.

- » Connect and secure the cables.

Tab. 6 3600 MPSU cable connections

Cable	From	To	Part of
Magnet ramp lead	MPSU X3 MPSU X4	Magnet Magnet	Accessory kit

Fig. 128: MPS 3600



7.4.7 Depressurization of the Helium Vessel (Magnet)

**WARNING****Cryogenic liquids**

Failure to observe the following safety warning may result in cold burns.

- » Do not work on the cryostat or disconnect any of the fittings without first depressurizing the helium vessel.

The helium vessel will normally be pressurized to approx. 40 mbar (0.6 psi) overpressure (a different pressure is possible with mobile magnets).

Before depressurizing the helium vessel, ensure that protective clothing, face mask and gloves are worn. Also check that the area is well ventilated.

The helium vessel is depressurized via a tube and ball valve which will bypass the over-pressure valve assembly.

NOTICE**Low pressure in the helium vessel**

Before opening the bypass, the pressure gauge reading must be in the yellow or green sector or at least 0.1 psi in the helium vessel.

- » To avoid ice build-up in the magnet, the bypass must never be opened when the pressure gauge reads negative (red sector). If this is the case, the EIS can be energized until the pressure gauge indicates at least a positive reading. Refer to (→ Energizing the EIS circuit for pressure build up / Page 176).

- If the system pressure is low check for leaks to locate the possible cause for low pressure before you continue with the turret and venting (see (→ Magnet System Leakage Check / Page 181).
- Open the depressurization bypass valve (→ Fig. 129 Page 129), (→ Fig. 130 Page 129).
 - » Helium gas will flow through the vent line and into the quench tube (venting system).

**WARNING****Risk of asphyxiation**

If not observed, person working with cryogen's becomes dizzy or loses consciousness

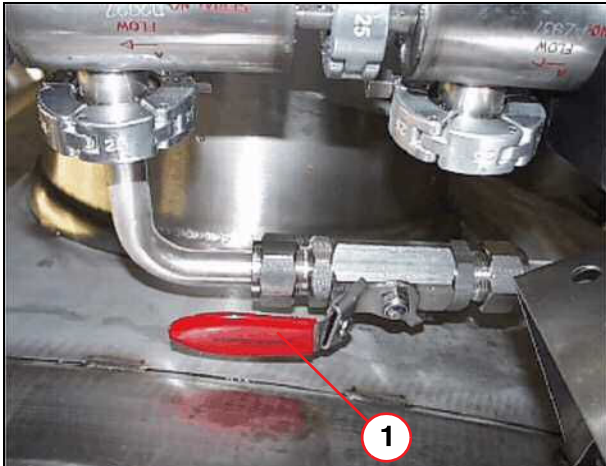
- » Helium gas must not be vented into the magnet room.



The magnet turret and the quench pipe will get very cold as a result of the escaping cold gas.

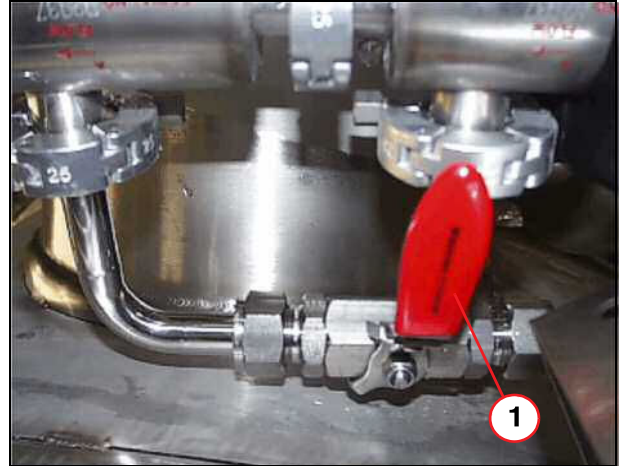
Wear personal safety equipment.

Fig. 129: Depressurization valve - open position



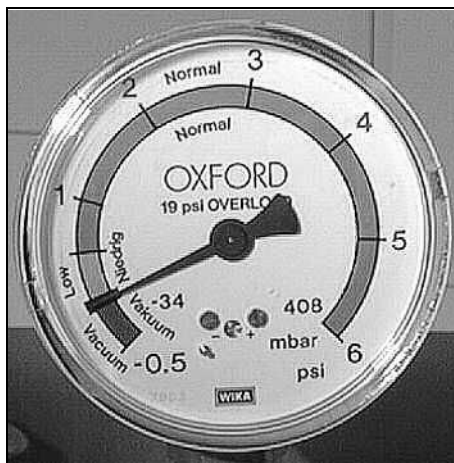
(1) Bypass valve open

Fig. 130: Depressurization valve - closed position



(1) Bypass valve closed

Fig. 131: System at atmospheric pressure (fully depressurized)



Allow the system to depressurize until the gauge reads zero psi (black region). The magnet is now at atmospheric pressure and the turret baffle may be opened.



CAUTION

Depressurized helium vessel

If the depressurization bypass valve is not closed immediately, ice will form in the helium vessel and neck tube, (→ Fig. 129 Page 129), (→ Fig. 130 Page 129).

» Do not leave the system depressurized longer than necessary.

- You can now open one of the turret openings into the LHe vessel.

- Immediately close the depressurization valve after opening one of the turret openings.

7.4.8 Inserting the Enhanced current lead (ECL)

The ECL (Enhanced Current Lead) is a hinged design of a current probe with hard-wired current leads. The leads are 1 m long and are connected to the top of the magnet.

WARNING

Attention, if the magnet quenches while inserting or removing the ECL, oxygen in the room will be displaced by helium. Risk of suffocation or cold burns.

If not observed, cold burns and suffocation can occur.

- » Immediately leave the RF cabin and as far as possible, close the door and secure the environment, until the gas flow from the magnet subsides.
- » Before entering the RF cabin again, check the oxygen level with an oxygen monitor.



Adhere to the cryogenic safety procedures during the following operations.

CAUTION

High pressure in the helium vessel

Cryogenic liquids

Failure to observe the following safety warning may result in cold burns.

- » Do not work on the cryostat or disconnect any of the fittings without first depressurizing the helium vessel.

WARNING

Current in leads

If ignored, death or serious physical injury can occur.

- » The MPS and ramp cables must remain connected to the ECL as long as it is inserted in the magnet. This applies to all procedures - including insertion or removal of the ECL. when the magnet is at field.

- Connect the 0.1 psi valve at the vent port, using a clamp and PTFE seal. Check the flow direction. The arrow must point toward the quench tube.

**CAUTION**

0.1 psi valve direction (→ Fig. 133 Page 133).

- » **If fitted with flow in the wrong direction, the magnet will quench**

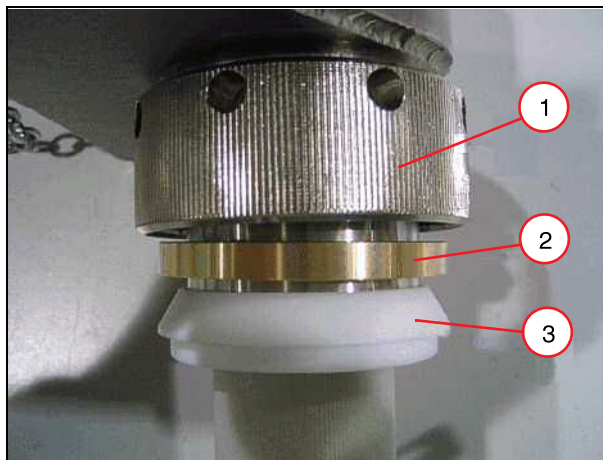
- Unpack the ECL and connect the vent line to the ECL using a high load seal and clamp. The 90 degree elbow should be connected to the ECL, (→ Fig. 133 Page 133).



When using it for the first time at a site, the contact springs of the inner and outer ECL contact must be replaced. .

- During the installation, the ECL vent line should remain connected to the ECL at all times with a high load seal and clamp.
- If cold and frosty, warm up the ECL ring nut and the baffle fitted on the turret.
- Immediately close the depressurization valve after opening any turret openings, (→ Fig. 129 Page 129), (→ Fig. 130 Page 129).
- Slide the system ring nut, brass compression ring, and PTFE seal onto the ECL, (→ Fig. 132 Page 131).

Fig. 132: ECL prepared with ring nut, brass compression ring, and PTFE seal



- (1) Ring nut
- (2) Brass compression ring
- (3) PTFE seal

New style fitment of ECL seals, fitted with:(→ 1/Fig. 132 Page 131)
(→ 2/Fig. 132 Page 131),
(→ 3/Fig. 132 Page 131),

When fitting, check to ensure they are assembled onto the ECL in the correct order and the orientation of the PTFE seal and compression ring is correct as shown in(→ Fig. 132 Page 131) .

**CAUTION**

**Inserting the ECL too quickly with the magnet at field
The magnet may quench**

- » **Open the ECL vent port and insert the ECL slowly**

**CAUTION**

Top of the ECL will get cold.

Failure to observe the following safety warning may result in cold burns.

- » Wear gloves for protection



If a strong mechanical resistance is felt, call the hotline. A large amount of ice may be present. If anything but minimal force is needed to push the ECL down, withdraw it and try to de-ice the port by warming and re-inserting the probe. If the ECL is removed, the port must be closed immediately using the baffle.



Do not attempt to force anything down the port to dislodge the blockage. This will only result in ice dropping into the socket - a condition which is more difficult to clear because it is in the helium reservoir.

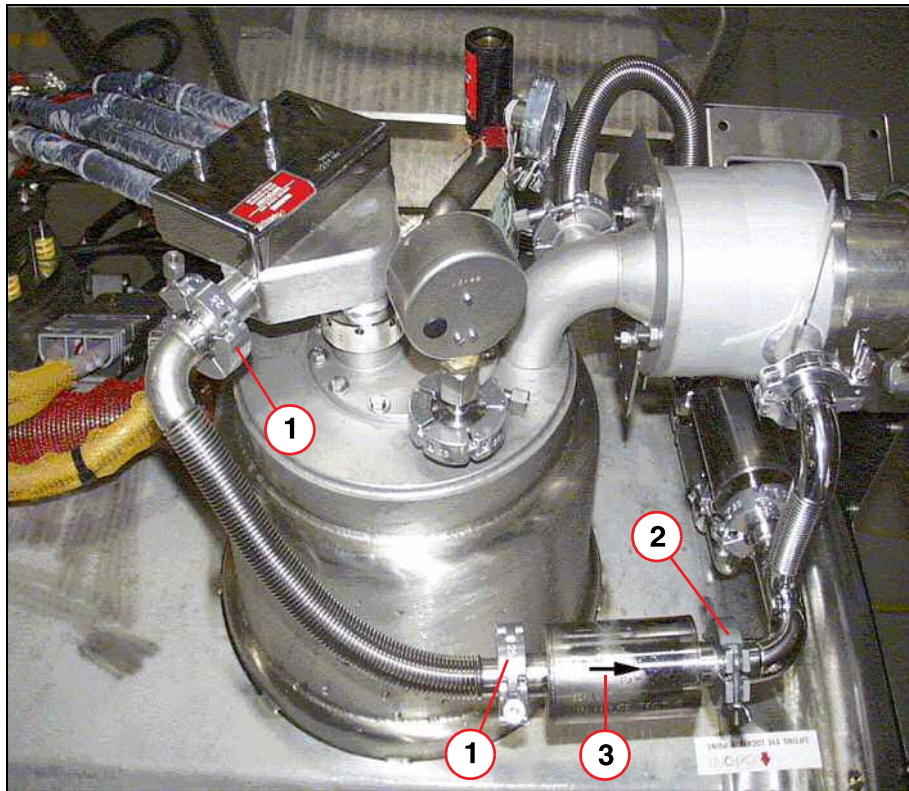
- Ensure that the ECL vent line is not capped. Remove the baffle set from the current lead port. Remove the copper gasket and destroy it (the gasket must not be used again).
- Before the gas flow subsides, insert the ECL into the port. Move it down until the middle of the upper hinge is halfway in. Hold it down in this position until the gas flow out of the vent pipe subsides. This indicates that the ECL has cooled down enough.



For low ceiling installations it may be necessary to bend the ECL at the hinges provided. Always ensure that the hinge sleeves are positioned over the hinge before the ECL is inserted into the port.

- Insert the ECL fully into the magnet and secure it by screwing down the ring nut by hand. If the ring nut cannot be screwed down use the heat gun to warm up the metal and try again. Finally tighten with "C" spanner.

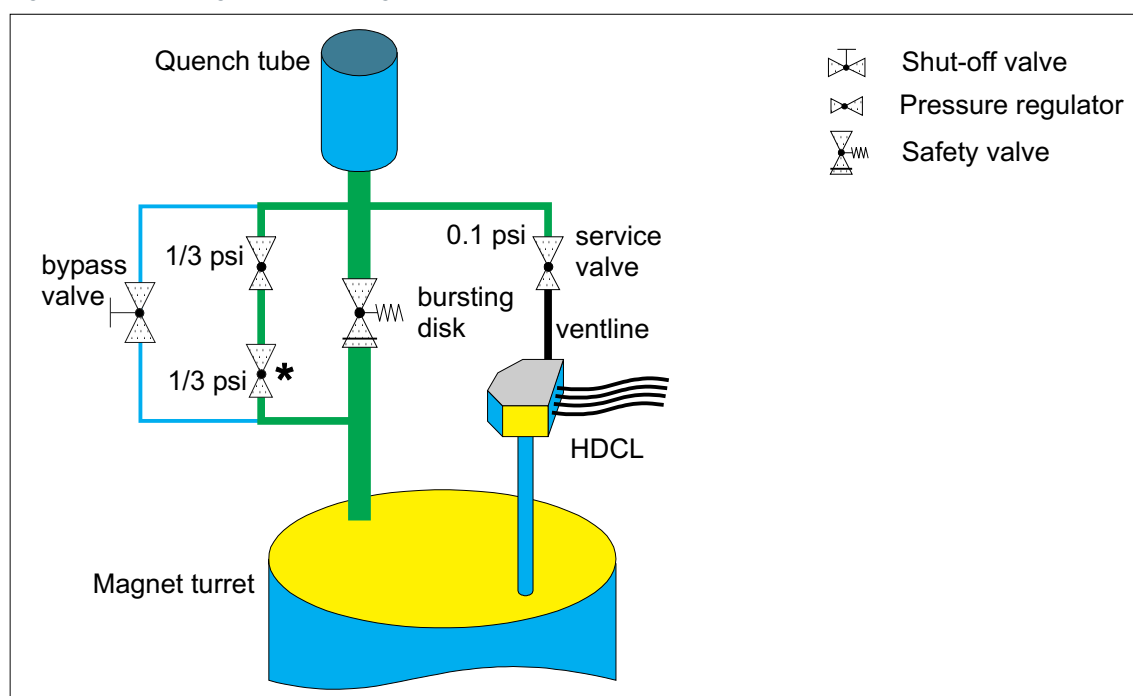
Fig. 133: Turn the vent line so that it can be connected to the 0.1 psi valve



- (1) High load clamps and seals
- (2) Low load clamp and seal
- (3) Flow direction of 0.1 psi valve

- Connect the vent pipe to the 0.1 psi valve already attached using a high load clamp and seal.
- The system venting with the ECL fitted should now be as shown in (→ Fig. 133 Page 133) and (→ Fig. 134 Page 134)
- Ensure that the ECL seals are fitted in the right order and seated correctly.

Fig. 134: Block diagram of the magnet and service valves

**NOTICE**

System pressure will continue to vent through the ECL.

» System will only repressurize when ECL is removed and the baffle refitted.

- » The pressure in the magnet should now rise to approx. 0.1 psi (yellow range) within a few minutes.
- If the pressure rises too high, check to ensure that the 0.1 psi valve is not stuck. Warm valve with heat gun until it is defrosted.
- If the pressure remains at zero, check for leaks around the ECL seals and its venting. For further troubleshooting, refer to (→ Pressure Raising Procedure for a Magnet NOT at Field / Page 183).



The ECL may be left venting through the 0.1 psi valve if further ramping or shimming is needed. Otherwise, remove the ECL.

**CAUTION**

Do not leave the ECL inserted for long periods

Risk of icing

» Remove ECL overnight or if magnet is not ramped for 4/5 hours

**Risk of quench**

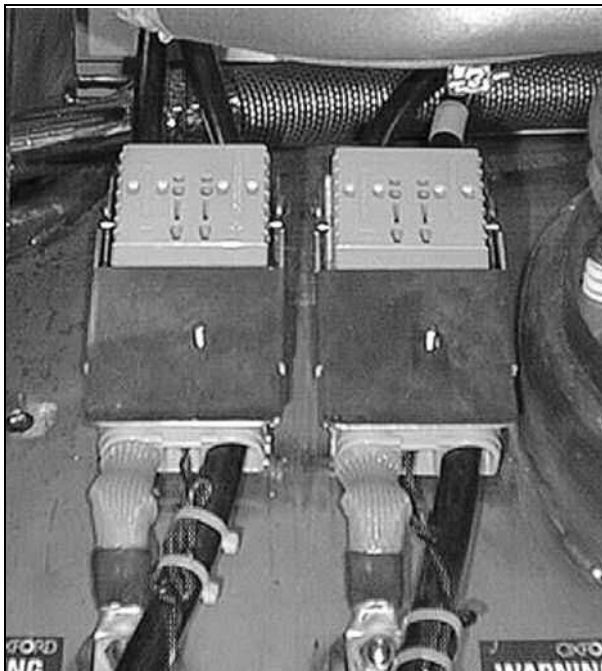
The ECL vent must NEVER be blanked.



After each operation has been completed on the turret, in particular fitting or removing the ECL, new copper seals must be used and checked for leaks using a suitable leak detector or solution.

- Connect the ECL cables to the top of the magnet. Ensure that the connection is properly made, and lock the lever with the nut provided.

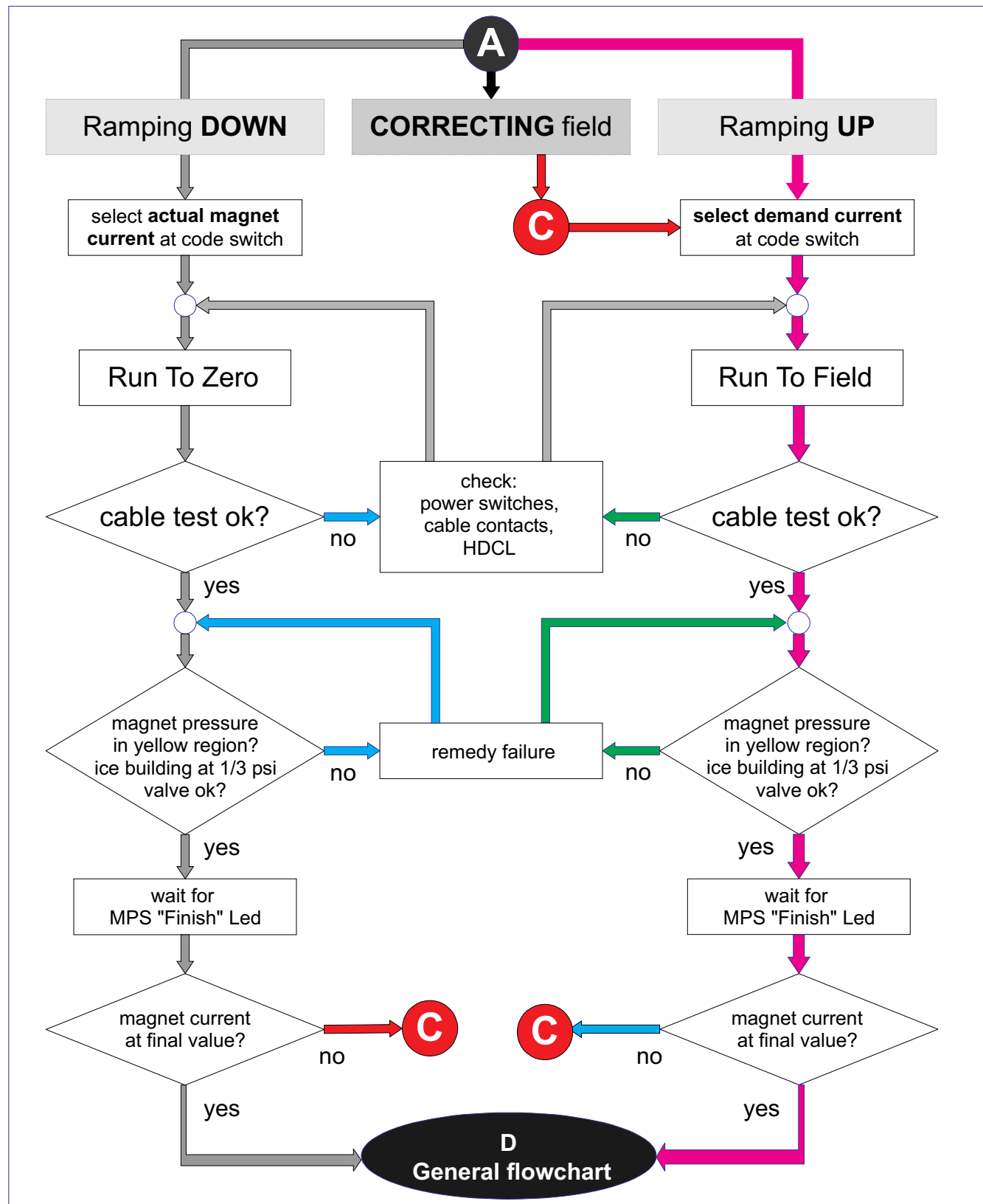
Fig. 135: ECL current leads connected at the magnet



- Verify that the MPS current leads are connected to the MPS and to the lower connectors at the magnet.

7.4.9 Ramping - flowchart

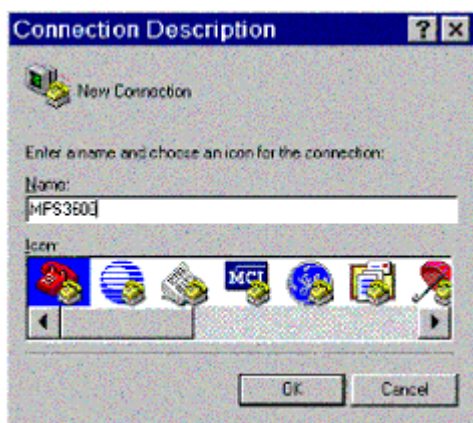
Fig. 136: Ramping flowchart



7.4.10 Set up of the MPSU

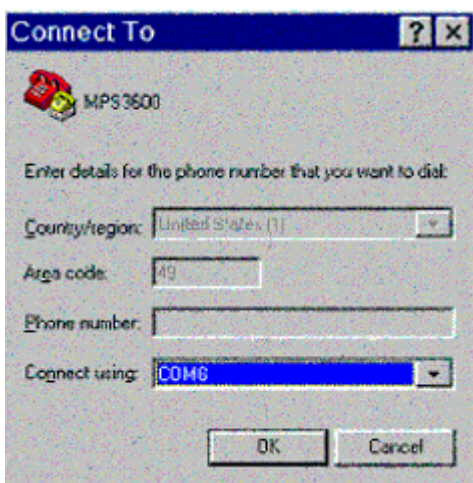
- Turn on F28 in the ACC. Turn ON the MPSU using circuit breaker CB1 on MPSU.
- Start the Hyperterminal (Password for "Advanced User" refer to speedinfo "SyngoMR password")
 - » Programs / Accessories / Communication
- Perform the following settings.

Fig. 137: Hyperterminal



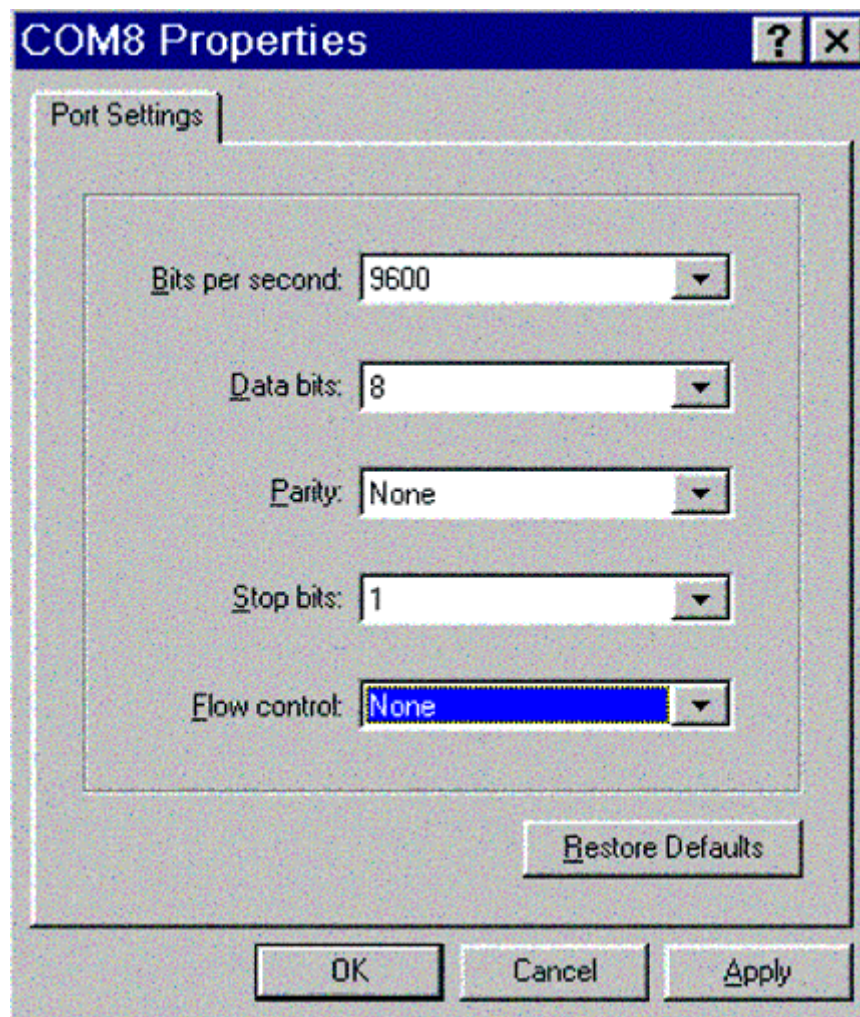
- Connect MPS using COM8.

Fig. 138: Port connection



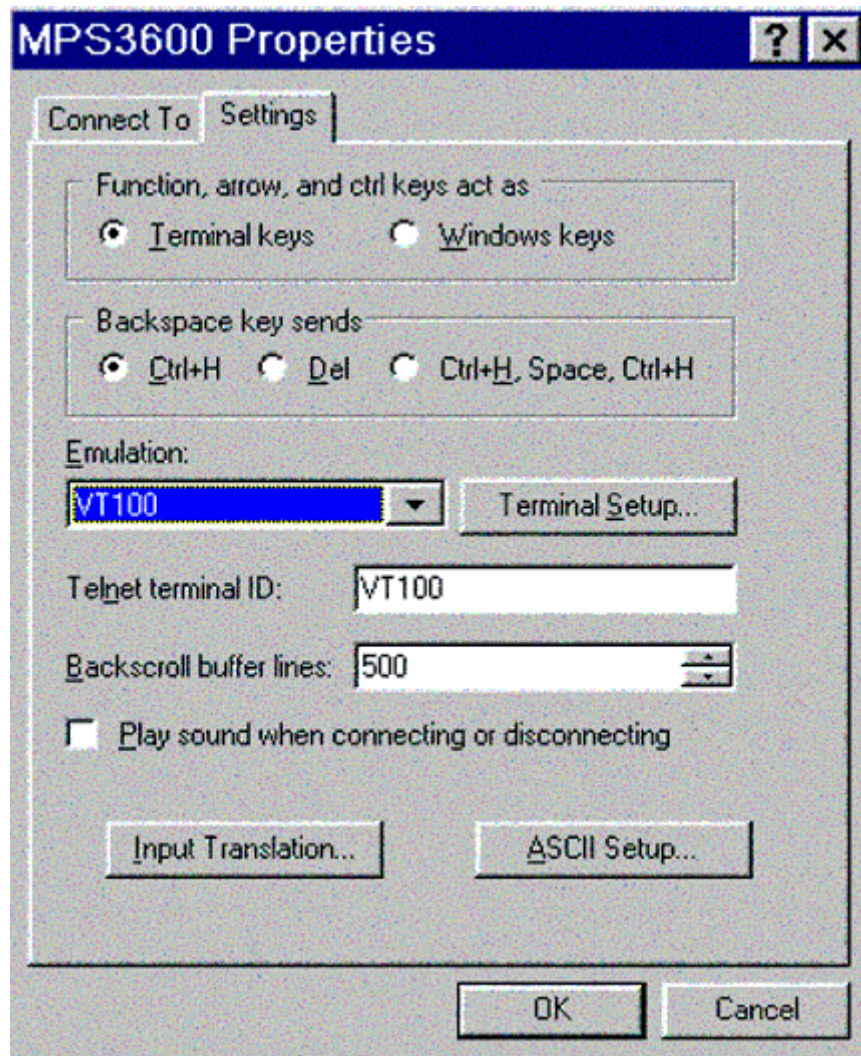
- Set the Hyperterminal Application as follows:

Fig. 139: COM8 Properties



- Set Emulation to VT 100 in File > Properties > Settings.

Fig. 140: Terminal setting

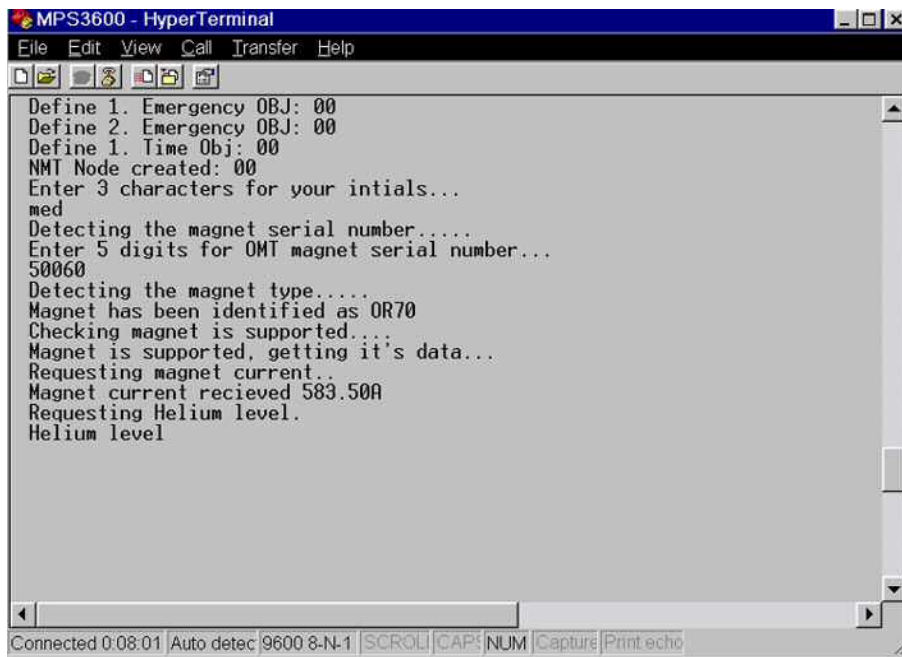


The last ramp current is stored in the MSUP. This will be displayed when the MPSU is switched on. If the MSUP has been reset, ensure that there is a valid helium level reading before switching on the MPSU.

- Turn the MPSU OFF and ON using the circuit breaker CB1 on MPSU.

The following screen is displayed:

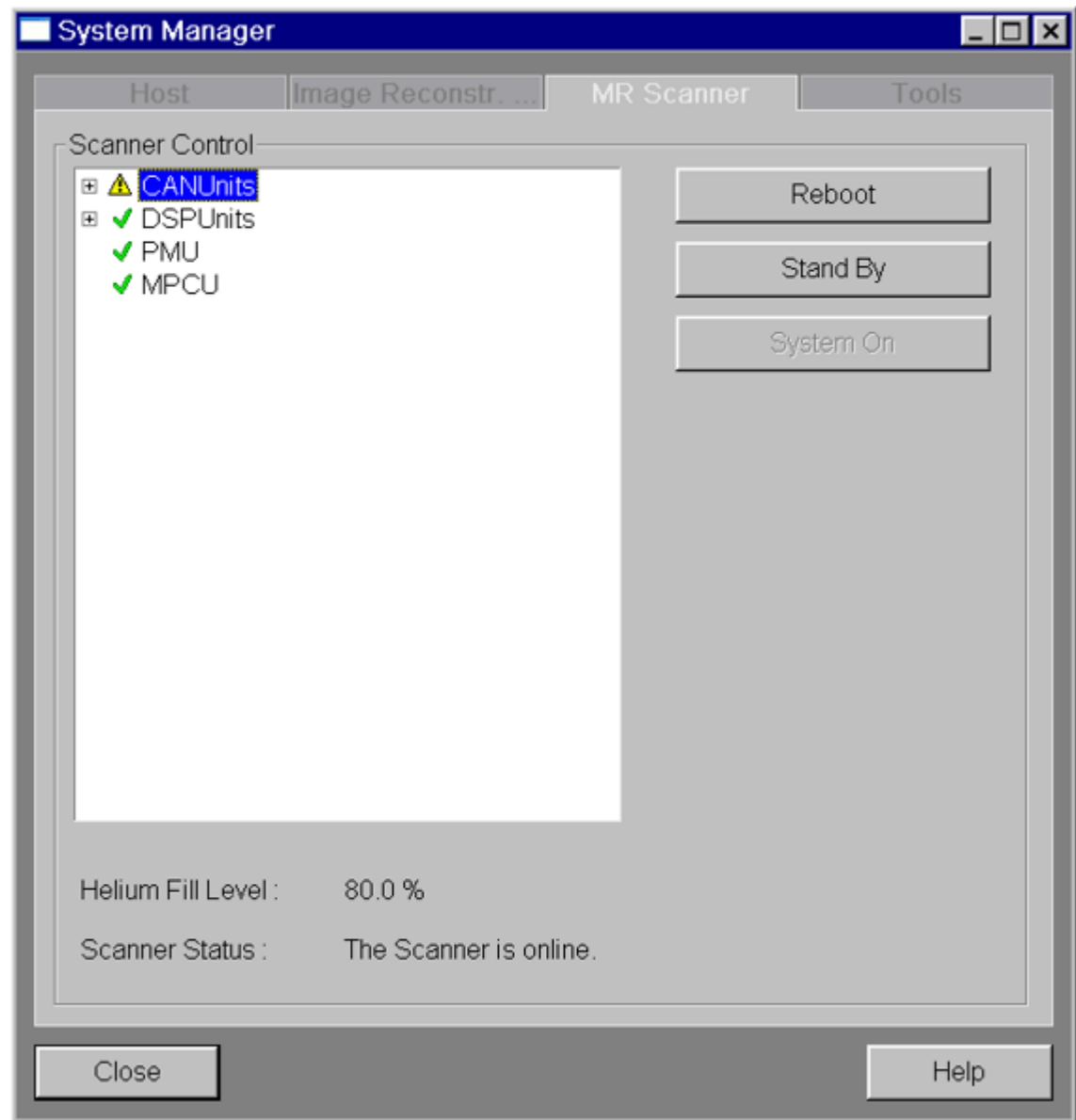
Fig. 141: HyperTerminal



```
MPS3600 - HyperTerminal
File Edit View Call Transfer Help
Define 1. Emergency OBJ: 00
Define 2. Emergency OBJ: 00
Define 1. Time Obj: 00
NMT Node created: 00
Enter 3 characters for your initials...
med
Detecting the magnet serial number.....
Enter 5 digits for OMT magnet serial number...
50060
Detecting the magnet type.....
Magnet has been identified as OR70
Checking magnet is supported....
Magnet is supported, getting it's data...
Requesting magnet current..
Magnet current recieved 583.50A
Requesting Helium level.
Helium level
Connected 0'08:01 Auto detec 9600 8-N-1 SCROLL CAPS NUM Capture Print echo
```


- Reboot the MR system

Fig. 142:



- » The reboot will update the MPS loadware.
- » The MPSRampLog.dat file will be copied to c:\medcom\log after the magnet ramp procedure is finished.



Reboot the MR system with the MPSU switched on.

Always reboot the MR system after the MPSU is switched off and on again.

Rename the logfile, e.g. "MPSRampLog1.dat" after the ramp is finished.

If not renamed, the file will be overwritten by the next ramp logfile.

Repeat booting of the MR system if the following warning message still persists: "MPS not operational".

- Enter 3 characters for your initials, then press <Enter>
- Enter the magnet serial number.



If the screen does not appear, make sure that the correct Com port and settings have been selected. If they have, turn the MPSU off and then back on, using the circuit breaker CB1.

The MPSU will now retrieve the magnet type, last current set, the He level, as well as date and time from the Magnet Supervisory Unit.

Once the data required have been obtained, the MPSU will initiate a self-test. The self-test LED indicator will light up. The self-test runs up to 700A, with the output shorting contactor closed.

Upon completion of the self-test, the operational screen is displayed:



Verify that the correct Magnet type is displayed.

7.4.10.1 Commands

Tab. 7 Commands

Command	Sequence of Settings
To set the field current	S<space>514.05<ENTER> The numerical number may be varied as required.
To run to field	F <Return>
To run to zero	Z <ENTER>
To abort current action	A <ENTER>
To refresh the screen	O <ENTER>
To go to the error screen	G <ENTER>

7.4.11 Run to field

7.4.11.1 Run to field prerequisites:

CAUTION

Helium level

Magnet can quench

- » Ensure that the helium level is higher than 50% (initially ramping must be above 70%) before ramping up the magnet.



The quench valve must be completely defrosted prior to ramping. The ECL may be left venting through the 0.1psi valve if further ramping or shimming is needed. Otherwise, remove the ECL (refer to (→ Removing the Enhanced current lead (ECL) / Page 148).

CAUTION

Insufficient waiting time between filling and ramping

Risk of quench

- » Allow one hour after filling and before ramping

CAUTION

No sufficient cooling through the ECL

Risk of a quench

- » If the system 1/3 psi vent valve frosts up during self-test, stop ramping. This indicates that cooling through the ECL is not sufficient. The 1/3 psi valve should be warmed up before continuing.

WARNING

Current in leads is indicated by a flashing LED on MPS

If ignored, death or serious physical injury can occur.

- » The MPS and ramp cables must remain connected to the ECL as long as it is inserted in the magnet. This applies to all procedures - including insertion or removal of the ECL. when the magnet is at field.



During magnet ramping, the operator must be within clear view of the MPS control panel to monitor and report any conditions or failures.

- Wait at least 1 hour after completing any previous ramp or aborted ramp (including cable checks).

7.4.11.2 Setting the current

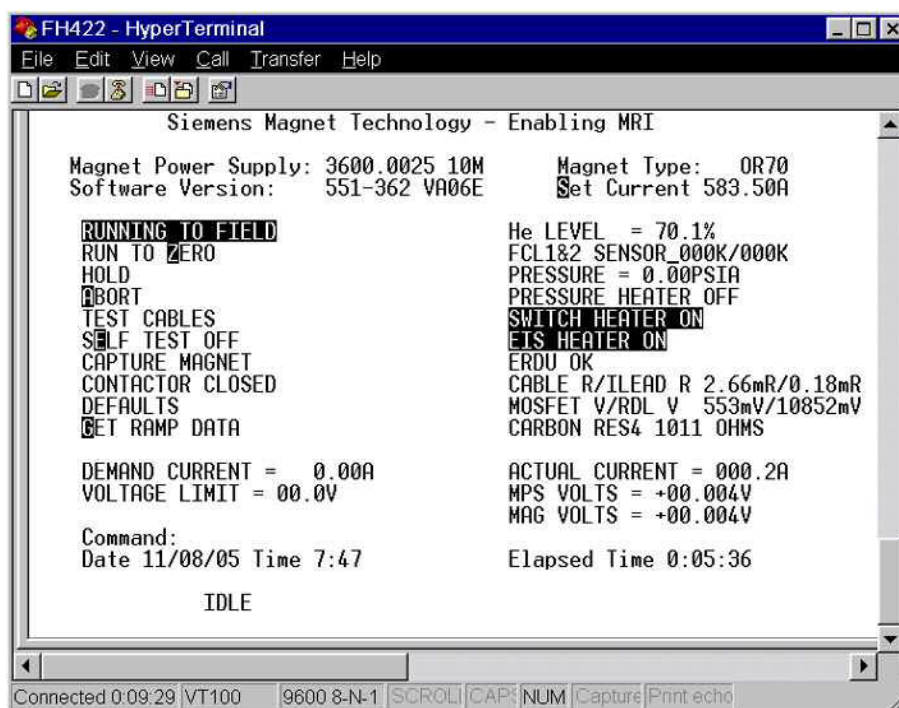
Setting the current is necessary for the first ramping after the upgrade to Trio, A Tim System.

Enter the magnet current via the HyperTerminal:

» S<space>xxx.xx<ENTER>

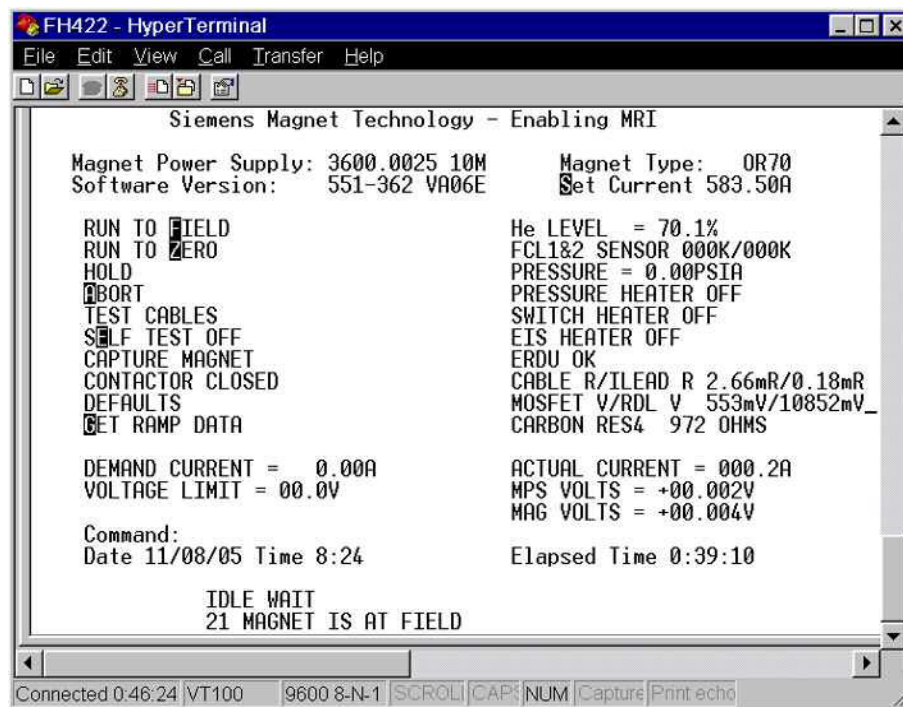
7.4.11.3 Run to field

Fig. 143: Running to field (example of a Symphony screen shot)



- Enter the command F <ENTER> for ramping up the magnet.
- Checks will be carried out and the MPSU will perform an automatic ramp to field.

Fig. 144: Magnet is at field (example of a Symphony screen shot)



7.4.12 Run to zero

7.4.12.1 Run to zero prerequisites:



The quench valve must be completely defrosted prior to ramping. The ECL may be left venting through the 0.1psi valve if further ramping or shimming is needed. Otherwise, remove the ECL (refer to (→ Removing the Enhanced current lead (ECL) / Page 148) .



During magnet ramping, the operator must be within clear view of the MPS control panel to monitor and report any conditions or failures.



CAUTION

Insufficient waiting time between filling and ramping

Risk of quench

- » Allow one hour after filling and before ramping

**CAUTION**

No sufficient cooling through the ECL

Risk of a quench

- » If the system 1/3 psi vent valve frosts up during self-test, stop ramping. This indicates that cooling through the ECL is not sufficient. The 1/3 psi valve should be warmed up before continuing.

**WARNING**

Current in leads is indicated by a flashing LED on MPS

If ignored, death or serious physical injury can occur.

- » The MPS and ramp cables must remain connected to the ECL as long as it is inserted in the magnet. This applies to all procedures - including insertion or removal of the ECL. when the magnet is at field.

- Wait at least 30 minutes after completing any previous ramp, or aborted ramp (including cable checks).
- Wait at least two hours after filling in helium.

7.4.12.2 Run to zero

Fig. 145: Ramping the magnet

```

MPS3600 - HyperTerminal
File Edit View Call Transfer Help

Siemens Magnet Technology - Enabling MRI

Magnet Power Supply: 3600.0025 10M      Magnet Type: 0R70
Software Version: 551-362 VA06E        Set Current 583.50A

RUN TO FIELD      He LEVEL = 68.8%
RUN TO ZERO      FCL1&2 SENSOR 000K/000K
HOLD              PRESSURE = 0.00PSIA
ABORT             PRESSURE HEATER OFF
TEST CABLES      SWITCH HEATER OFF
SELF TEST OFF    EIS HEATER OFF
CAPTURE MAGNET   ERDU OK
CONTACTOR CLOSED CABLE R/ILEAD R 0.00mR/0.00mR
DEFAULTS         MOSFET V/RDL V 554mV/10713mV
GET RAMP DATA   CARBON RES4 972 OHMS

DEMAND CURRENT = 0.00A      ACTUAL CURRENT = 000.2A
VOLTAGE_LIMIT = 00.0V      MPS VOLTS = +00.004V
                           MAG VOLTS = +00.004V

Command:
Date 11/08/05 Time 8:45    Elapsed Time 0:00:00

IDLE

Connected 0:08:45 | Auto detect | 9600 8-N-1 | SCROLL | CAP: NUM | Capture | Print echo
  
```

- Enter the command Z<ENTER> for ramping down the magnet. Checks will be carried out and the MPSU will perform an automatic ramp to zero.

7.4.13 Ramping the magnetic field to the required field strength

To avoid magnet quenches during short ramp cycles, the field must be reduced to zero first and then set to the calculated final magnet current.

- Check the actual frequency (**f** actual).
- Calculate the **final current** using the **highest frequencies possible** listed in the formulas below: The formula takes the tolerances of the different magnet power supplies in consideration

Fig. 146:

$I_{\text{final}} = \frac{f_{\text{nominal}} \times I_{\text{demand}}}{f_{\text{actual}}}$	
$I_{\text{final}} = \frac{40,53[\text{MHz}] \times I_{\text{demand}}}{f_{\text{actual}}}$	$I_{\text{final}} = \frac{63,68[\text{MHz}] \times I_{\text{demand}}}{f_{\text{actual}}}$
<i>MAGNETOM Harmony</i>	<i>MAGNETOM Symphony</i>

Tab. 8 Minimal frequency

	1 Tesla	1.5 Tesla
Lower limit of frequency	40.35 MHz	63.50 MHz
Upper limit of frequency	40.55 MHz	63.70 MHz



Before reaching the lower limit of the frequency range, the magnet must be scheduled for re-ramping.

Field decay for Harmony: approx. 3 kHz / month

Field decay for Symphony: approx. 4.6 kHz / month

7.4.14 Work after ramping the magnet

7.4.14.1 Downloading the MPS log file

The Magnet Peak Current is recorded. If the magnet should quench, the recording will tell you at what current level the magnet quenched.

To download the log file:

- **G** <Return>, this will ensure that you are in the error screen.

- L <Return>, the data screen appears.

The data screen contains seven columns. Each line displays the data stored at a specific period of time during magnet ramping.

- Select the following in the Hyper terminal:

"Capture Text"

Type the location and name of the log file, e.g. C:\Medcom\Temp\MR25123_51099log.txt

- Press the **"Start"** button

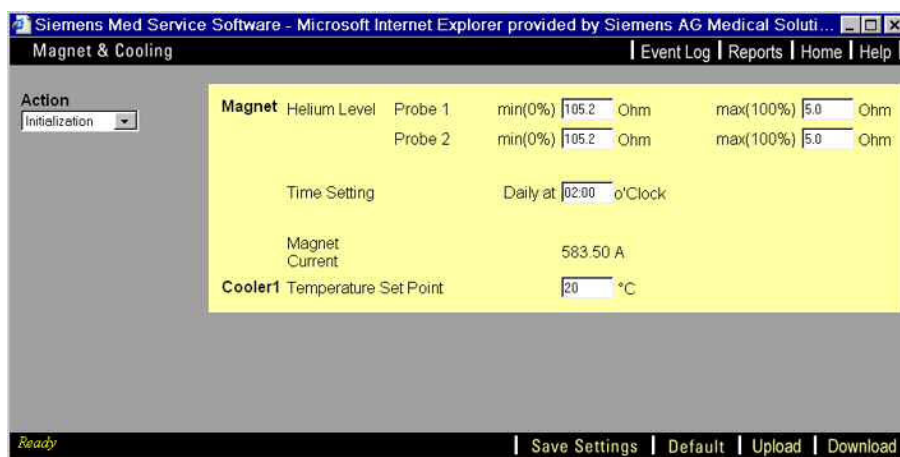
» Send the log file to the following email address:

To: magnet.hsc@siemens.com

7.4.14.2 Saving the magnet current to the HOST

- Access the service menu in the SeSo at MRC.
- Select menu item "Magnet & Cooling."
- Select "Initialization" in the "Action" menu

Fig. 147: Initialization



- Click "Download" to save the magnet current value to the HOST.

7.4.15 Removing the Enhanced current lead (ECL)

NOTICE

Removing the ECL

- » The MPS must remain connected until after the ECL has been removed. The ECL vent port **MUST NOT** be blanked.

**WARNING**

Attention, if the magnet quenches while inserting or removing the ECL, oxygen in the room will be displaced by helium. Risk of suffocation or cold burns.

If not observed, cold burns and suffocation can occur.

- » Immediately leave the RF cabin and as far as possible, close the door and secure the environment, until the gas flow from the magnet subsides.
- » Before entering the RF cabin again, check oxygen level with an oxygen monitor.

7.4.15.1 Safety checks

Before removing the ECL, ensure that no current is flowing in the magnet leads. Short the contactor circuit by performing the following procedure:

- Ensure that circuit breaker F28 on top of the ACC and CB 1 on the MPS are turned on (the lamp in the ON key is illuminated green).
- Check if the red LED "Warning - Current in leads" at the MPS control panel is **not** illuminated. If it is illuminated, hazardous currents may be present and the procedures in (→ Power failure during magnet ramping / Page 153) must be carried out.

**CAUTION**

Do not remove the ECL with current in leads

This can result in damage or personal injury.

- » Do not remove the ECL, if the red LED "Warning - Current in leads" is illuminated. The MPS must be used to fully discharge the magnet leads by following the procedure described in (→ Power failure during magnet ramping / Page 153)

**WARNING**

Current in leads

If ignored, death or serious physical injury can occur.

- » The MPS and ramp cables must remain connected to the ECL as long as it is inserted in the magnet. This applies to all procedures - including insertion or removal of the ECL. when the magnet is at field.

! CAUTION

Disconnecting the ECL during a power failure.

This can result in damage or personal injury.

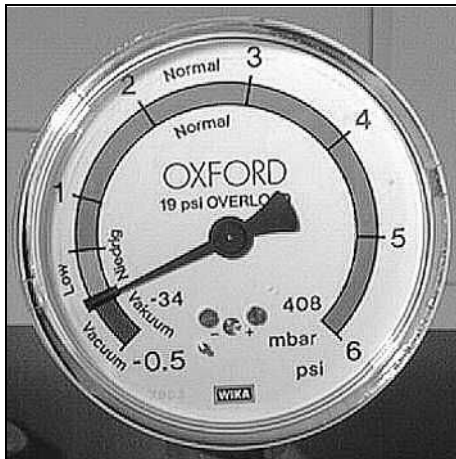
- » Never remove or disconnect the ECL during a power failure. There will be no warning of hazardous currents.

NOTICE

Negative pressure in the helium vessel

- » Never open the ECL port with negative pressure in the helium vessel. Refer to (→ Energizing the EIS circuit for pressure build up / Page 176).

Fig. 148: System at atmospheric pressure (fully depressurized)



- Depressurize the magnet (if it is pressurized) to zero psi as described in (→ Depressurization of the Helium Vessel (Magnet) / Page 128).

- Remove any water/ice build-up from around the current lead entry port using a hot air gun and paper towels.
- Shut off the switches at the MPS and switch off circuit breaker F28 on top of the ACC.

! CAUTION

Very cold gas is being emitted from the entry port.

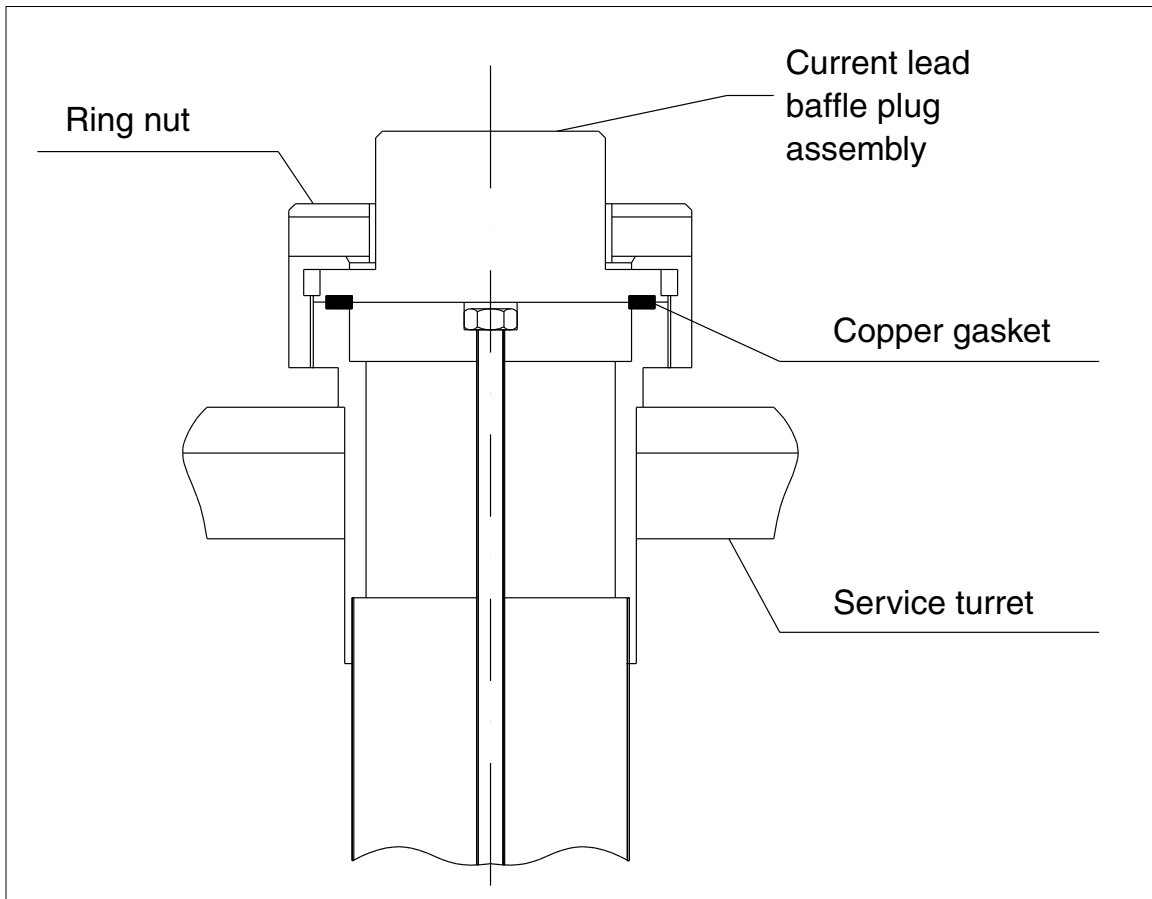
Failure to observe the following safety warning may result in cold burns.

- » Wear gloves for protection. Certain operations must be performed quickly. Ensure that the system is open for the minimum time possible.

- Prepare the baffle plug with a **new** copper gasket.
- Immediately close the depressurization valve after opening any turret openings. Lock the handle with the specially fitted bracket.

- Undo the ring nut, while holding the ECL in place. Pull the ECL out quickly to prevent air from entering the helium reservoir, which may form ice on the main magnet connector.
- Immediately insert the baffle plug, taking care to locate it properly on the copper gasket.
- Refit the ring nut and tighten it using the 'C' spanner provided without using excessive force.
- Warm up and remove the ring nut and seals from the ECL.

Fig. 149: Baffle for the current probe port



- Disconnect the ECL vent line from the 0.1 psi valve.
- Remove the 0.1 psi valve from the quench valve, and blank the open vent port.
 - » The system will develop some under-pressure at first and should normally start to repressurize to approx. 40 mbar (0.6 psi) (a different pressure is available at mobile magnets) after one day. Failure to do so may indicate a leak at the turret. If you suspect a leak on the ECL port, remove the ring nut and check the condition of the copper seal to ensure that it is located correctly and shows an even ring indent on both sides where the knife edges cut into it. If the knife edges are damaged, contact the hotline; if they are ok, carefully reseal the port using a new copper gasket.
- Repressurize system.

- Perform a leakage check (refer to (→ Magnet System Leakage Check / Page 181).



CAUTION

Low system pressure

Icing of neck tube.

- » Check accessible seals with a leak detector gun or suitable leak detection solution (provided with the MPS kit). There may be a leak in one or more of the turret seals. Check/correct the quench valve bursting disk/poppet seal.
- » Refer to (→ Energizing the EIS circuit for pressure build up / Page 176).

- Disconnect the current leads from the magnet (unlock and release the connections at the upper and lower magnet connectors).

NOTICE

The MPS cables should be connected to the MPS only when ramping the magnet.

- » Never connect the MPS cables during patient operation. All cables must be disconnected and removed from the examination room.

- Warm and dry the ECL and refit protective cap to connector. Store it along with the 0.1 psi valve, vent line and fittings into the transport box.

7.4.15.2 Packing

Fig. 150:



- Warm up and dry after use. Fit the plastic plug protection cap into the connector end.



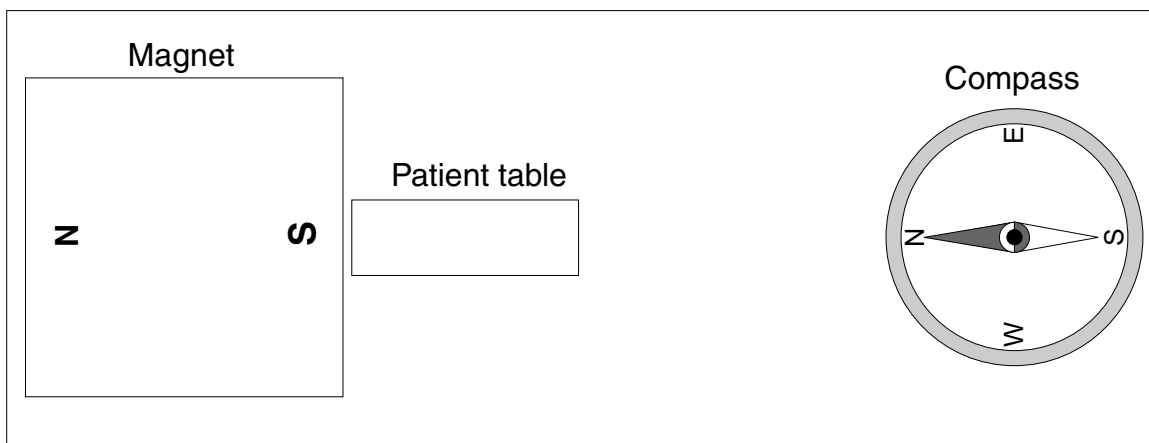
Check image orientation after ramping the magnet.

7.4.16 Checking the field direction with a magnetic compass

- Align the compass to geographic north outside the building, well away from all magnetizable structures:
 - Note which end of the needle points to **geographic north** (geographic north pole coincides with **magnetic** south pole). In the following, this G17 end of the needle will be referred to as the "N end of the needle".

- Enter the RF room with the compass, continuously observing the compass needle. Keep the compass away from magnetized elements, since they could demagnetize the compass needle.
- Proceed as described above. The "N end of the needle" must point to the magnet. This ensures that the south pole of the magnet points to the patient table.
- Again align the compass to the geographic north outside the building, well away from all magnetizable structures:
 - Ensure that the **same** end, the "N end of the needle" points to **geographic north** as before. If this is the case, you can be sure that the compass wasn't demagnetized and the measurement was successful.

Fig. 151: Magnetic field check



7.4.17 Power failure during magnet ramping

A power failure when ramping the magnet up or down will cause the magnet power supply to trip out of RUN mode, closing the output shorting contactor. The magnet current will then run very slowly down through the magnet leads and shorting contactor.

WARNING

Current in leads

If ignored, death or serious physical injury can occur.

- » The MPS must remain connected to the ECL during and following a power failure. The magnet leads and the shorting contactor are not indicated. Take no action until the power has returned and the following procedure is carried out.



After a power failure, there is NO indication on the power supply until the mains power is restored.

When power returns, or after a momentary power failure, the red LED "Warning - Current in leads" will be illuminated as a warning that current is still flowing in the magnet, magnet leads and shorting contactor circuit.

- Press S1 and confirm that mains power has returned (the green lamp in S1 should be illuminated).
- Check that the MPS has powered up in the OFF state by verifying that the LED in the **ON** button is extinguished.
- Check whether the red LED "Warning - Current in leads" is illuminated or not.
 - If it is not illuminated, it is safe to switch off the MPS and circuit breaker F5 and proceed with removal of the ECL.
- If it is illuminated, press the **ON** button on the MPS.
- Then select **Run to Zero** or **Run to Field** to continue ramping under MPS control.
- Wait until the green LED on the MPS illuminates in **Finish**, then press the **OFF** button.
- Check that the red LED "Warning - Current in leads" is not illuminated before switching off the MPS.
- It is now safe to remove the ECL and disconnect the magnet leads (follow the procedure (→ Removing the Enhanced current lead (ECL) / Page 148) .

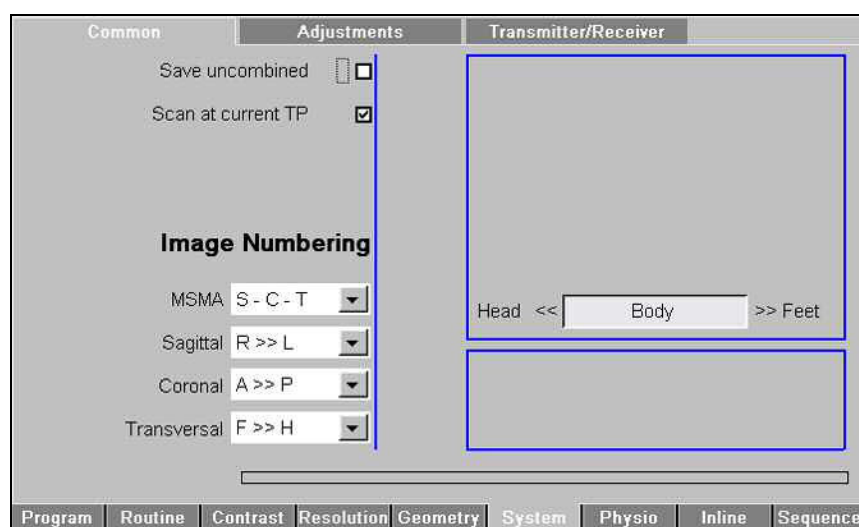
7.4.18 What to do if the frequency adjustment failed



This procedure is valid for the local coil and the Array shim device!

- Register a patient
- Open the protocol "se_15b130"
- Check if "Scan at current TP" is active

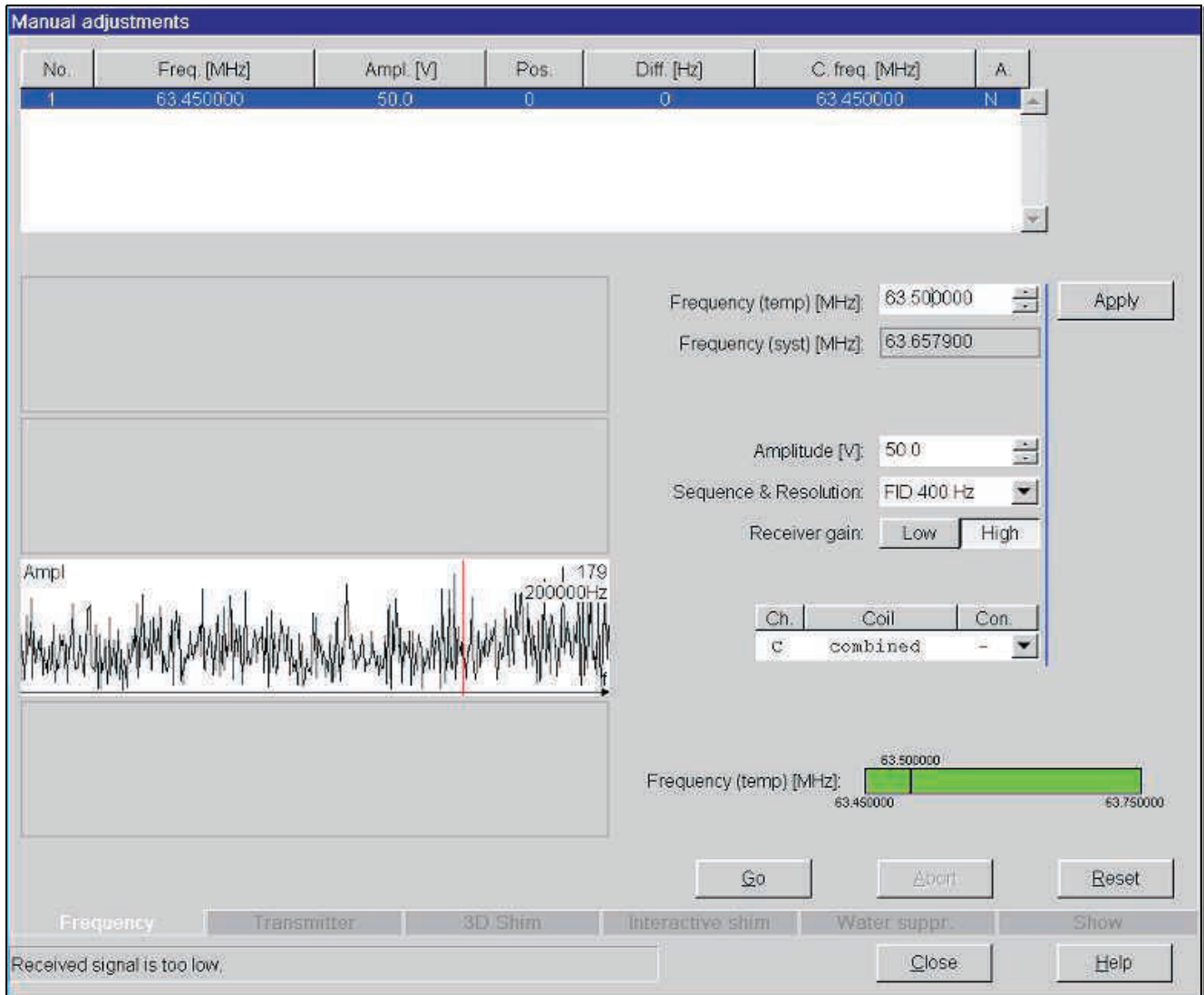
Fig. 152: Manual Adjustments



- Open manual adjustments

- Enter frequency (temp) manually, start from the lowest frequency 63.450 000 MHz

Fig. 153: Manual adjustments



- Select **FID 400Hz** of Sequence and Resolution



The FID 400 Hz uses a frequency bandwidth of +/- 100kHz.

- Click **Go** - button
- Continue in 50 kHz steps from lowest to the highest frequency.
 - » If necessary calculate the new magnet current and ramp the magnet to the required frequency

7.4.19 Check the mechanical adjustment of the patient table



Wear protective gloves for the mechanical adjustment of the patient table.

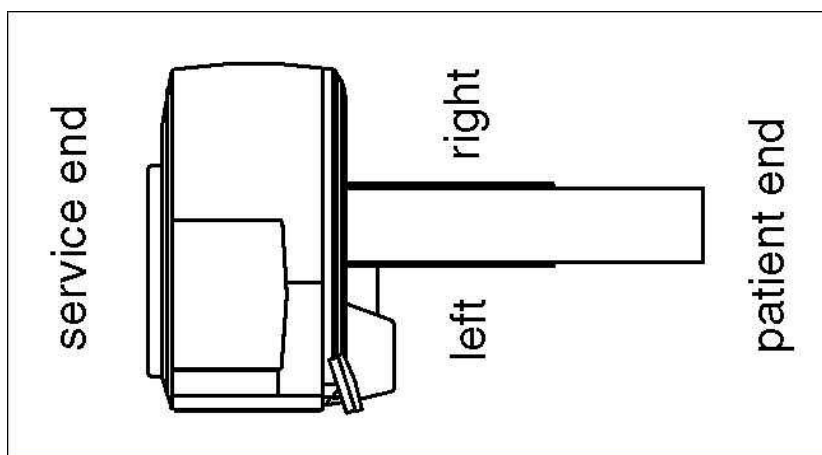


The mechanical adjustment of the patient table may be influenced by the magnetic field. For this reason, the final mechanical patient table adjustment has to be performed with the magnet at field.



The directions of the adjustments e.g. left, right, relate to the front view (patient end) of the magnet.

Fig. 154: View directions from patient end to the magnet



7.4.19.1 Check the free horizontal reference moving of the patient table



To prevent problems when referencing the patient table along the vertical axis, ensure that the table can find the reference point within the 3-8 mm clearance between the support frame and magnet cover without any obstruction.



Perform this test prior to starting up the patient table by referencing the highest, center, and lowest positions. Also, ensure that the tabletop does not collide with the covers.

- Visually examine the covers along the seam to the table. Ensure that they are LEVEL.
- Monitor vertical movement(s) at the narrowest position between the table and covers. There should be a clearance of **3-8 mm**, remain at this position.
- Power off the table, manually (F22) move the tabletop to the outer mechanical end stop.
- Power on the table and monitor the subsequent automatic tabletop movement for referencing: During the reference phase, the tabletop moves toward the magnet or

cover. This should result in the table reference position at the front of the support frame.

- **Check:** The front of the table should be flush with the front of the support frame.



The table should NOT touch the cover. Shortly before reaching the cover, the table should move away from the magnet.

- Reboot



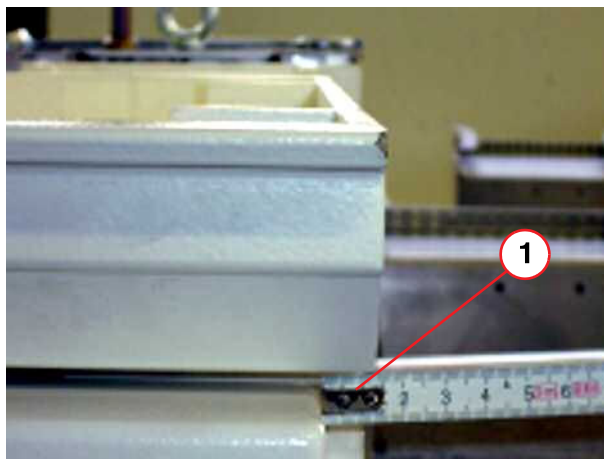
If the clearance between the table and the magnet cover is not between 3-8 mm, the patient table needs to be adjusted to the magnet.

Checking the reference switch S10 of the patient table



After each readjustment, circuit breaker F22 (ACC RF-Room) has to be switched off and then on again to ensure that the new reference position is transferred.

Fig. 155: Adjustment value for threaded pin (example)

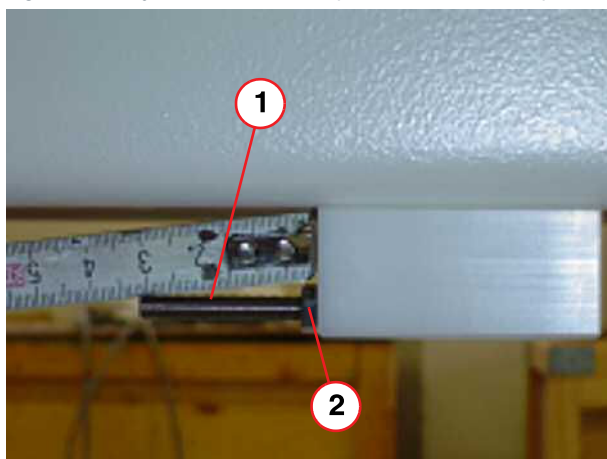


(1) Table excess length

Checking and setting the reference switch S10:

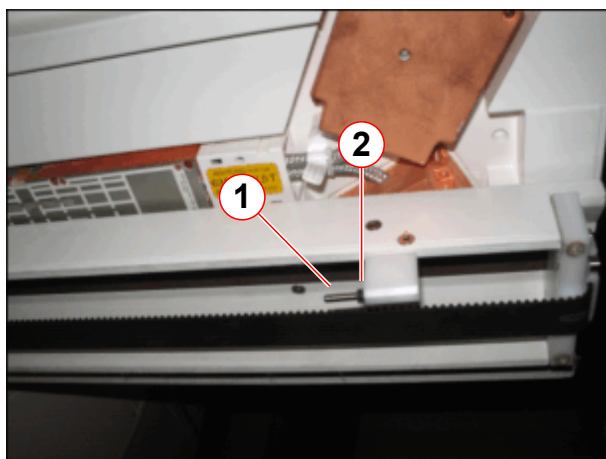
- Move the table into the home position by pressing the home position key.
- Measure the projection of the patient table.
- Move patient table all the way through the magnet.

Fig. 156: Adjustment value required for threaded pin



- (1) Threaded pin
- (2) Counter nut

Fig. 157: Reference point adjustment (partial drive unit)



- (1) Threaded pin
- (2) Locknut

- Turn the threaded pin (1) by the adjustment measured and tighten the counter nut (2).
- Move the patient table forward (home position).
- Switch off fuse F22 (RF room) in the ACC.
- Switch on fuse F22 (RF room) in the ACC (reference search).
- Check whether the patient table is aligned in vertical direction with the support arm. Also ensure that the patient table does not collide with the panels when moving it down (distance 3-8 mm).

Readjust, if necessary.

7.4.19.2 Checking the distance between patient table and cover

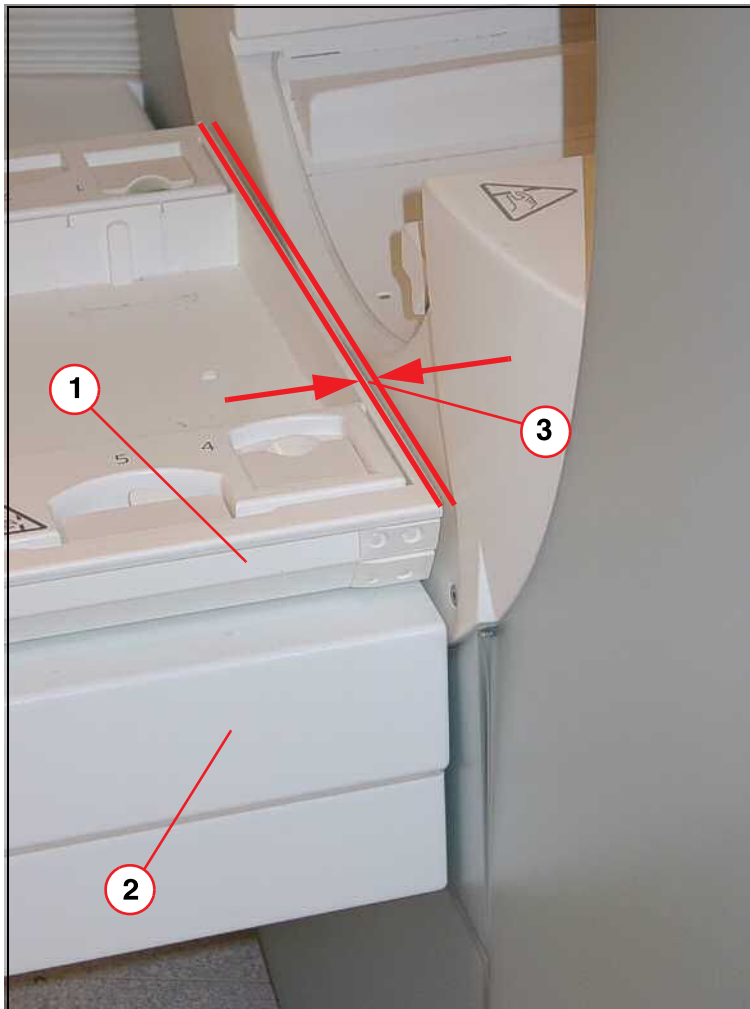


The translucent front cover may not be exactly vertical. Therefore, the translucent front cover and table may not be in parallel. In this case, do not try to compensate for the angle between the table and the cover using the X-radial adjustment procedure because this will also change the travel path of the tabletop inside the magnet bore.



The following pictures provide general information on how to perform the distance checks. The cover design differs slightly among Tim systems.

Fig. 158: Clearance, vertical gap



- (1) Tabletop
- (2) Support frame
- (3) Clearance < 8 mm between table and magnet cover

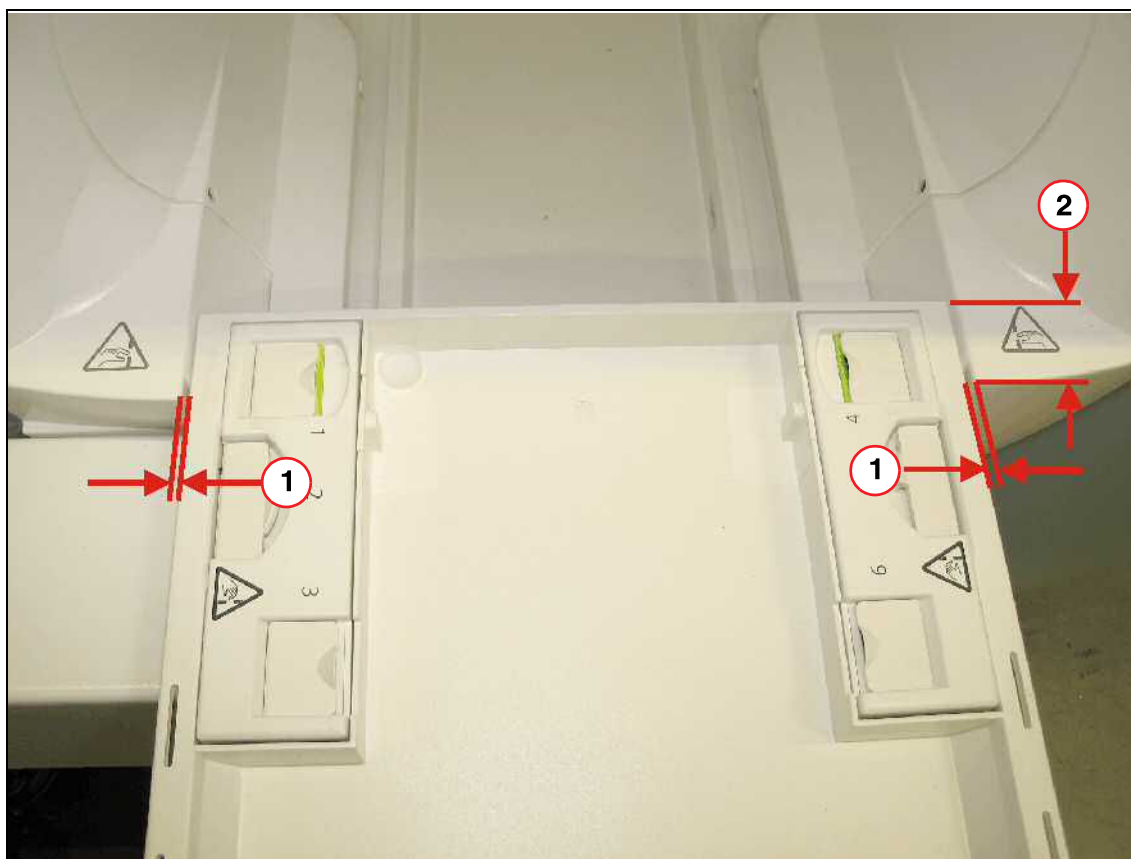
1. Move the patient table up to the highest position and then lower it about 80 mm.
2. Measure the distance of the vertical gap (→ 3/ Fig. 158 Page 159) between the tabletop and the magnet cover.
 - » The distance between the patient tabletop and the magnet cover (→ 3/ Fig. 158 Page 159) has to be within the range of **3-8 mm**.
 - » If the distance is not within the range, adjustment of the patient table is required.

The patient table should run in parallel to the Body Coil / covers with a sufficient gap across the entire patient board to avoid scratching.



There is always a clearance between the exchangeable tabletop and the table carrier, as well as between the table carrier and the slide way. For this reason, the exchangeable tabletop has to be pushed as described.

Fig. 159: Case 1: Checking the distance between patient table and cover (patient end)



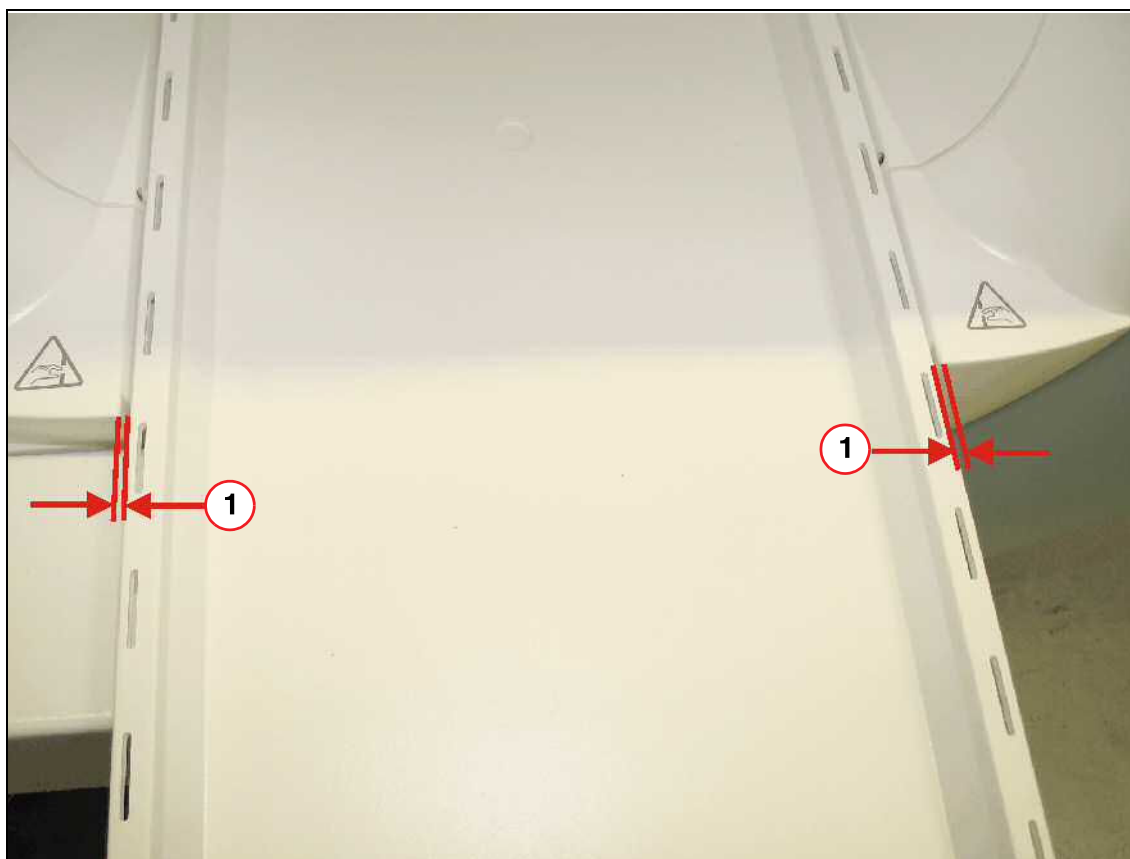
(1) Distance to be measured

(2) Approx. 50 mm

Case 1:

1. Drive the patient table approx. 50 mm (Symphony, A Tim System: approx. 400 mm) into the magnet.
2. Push the tabletop (exchangeable tabletop) to the **left** side and check the distance of < 8 mm between the patient table and cover on the **right** side, at the front of the magnet.
3. Repeat the procedure by pushing the tabletop to the **right** side and check the distance of < 8 mm on the **left** side, at the front of the magnet.
 - » The distance in both tests has to be within the range of < 8 mm.
 - » If the distance is not within the range of < 8 mm, adjustment of the patient table is required.

Fig. 160: Case 2: Checking the distance between patient table and cover / body coil (patient end)

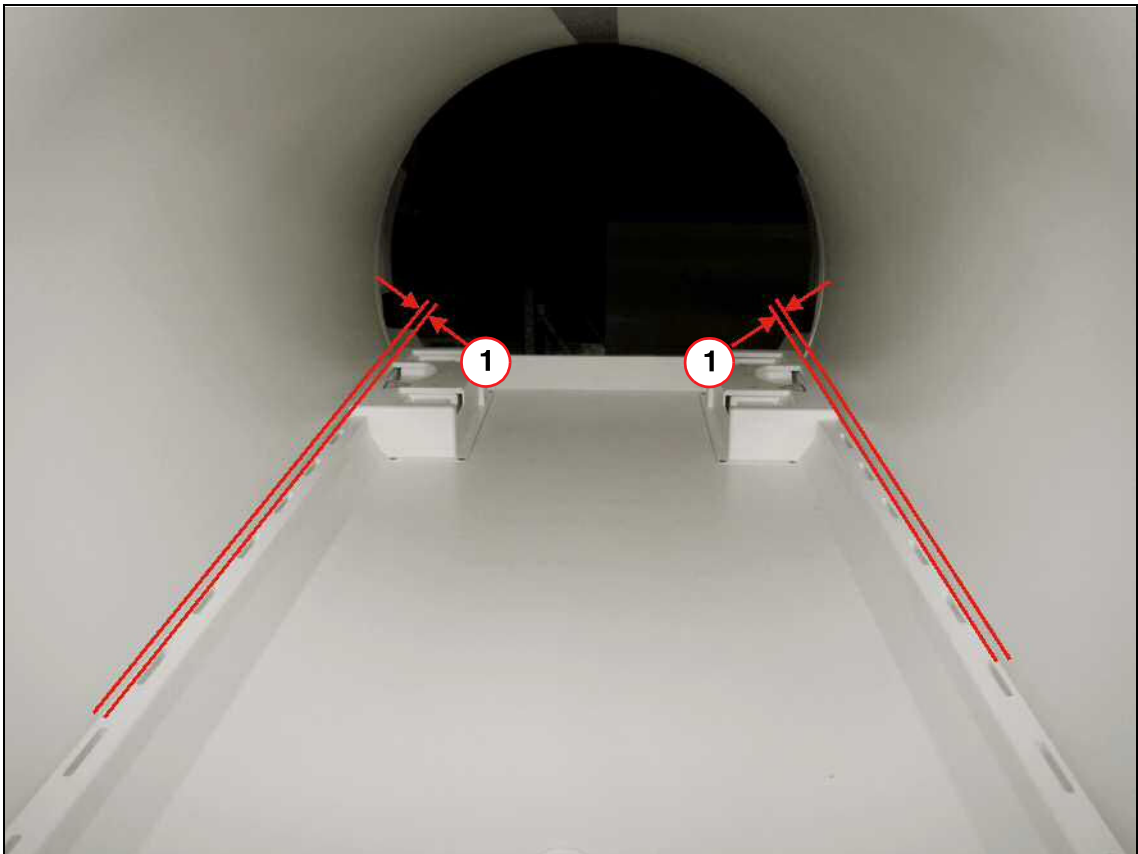


(1) Distance to be measured

Case 2:

1. Drive the patient table approx. 1200 mm into the magnet.
2. Push the tabletop (exchangeable tabletop) to the **left** side and check the distance of < 8 mm between the patient table and cover on the **right** side, at the front of the magnet.
3. Repeat the procedure by pushing the tabletop to the **right** side and check the distance of < 8 mm on the **left** side, at the front of the magnet.
 - » The distance in both tests has to be within the range of < 8 mm.
 - » If the distance is not within the range of < 8 mm, adjustment of the patient table is required.

Fig. 161: Case 3: Checking the distance between patient table and Body Coil / cover (service end)



(1) Distance to be measured

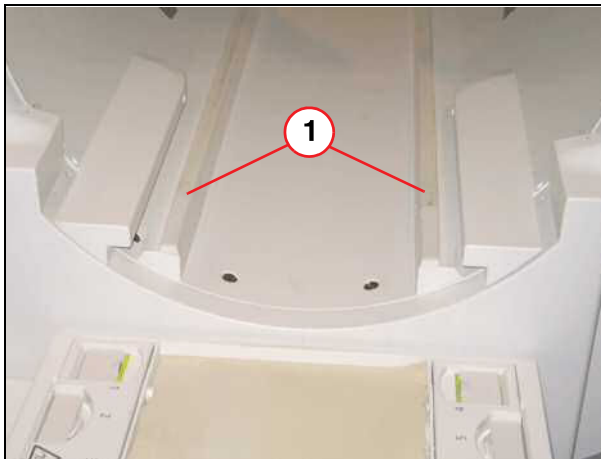
Case 3:

1. Drive the complete patient table into the magnet.
2. Check the distance of < 8 mm between the patient table and cover / body coil on both sides along the entire length of the tabletop.
 - » The distance has to be within the range of < 8 mm.
 - » If the distance is not within the range of < 8 mm, adjustment of the patient table is required.

7.4.19.3 Checking the travel path of the patient table

- Check the travel path of the patient tabletop on the slide rails while moving in or out. Check to see that the tabletop moves quietly and in parallel in both directions. It should be free of warping. Perform the following adjustments as described.

Fig. 162: Slide rails, patient end, example Espree



(1) Slide rails

The following chapters describe how to set the patient table in the various axes.



The final adjustment is noticeable only after all screws and nuts have been tightened again.



The travel path of the patient table on the slide rails has to be checked under load as well. For example: using a person or a weight.

7.4.19.4 Removing the covers for patient table adjustment

Remove the following covers:

- Service access covers
- Control panels
- Front ring segments
- Lifting column cover
- Lower front cover

Fig. 163: Patient table



(1) Support foot

- Loosen the support of the patient table.

7.4.19.5 Setting the patient table

The following chapters describe how to set the patient table in the various axes.



CAUTION

Consideration

Orange market screws and washers are extremely magnetic.

- » Do not remove/replace while magnet is ramped.
- » Do not exchange with non magnetic parts.

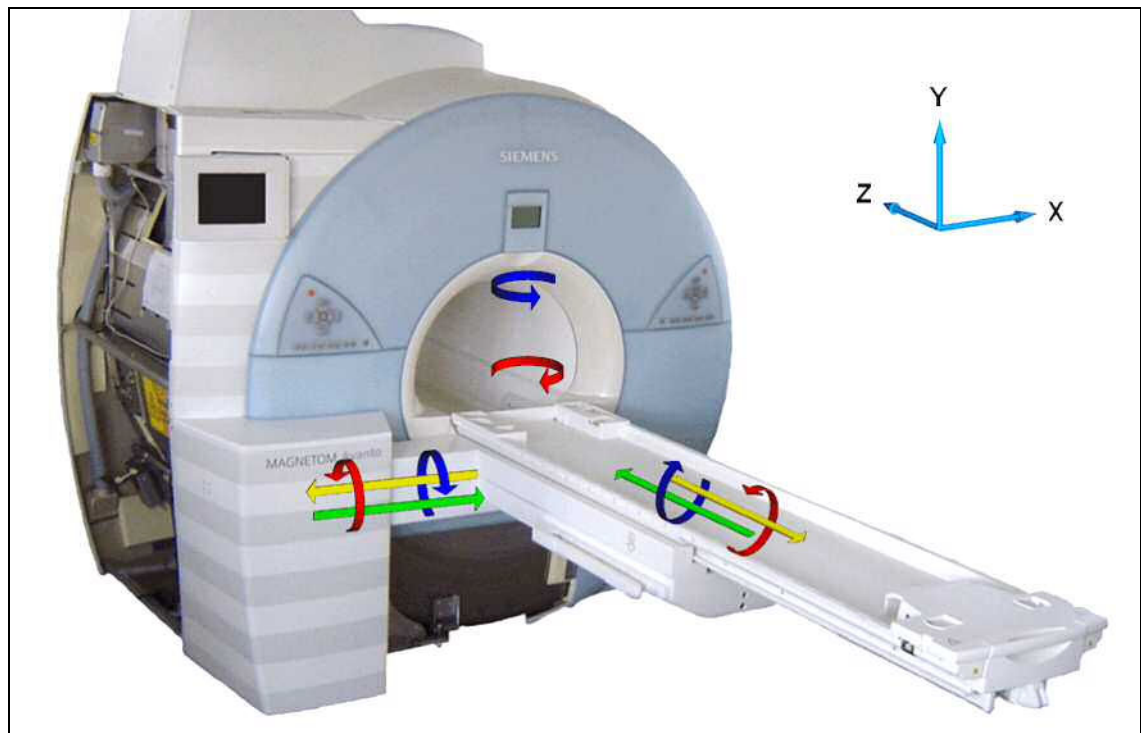


The final adjustment is noticeable only after all screws and counter nuts have been tightened again.



The travel path of the patient table on the slide rails has to be checked under load as well. For example: using one person or a weight.

Fig. 164: Patient table adjustment overview (example Avanto)



7.4.19.6 Setting the Z-axis

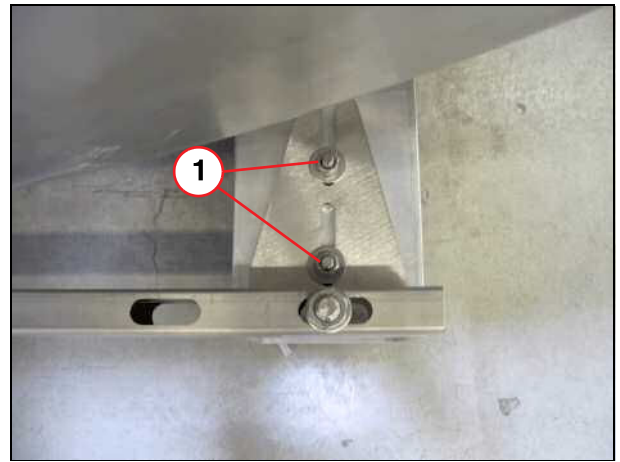
- Position the patient table at the most narrow location.
- Loosen the nuts (1).

Fig. 165: Patient end, left side



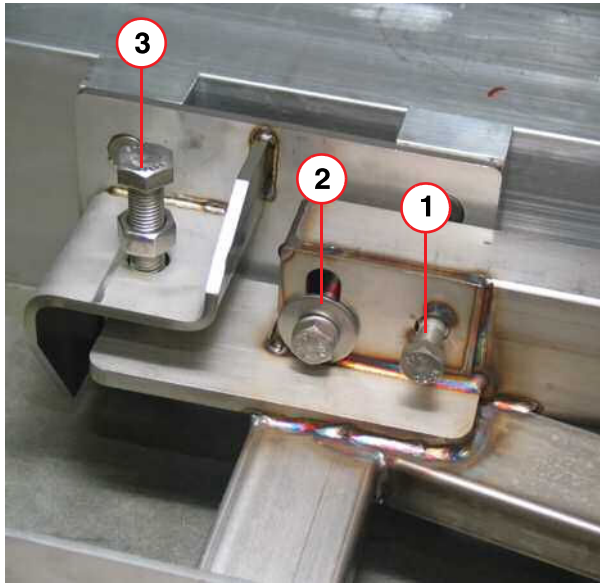
(1) Nuts

Fig. 166: Patient end, right side



(1) Nuts

Fig. 167: Back adjustment

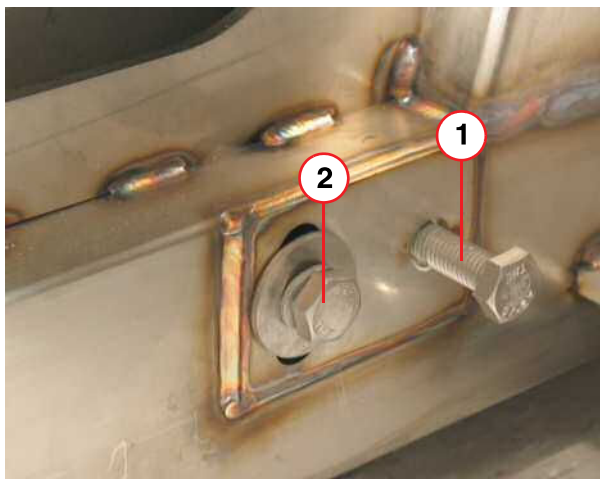


- (1) S1
- (2) S2 - pull
- (3) S3 - push

At the left side of the magnet:

- Loosen screw S2.

Fig. 168: Front adjustment



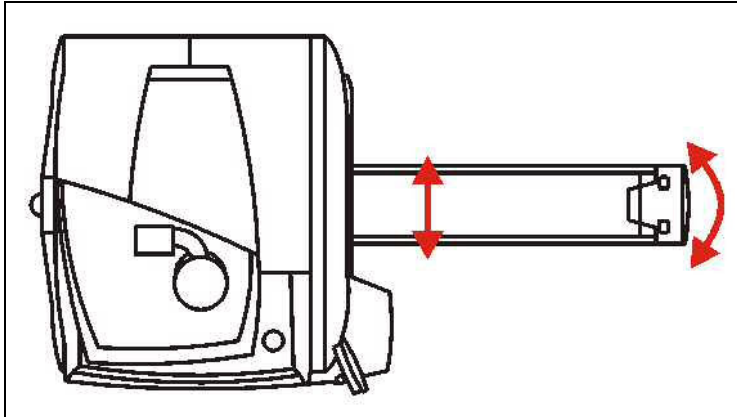
- (1) S4 - push
- (2) S5 - pull

- Loosen screw S4.

- Adjust the distance between the patient table and the magnet cover to **3-8 mm**.
In the Z-axis, the patient table must be moved manually (or with the auxiliary tool).
- Tighten the loosened screws and nuts.
- Tighten the patient table support again.
- Check whether the distance is **3-8 mm**.

7.4.19.7 Setting the X-axis and Y-axis radial

Fig. 169: Top view, X-axis and Z-axis radial



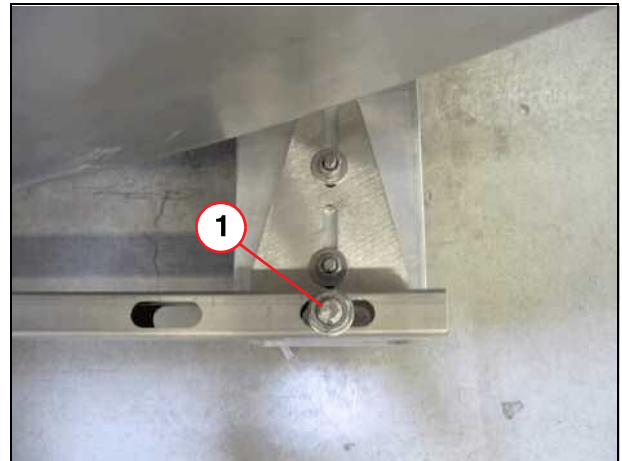
- Loosen the two nuts (1).

Fig. 170: Patient end, left side



(1) Nut

Fig. 171: Patient end, right side



(1) Nut

Adjustment:

By loosening the screws (→ 2/Fig. 172 Page 168) and (→ 2/Fig. 173 Page 168)) and tightening the screws (→ 1/Fig. 172 Page 168) and (→ 1/Fig. 173 Page 168)) you will enlarge the distance of the table frame to the box section.

- » The patient table moves to the left side on the X-axis as seen from the front.

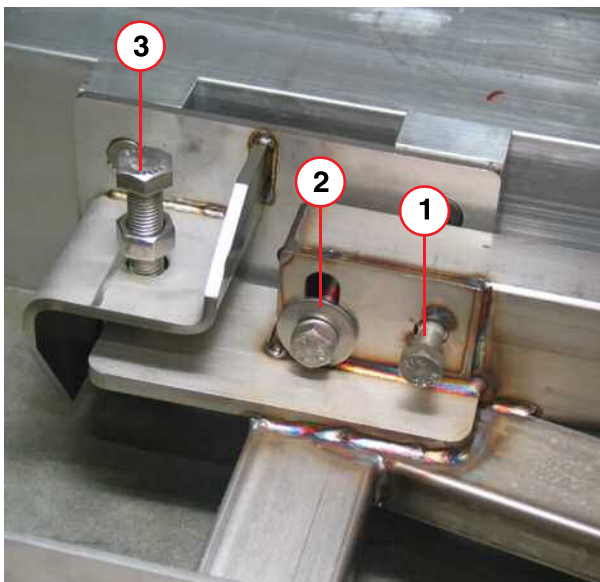
By loosening the screws (→ 1/Fig. 172 Page 168) and (→ 1/Fig. 173 Page 168)) and tightening the screws (→ 2/Fig. 172 Page 168) and (→ 2/Fig. 173 Page 168)) you will reduce the distance of the table frame to the box section.

- » The patient table moves to the right side on the X-axis as seen from the front.

If you change the position only on one side, either the left or right side, you will not change the distance of the table frame to the box section in the parallel plane.

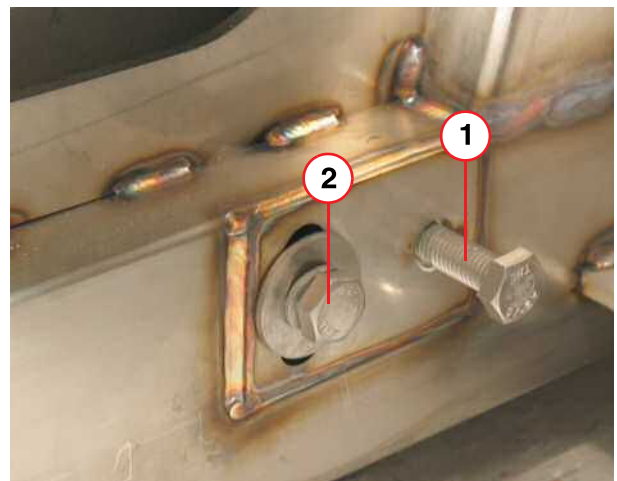
- » The patient table moves radially on the Y-axis and will not slide in parallel to the slide rails.

Fig. 172: Back adjustment



- (1) S1
- (2) S2 - pull
- (3) S3 - push

Fig. 173: Front adjustment



- (1) S4 - push
- (2) S5 - pull

- Tighten the support foot of the patient table and all screws and nuts.

7.4.19.8 Setting the X-axis radial

Fig. 174: Side view, X-axis radial

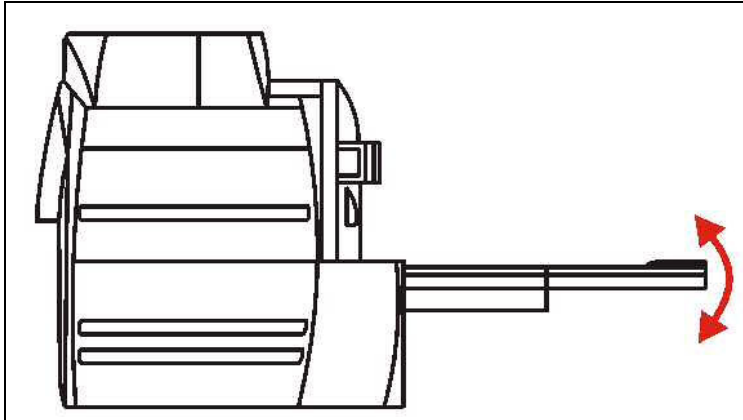
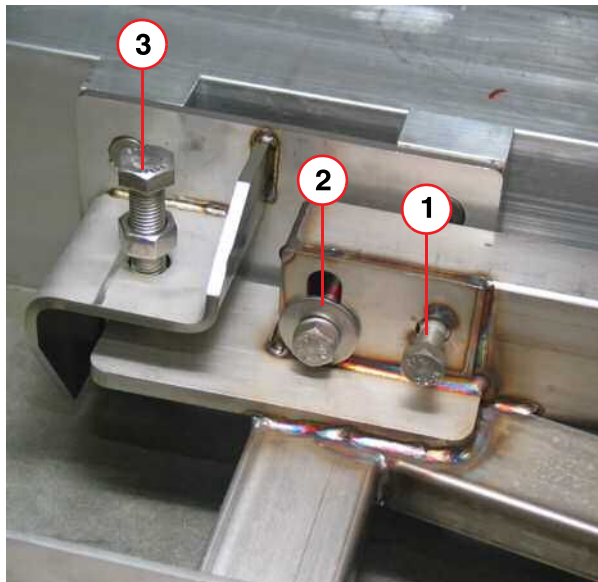


Fig. 175: Back adjustment



- (1) S1
- (2) S2 - pull
- (3) S3 - push

- Loosen the screw (2)

Turn the screw CW or CCW (3).

- » Patient table rotates radially around the X-axis.

- Tighten all screws and nuts.

7.4.19.9 Setting the Y-axis

Fig. 176: Side view, Y-axis

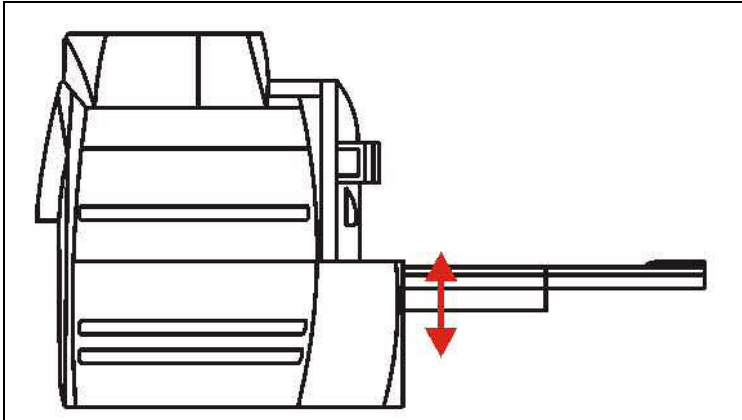
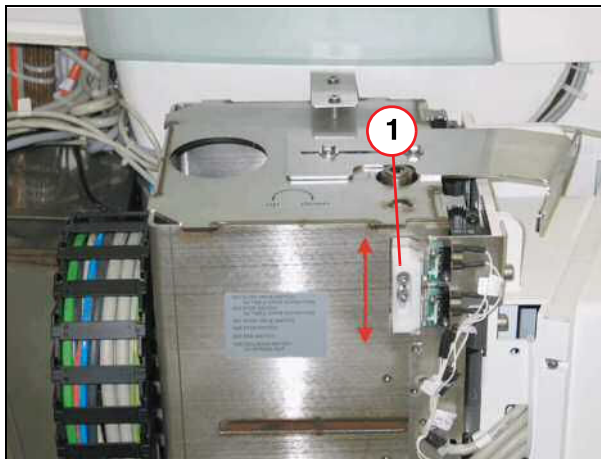


Fig. 177: End stop for micro-switch S20 and S21



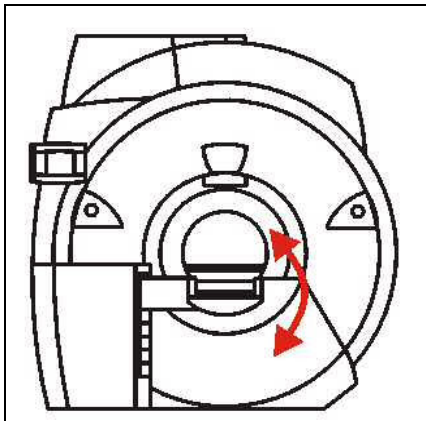
(1) Switch cam

1. Loosen the two screws at the end stop (1).

- By changing the (up-down) of the end stop (1) for the micro-switch, the end position (top) of the patient table is changed (distance between tabletop and the two slide rails).
- The end position (to) changes proportionally to the end stop.
- Tighten the screws.

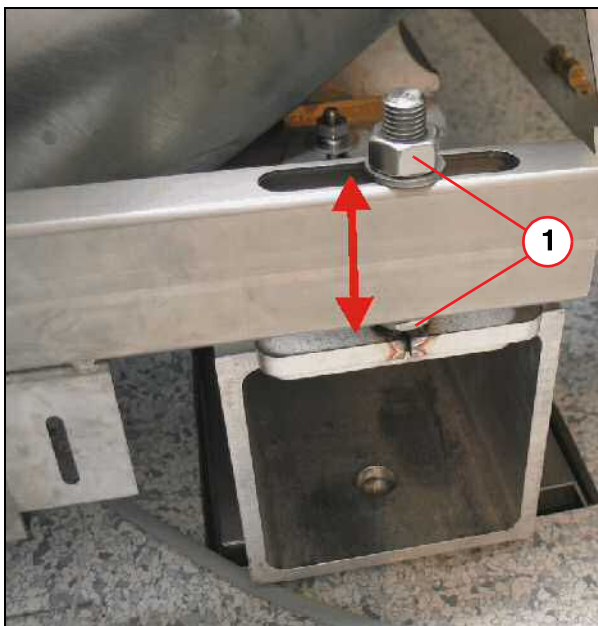
7.4.19.10 Setting the Z-axis radial

Fig. 178: Front view, Z-axis radial



- By moving the nuts (1), the patient table is changed in relationship to the slide rails (upward: the distance to the right slide rail is increased; downward: the distance to the right slide rail is reduced).

Fig. 179: Patient end, right bottom side



(1) Nuts

- Tighten the nuts.

7.4.19.11 Attach the covers

- Attach the removed covers (for adjusting the patient table) at the magnet again.

7.4.20 POST MAGNET QUENCH

If the magnet quenches, the magnet will lose all the helium contained in the vessel and will go sub-atmospheric. (negative pressure)

After approx. 30 minutes, the quench pressure will have subsided and the gas flow is reduced to normal levels. The turret, valves and venting should now be warmed up using a heat gun

The system must then be refilled with helium immediately. It is a requirement with the magnet that to fill the magnet with zero or 'warm carbon resistor temperatures' the syphon must be removed and a longer extension fitted to allow for 'bottom filling' for the post quench fill.

To enable the removal of the syphon, the system must be pressurized using the 'Pressurization heater' or the EIS circuit.

Once helium has started to collect (after 1st dewar) and the system is holding pressure, the ECL should be removed.

NOTICE

Negative pressure in the magnet helium vessel

- » **Do not remove the ECL if negative pressure is observed. Leave in system and continue helium fill.**

Continue helium filling until required ramp levels are reached. The longer syphon extension leg must be replaced again with the standard leg.



Higher static boil-off will occur if the long syphon extension leg is left fitted.

NOTICE

Leaks can occur if using the old copper seal.

- » **Fit new copper seal to the syphon and ECL baffle when refitting.**

After the helium fill, it is good practice to warm up the valves, turret and venting again.

If pressure is positive, remove the ECL. Warm up, dry off and inspect. Ensure baffle is fitted using a new copper seal.

If pressure is not seen to be building to normal operating levels, inspect the quench valve bursting disk and O ring. Replace if broken or damaged

Build system pressure using the shorting plug on the EIS circuit or the 'pressurization heater'.

When pressure has returned to normal, the turret must be leak checked using a suitable leak check sniffer or solution.

7.5 Shimming the magnet

7.5.1 Shim Procedure

The magnet is now prepared for shimming. Follow the procedure below only if the magnet is to be shimmed.



Reinstalled systems and local moved systems must be shimmed from bare (all existing iron must be removed from the shim trays).



If the system is left overnight between shimming iterations, the current probe should be removed and the ECL baffle refitted.



Before plotting the field, wait about 30 minutes to one hour to let the field stabilize.

The shim procedure is part of the service software ("SeSo"). All relevant work steps are to be performed "on-line" (on the screen). Online Help will explain the individual work steps.

- Start Numaris at the MRC.
- Start the "NUMARIS Service Software".
- Enter your password and select the menu item "**Shim**".
- Perform the shim process by following the instructions on the screen.



Use hard packers only. Rubber pads are no longer used.

Depending on the space between the shim plates and the cove, use hard packers to keep the shim plates from moving in the shim trays.



After shimming is done, continue with (→ Removing the Enhanced current lead (ECL) / Page 148).

7.6 Tim Whole Body Suite (optional)



If the system is provided with the option “Tim Whole Body Suite”, perform the following steps as described in (→ Whole Body / MR00-000.814.31), Technical Documents Volume 2, Options register.

7.7 Spike Check



Perform a QA Expert Spike Check with the Head Coil (without phantom) in center position and ± 600 mm off-center position, before assembling the cover (loose cables, screws, cover frame could cause spikes).

7.8 Appendix

7.8.1 Energizing the EIS circuit for pressure build up

After opening the turret port(s) for service, the system pressure can be restored more quickly by energizing the EIS circuit. It is essential that slight **positive pressure** exists in the system prior to performing any operations that require you to open any of the turret ports (e.g. insertion/removal of the ECL).

- Switch ON the EIS circuit using the remote display unit.
- The manual samples may be discontinued when sufficient pressure has been reached. The pressure should measure at least 0.3 psi (on the compound gauge this corresponds to the pointer being in the yellow sector and closer to the green than the red sector). When the system has reached sufficient pressure, it can then be depressurized again for insertion/removal of the ECL etc., if necessary.
- EIS OFF has to come up on the remote display.

7.8.2 Footnotes

The following footnotes apply to the entire chapter.

7.8.2.1 LHe Level

The initial installation ramps **MUST** be performed with helium levels of 70% or above.

When helium levels drop to 50% it is recommended that a LHe fill is performed.

The minimum operating LHe level is 30 %.

If helium levels drop to 30%, the magnet **MUST** be ramped down.



CAUTION

Low helium level

Magnet can quench

- » Ensure that the helium level is higher than 50% (initially ramping 70%) before ramping up the magnet.



After ramping the magnet, the reservoir for water of condensation should be drained

7.8.3 ECL contact checks

The ECL resistance levels for OR70 magnets is set to 130 mV.

If during the lead check 130 mV or less is not achieved, the power supply will trip out.

The following is a list of actions which can be taken to reduce the lead check voltage.

- » Connect a digital multimeter to the +/- socket terminals on the back of the MPS. This will give an accurate voltage drop during ramping, instead of reading the front DVM display.
- » Ensure that all gas flow is through the ECL.



The lead check assists in generating boil-off, so immediately after the first a second attempt will start with a higher boil-off and should give a lower voltage. Further attempts will not improve the readings.

- » Ensure that the lead is fully engaged with the magnet connector. There should be a gap of not more than 12 mm between the bottom of the ECL housing and the top of the ring nut on the port
- » The connection of the contact springs to the magnet connector can sometimes be improved by rotating the ECL. Two to three attempts may be required since this does not always improve the situation. No upward force may be applied.
- » If the contacts need to be replaced, refer to (→ ECL Inspection and Repair / Page 102).
- » Ensure that the volt sense lead contacts in the Anderson connector are mating well. If there is no contact for the volt sense leads, the MPS will measure the voltage across the complete cable set. If there is poor contact, the MPS will measure a voltage between the full cable voltage and the voltage across the top of the ECL. Moving the Anderson connectors may improve the contact, but cannot be attempted during the lead check. The resistance of the volt sense circuit should be in the range of 0.1 to 0.2 ohms.
- » Disconnect the Anderson connectors and ensure that the volt sense contacts are approximately in the center of their housing and have some free movement. If this is not the case, contact the MR-hotline.
- » Remove the ECL and completely dry the bottom contacts. Check that the contact springs are all present and still in good condition. If any damage can be seen to the springs, continue by checking for ice.
- » Check that the magnet connector is free of ice by using a torch and acrylic window, looking down the ECL port. Hold the acrylic window tightly over the port for several minutes until the gas clears. If a build up of ice is visible, contact the Up-time Service Center.

7.8.4 ECL Height Engagement Check

This check can be used to measure the "expected" height position of the ECL to ensure that full engagement to the magnet connector has been achieved. The procedure should be performed if the milli-volts are above 130 mV to ensure full ECL engagement.

- » Depressurize the system to almost zero pressure by opening the depressurization bypass valve. Check that zero pressure is indicated on the pressure gauge.
- » Push the ECL blanking cap (623-13260) onto the end of the ECL, .
- » Clean the ECL metal collar so that a new mark will be visible.
- Generate a positive pressure in the system by sorting the EIS switch heater circuit.
- Remove the ECL baffle assembly.



CAUTION

Very cold gas is being emitted from the entry port.

Failure to observe the following safety warning may result in cold burns.

- » **Wear gloves for protection. Certain operations must be done quickly. Ensure that the system is open for the minimum time possible.**

- Close the system depressurization bypass valve.
- Look down the ECL port and inspect for ice using the acrylic window and torch. If ice can be seen, refit the ECL baffle and contact the Uptime Service Center.
- Slide the ring nut, brass retaining ring and PTFE seal onto the ECL.
- Insert the ECL slowly until the end cap is touching the magnet connector. Tighten the ring nut by hand.
- Mark a line with a marker pen onto the sleeve of the ECL level with the top of the ring nut.
- Remove the ECL, ring nut and seals. Refitting the baffle assembly.
- Switch off the EIS circuit.
- Warm up the ECL and remove the blanking cap. Dry the ECL completely.
- Mark a 2nd line 43mm above the original mark onto the ECL sleeve.
- Depressurize the system and remove the baffle assembly.
- Refit the ring nut, brass retaining ring and PTFE seal.
- Slowly insert the ECL again. Tighten by hand plus a quarter turn with the "C" spanner. Warm the ring nut up if it becomes stiff or frozen.
- If full engagement has been achieved, the 2nd line will be level with the top of the ring nut

- If the line is level or below the ring nut, full engagement has been made. Continue with the "ramp procedure"



If there is a difference with the 2nd line being above the ring nut by more than 2/3 mm, loosen the ring nut and attempt to insert the ECL further down. This may require a side-to-side twisting action.

Fig. 180: ECL with end cap fitted to connector



7.8.5 Pressure Gauge Calibration

With high pressures produced in the magnet system during shipment, the system pressure gauge can become uncalibrated. A simple check and recalibration is necessary and can be made with the system depressurized to zero pressure.

7.8.5.1 Calibration Check and Repair

The following procedure must be performed quickly once the system has been depressurized to ensure that air ingress does not occur.

- Open the depressurization bypass valve.
- When the system pressure has equalized to atmospheric pressure, check that the needle of the pressure gauge is reading zero. (needle on the black line).
- If the gauge is reading zero, no further action is required.
- If the pressure gauge needle is not reading zero, an adjustment will be required.
 - Remove the plastic lens cover by inserting a small flat bladed screwdriver into the notch in the side of the lens and carefully lever off.
 - Insert a small 2mm flat screwdriver into the adjustment hole at the bottom of the gauge face marked +/- and adjust the needle until it correctly reads zero.
 - Refit the plastic lens cap.

7.8.5.2 Pressure Gauge Leak Check

The pressure gauge can become loose when turned by hand for better visibility of the needle. If the gauge is found to be loose, correct it by tightening and checking for leaks.

- Tighten the compression seal between the gauge's brass coupling and the stainless steel interface bush. Use a 24mm spanner on the stainless steel coupling and a 21mm

torque spanner on the brass coupling. A pressure of 30NM is required to lock the gauge.



If overtightened, internal damage can occur to the gauge.
Use specified torque setting.

- Check the NW 25 high load clamp and seal for tightness. If the gauge rotates on this seal, tighten the NW 25 clamp.
- With the system at normal operating pressure (needle in the green band), apply a leak check fluid around the NW25 seal. If a leak is found, depressurize the system and remove the gauge assembly. Replace the high load copper seal, refit the gauge and re-tighten.
- With the system at normal operating pressure (needle in the green band), apply a leak check fluid around the threads of the brass gauge coupling. If a leak is found, re-tighten the joint. If this does not stop the leak, replace the seal.



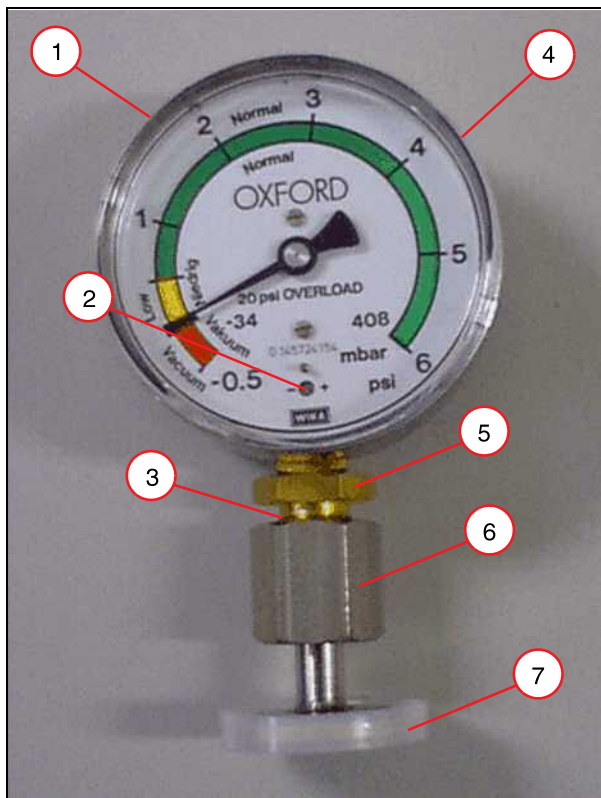
CAUTION

Air ingress.

Icing of magnet system causing blockage in neck tube.

- » Close the depressurization bypass valve.

Fig. 181: Calibrating the system pressure gauge



- (1) Notch for lens cover
- (2) Zero adjustment screw
- (3) Leak check with fluid around this joint
- (4) Gauge body
- (5) Brass coupling
- (6) Stainless steel interface
- (7) NW 25 flange

7.8.6 Magnet System Leakage Check

The magnet system must be checked for leakage after work has been performed during a service visit or a helium fill visit, if the system turret and venting have been subjected to extreme temperatures.

The magnet system must remain pressurized at all times. Before leaving an installed system site, a full leakage check must be performed and the system observed as returning back to its normal operating pressure.



WARNING

There must always be a positive pressure in the system.

Internal "icing" of the magnet may result, causing the magnet to quench and the helium vessel to rupture.

» Follow the leakage check procedures during all service operations.

If the system pressure remains at zero but there is no apparent leak, check the quench valve integrity. Refer to the (→ Magnet System / M6-015.840.17).

- For stationary systems, the normal operating pressure is 0.6 psi [40 mbar] at sea level.
- For mobile systems the normal operating pressure is 1.6psi [1.1 bar] at sea level.

NOTICE

The system operating pressure for mobile and stationary systems will vary with changes in local atmospheric pressures.

- » **Mobile systems taken to high altitudes (above 6000 feet) will see operating pressures up to 4.5 psi [300mbar].**

7.8.6.1 Procedure

An electronic helium leak detector should be used to search for leaks around all the joints on the service turret. (Follow manufacturer's instructions for correct use of the electronic detector.)

- The leak rate using an electronic device should not exceed 1×10^{-4} mbar-l/sec.



WARNING

Observe caution when using an electronic leak detector with the magnet at field. Accidents to personnel and equipment damage will occur.

- » **Check for magnetic parts before use.**

- If electronic equipment is not available, a suitable leak detection fluid must be used.
- Leak check fluid must be applied liberally by either brushing or squirting it onto the joints and seals. Leaks will be detected by observing bubbles where the fluid has been applied.
- The turret, quench valve and venting must be warmed to room temperature and dried before commencing the leak check. Remove condensation and water with paper towels.
- The specific areas to leak check are:
 - Clamps and seals on vent lines
 - ECL and syphon seals
 - System pressure gauge (see section (→ Pressure Gauge Leak Check / Page 179))
 - Turret top plate
 - Syphon end cap
 - The turret electrical connections
 - Flexible tubing

Where leaks are detected you should:

- Tighten or replace clamps, if damaged.
- Check that the correct type of seals are fitted.

- Check seal's faces for damage. Replace, if damaged.
- Tighten syphon end cap and the central retaining screw for the chain.
- Tighten Syphon / ECL seals. If leak persists, replace the copper seals.
- Check quench valve burst disk and "O" ring. Replace, if necessary.
- Check that bolted flanges are correctly torqued. (→ Tab. 9 Page 183)
- Seals covered in vacuum grease should be cleaned and inspected. Refit the seal or "O" ring using NO vacuum grease.

If problems persist with leaks to the turret or venting, contact your local Uptime Service Center.



Do not use any form of vacuum grease on O rings or seals.

It is recommended that these be wiped clean and checked for damage prior to fitting. Vacuum grease dries over time and can cause a seal to leak.



Replace used high load seals.

The high load seals (copper (Cu) or Heli-flex types) must be replaced with a new one each time the joint is broken.

7.8.6.2 Metric Torque Settings

Recommended torque settings for specific bolt sizes should be used. The overtightening of bolts can cause leaks and reduce the life of seals.

Use these torque settings for any bolts on the system turret, particularly when replacing the quench valve burst disc.

Tab. 9 Torque settings

Nominal bolt diameter (mm)	4	5	6	8	10	12
Torque setting (NM)	2.1	6.3	7.2	17.6	35	61

7.8.7 Pressure Raising Procedure for a Magnet NOT at Field

7.8.7.1 Introduction

This procedure explains how to maintain a positive gas pressure within the helium vessel and how the pressure can be restored quickly after a quench. It also explains how service operations can continue without the ingress of air, which will cause "icing".

If left unattended, the pressure within the helium vessel can go sub-atmospheric after a magnet quench. While every effort has been made within the venting design to ensure that all entries are leak tight, this does not legislate for broken burst disks or undetected leaks. It is important to have clear procedures for checking the condition of the magnet after a quench because any attempt at accessing the turret components while the pressure is below atmospheric will cause air to ingress into the neck where it will freeze.

This procedure is divided into sections depending on whether the quench was **Attended** (eg. quench during a ramp or fill) or **Unattended** (eg. quench overnight).

The system pressure gauge fitted to the venting is used as a reference throughout. The gauge is divided into green, yellow and red bands and during normal operation is interpreted as follows:

- **Green** - Normal operating pressure. All standard procedures may be followed.
- **Yellow** - Low pressure. A leak may be present and must be investigated and repaired before carrying out any operation on the magnet.
- **Black/Red** - The pressure in the helium vessel is sub-atmospheric and pressure must be restored before attempting to open any port on the turret.

 CAUTION

Very cold gas is being emitted from the entry port.

Failure to observe the following safety warning may result in cold burns.

- » Wear gloves for protection. Certain operations must be done quickly. Ensure that the system is open for the minimum time possible.

 CAUTION

Prevention of air ingress or "icing".

Air ingress will cause icing in the magnet. This could cause the magnet to quench or cause the system to overpressurize.

- » When opening the turret, especially the quench valve, use tissue paper to block the port, and keep the pressure builder turned on to promote LHE gas to boil off.

7.8.7.2 Materials and Tools required

- 20 to 24-volt DC, 2-amp switch mode power supply (provided with the MPS kit, refer to (→ Fig. 182 Page 185).
- Insulating gloves.
- Hot air gun and paper towel.

Fig. 182: Power supply kit for pressure build up at 0 field



Fig. 183: Connecting the power supply at the Magnet



7.8.7.3 Connecting the Pressure Builder

! WARNING

NEVER use the pressure builder psu with the magnet at field.
The magnet will quench.

» Read this procedure carefully to become familiar with its operation.

- Disconnect the lead from turret plug 'B' and connect the pressure builder adapter to plug B (refer to (→ Fig. 183 Page 185).
- Connect the power supply to the adapter and turn it on. This applies power to the magnet switch and EIS switch heaters and promotes helium gas boil-off.
- When normal operating pressure has been achieved (pressure gauge is in the green band and staying there), remove the pressure builder and refit the original cable to plug B.

7.8.7.4 Procedure for attended quenches (Liquid helium not available)

Refer to (→ Fig. 184 Page 190)

- Wait for the quench to subside. It will take around 20 minutes before the pressure on the gauge has dropped to around 0.5-psig [0.034 bar].
- When the pressure has dropped into the 'yellow' region of the pressure gauge, quickly disconnect the ECL 0.1 psi valve from the vent assembly and fit a blank to the exhaust port.

! CAUTION

Blanking the ECL vent port.

Under normal ramping procedures, blanking the ECL port may cause a quench.

» This is the only time the ECL should be blanked. If the system is sub-atmospheric, it is not possible to remove the ECL.

- Fit the pressure builder and switch it on.
- After a few minutes, the pressure should begin to rise. Switch off the power supply when the pressure is in the 'green' band.
- Thoroughly de-ice the turret, quench valve and vent valves with a heat gun.
- If the pressure does not increase (i.e. stays at or just above zero), the burst disk most likely ruptured. Refer to , "Replacement of the burst disk"
- While the pressure is at zero, check the pressure gauge calibration. Refer to (→ Pressure Gauge Calibration / Page 179).
- Maintain a flow of helium gas throughout this process by keeping the pressure builder connected and switched on.
- Continue monitoring the pressure gauge and re-activate the pressure builder if the pressure continues to drop. Do not allow the pressure to fall more than halfway into the 'yellow'.
- This process may take up to 3 hours until the helium vessel stops cryo-pumping and starts building pressure on its own.
- Remove the ECL and check inside the turret for ice. If ice is present, call the Uptime Service Center for further advice.
- Warm up and dry the ECL off as thoroughly as possible.
- Fit the ECL baffle set using a new copper seal.
- If the quench valve has not been removed for inspection, it should be done now.
- Repressurize the helium vessel into the 'green' band using the pressure builder.
- Check all turret connections for leaks using a leak check fluid or other approved leak detection methods. Repair any leaks found by re-tightening the connection. If the leak persists, replace the seal after the helium vessel has been depressurized. Repressurize the helium vessel and leak check again.
- If normal operating pressure is lower than before the quench, it is possible that the 1/3rd or 16 psia valve has developed a fault. Ensure a positive pressure is maintained by reconnecting the Supervisory and energizing the EIS. Call the Uptime Service Center for further advice.
- Disconnect the pressure builder and fit the normal lead to the turret connector 'B'.
- Refill with liquid helium as soon as possible.

7.8.7.5 Procedure for Attended Quenches (Liquid helium available)

Refer to (→ Fig. 185 Page 191)

- Wait for the quench to subside. It will take around 20 minutes before the pressure on the system gauge has dropped to around 0.5 psig [0.034 bar].
- If liquid helium is available, then refill immediately, transferring at least 250 liter or until there is a positive pressure in the system.
- When transfer is complete, de-pressurize the helium vessel, remove the ECL and check for ice. If ice is present, call the Uptime Service Center for further advice.
- Fit the ECL baffle set using a new copper seal.
- Switch on the pressure builder and perform the quench valve checks.
- While the pressure is at zero, check the pressure gauge calibration.

- Maintain a flow of helium gas throughout this process by keeping the pressure builder connected and switched on.
- Repressurize the helium vessel into the 'green' band using the pressure builder.
- Check all turret connections for leaks using a leak check fluid or other approved leak detection methods. Repair any leaks found by re-tightening the connection. If the leak persists, replace the seal after the helium vessel has been depressurized.
- Repressurize the helium vessel and leak check again.
- If normal operating pressure is lower than before the quench, it is possible that the 1/3rd or 16 psia valve has developed a fault. Ensure a positive pressure is maintained by reconnecting the Supervisory and energizing the EIS. Call the Uptime Service Center for further advice.
- Fill the helium vessel to at least 70% with liquid helium.

7.8.7.6 Procedure for Unattended Quenches

This procedure is divided into sections depending on pressure gauge indications.

7.8.7.7 Pressure gauge showing green

Refer to (→ Fig. 186 Page 192)

- Check all turret connections for leaks using a leak check fluid or other approved leak detection procedures. Repair any leaks found by re-tightening the connection.
- If the leak persists, replace the seal after the helium vessel has been depressurized.
- Depressurize the magnet and remove the ECL baffle.
- Inspect for ice within the turret. If ice is present, call the Uptime Service Center for further advice.
- Refit the ECL baffle set using a new copper seal.
- Replace any faulty seals identified during the leak test.
- Check the pressure gauge calibration.
- Maintain a flow of helium gas throughout this process by using the pressure builder or energizing the EIS.
- Repressurize the helium vessel into the 'green' band using the pressure builder or EIS.
- If the pressure gauge is now indicating a low pressure or is in the 'yellow', switch on the pressure builder and perform the quench valve checks.
- Maintain a flow of helium gas throughout this process by keeping the pressure builder connected and switched on.
- Repressurize the helium vessel into the 'green' band using the pressure builder.
- If normal operating pressure is lower than before the quench, it is possible that the 1/3rd or 16 psia valve has developed a fault.
- Ensure a positive pressure is maintained by reconnecting the Supervisory and energizing the EIS. Call the Uptime Service Center for further advice.
- Refill with helium as soon as it is available.

7.8.7.8 Pressure Gauge showing yellow

Refer to (→ Fig. 187 Page 193)

- Check all turret connections for leaks using a leak check fluid or other approved leak detection procedures. Repair any leaks found by re-tightening the connection.
- If the leak persists, replace the seal after the helium vessel has been depressurized.
- Depressurize the magnet and remove the ECL baffle.
- Inspect for ice within the turret. If ice is present, call the Uptime Service Center for further advice.
- Refit the ECL baffle set using a new copper seal.
- Perform the quench valve checks.
- While the pressure is at zero, check the pressure gauge calibration, if not already done. Maintain a flow of helium gas throughout this process by keeping the pressure builder connected and switched on.
- Repressurize the helium vessel into the 'green' band using the pressure builder.
- If normal operating pressure is lower than before the quench, it is possible that the 1/3rd or 16 psia valve has developed a fault. Ensure a positive pressure is maintained by reconnecting the Supervisory and energizing the EIS. Call the Uptime Service Center for further advice.
- Refill with liquid helium as soon as it is available.

7.8.7.9 Pressure Gauge showing Zero

Refer to (→ Fig. 188 Page 194)

- If the pressure gauge shows zero, either the burst disk has ruptured, there is some other gross leak, or coincidentally, the pressure has reached zero in its cycle to below atmospheric and recovery.
- Fit the pressure builder and switch it on.
- Inspect the turret for any open ports or loose connections and rectify where necessary.
- Perform the quench valve checks.
- While the pressure is at zero, check the pressure gauge calibration.
- Maintain a flow of helium gas throughout this process by keeping the pressure builder connected and switched on.
- Remove the ECL baffle and inspect the neck tube for ice.
- Inspect for ice within the turret. If ice is present, call the Uptime Service Center for further advice.
- Refit the ECL baffle set using a new copper seal.
- When ice has been removed, connect and turn on the pressure builder.
- Repressurize the helium vessel into the 'green' band using the pressure builder.
- Check all turret connections for leaks using a leak check fluid or other approved leak detection procedures.
- Repair any leaks found by re-tightening the connection. If the leak persists, replace the seal after the helium vessel has been depressurized.

- Repressurize the helium vessel and leak check again.
- If normal operating pressure is lower than before the quench, it is possible that the 1/3rd or 16 psia valve has developed a fault. Ensure a positive pressure is maintained by reconnecting the Supervisory and energizing the EIS. Call the Uptime Service Center for further advice.
- Refill with liquid helium as soon as it is available.

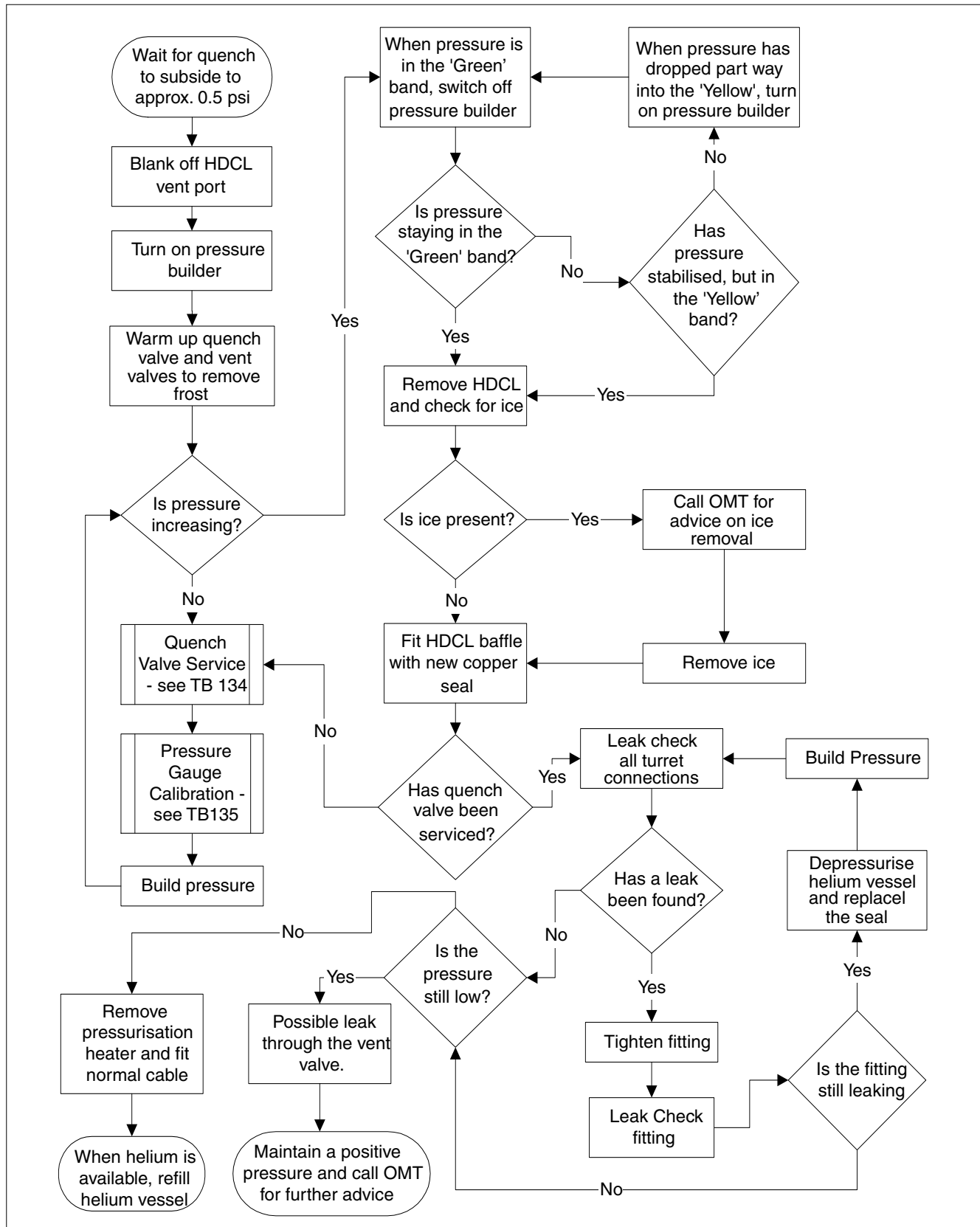
7.8.7.10 Pressure Gauge showing Red

Refer to (→ Fig. 189 Page 195)

- Fit the pressure builder and turn it on. After a few minutes, the pressure will begin to rise.
- When the pressure has risen to just above zero, perform quench valve checks.
- While the pressure is at zero, check the pressure gauge calibration.
- Maintain a flow of helium gas throughout this process by keeping the pressure builder connected and switched on.
- Remove the ECL baffle.
- Inspect for ice within the turret. If ice is present, call the Uptime Service Center for further advice.
- Refit the ECL baffle set using a new copper seal.
- When ice has been removed, repressurize the helium vessel into the 'green' band using the pressure builder and check the turret for leaks using a leak check fluid or other approved leak detection procedures.
- Repair any leaks found by re-tightening the connection. If the leak persists, replace the seal after the helium vessel has been depressurized.
- Repressurize the helium vessel and leak check again.
- If normal operating pressure is lower than before the quench, it is possible that the 1/3rd or 16 psia valve has developed a fault. Ensure a positive pressure is maintained by reconnecting the Supervisory and energizing the EIS. Call the Uptime Service Center for further advice.
- Refill with liquid helium as soon as it is available.

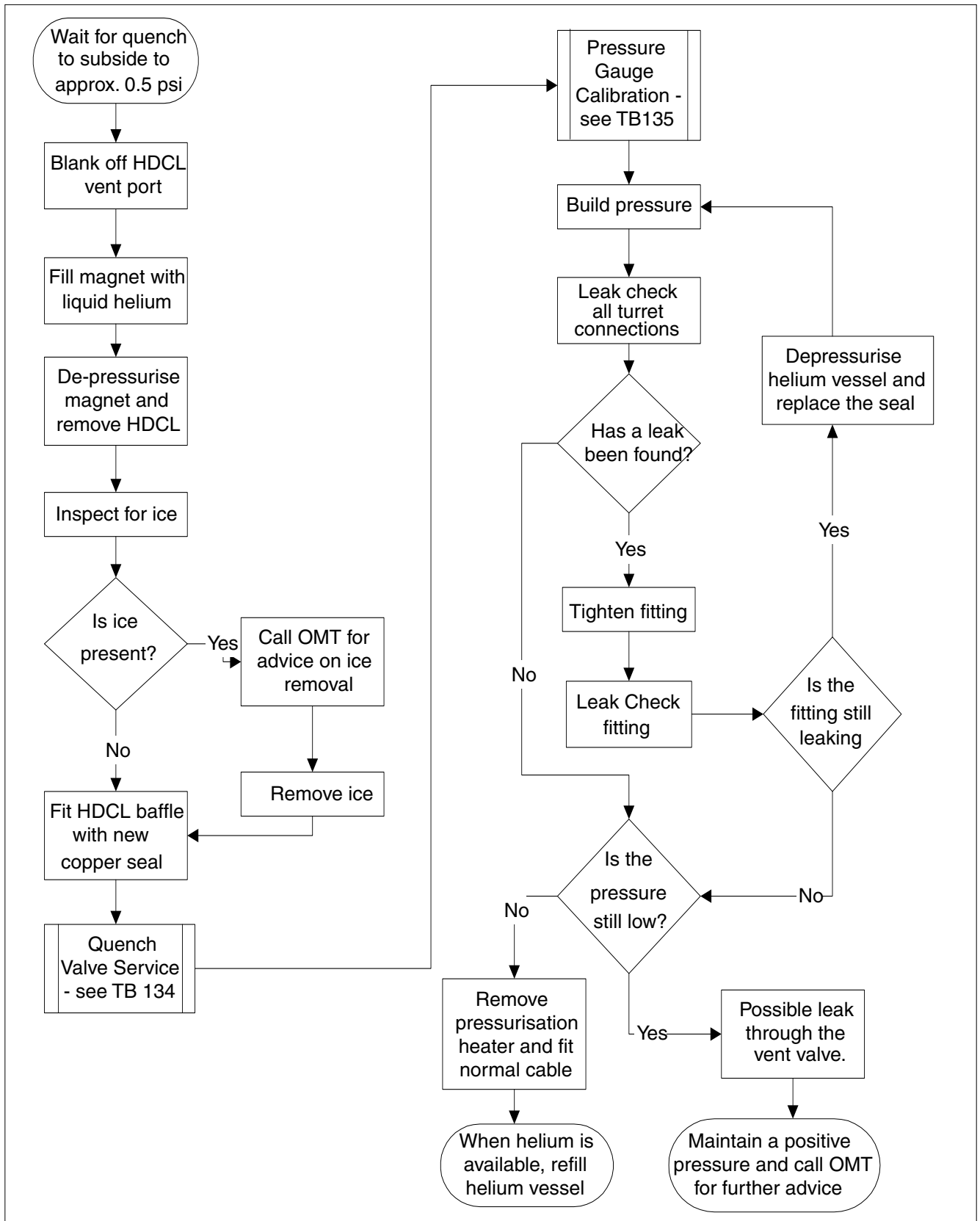
7.8.7.11 Flowchart for raising Pressure after Attended Quench

Fig. 184: No liquid helium available



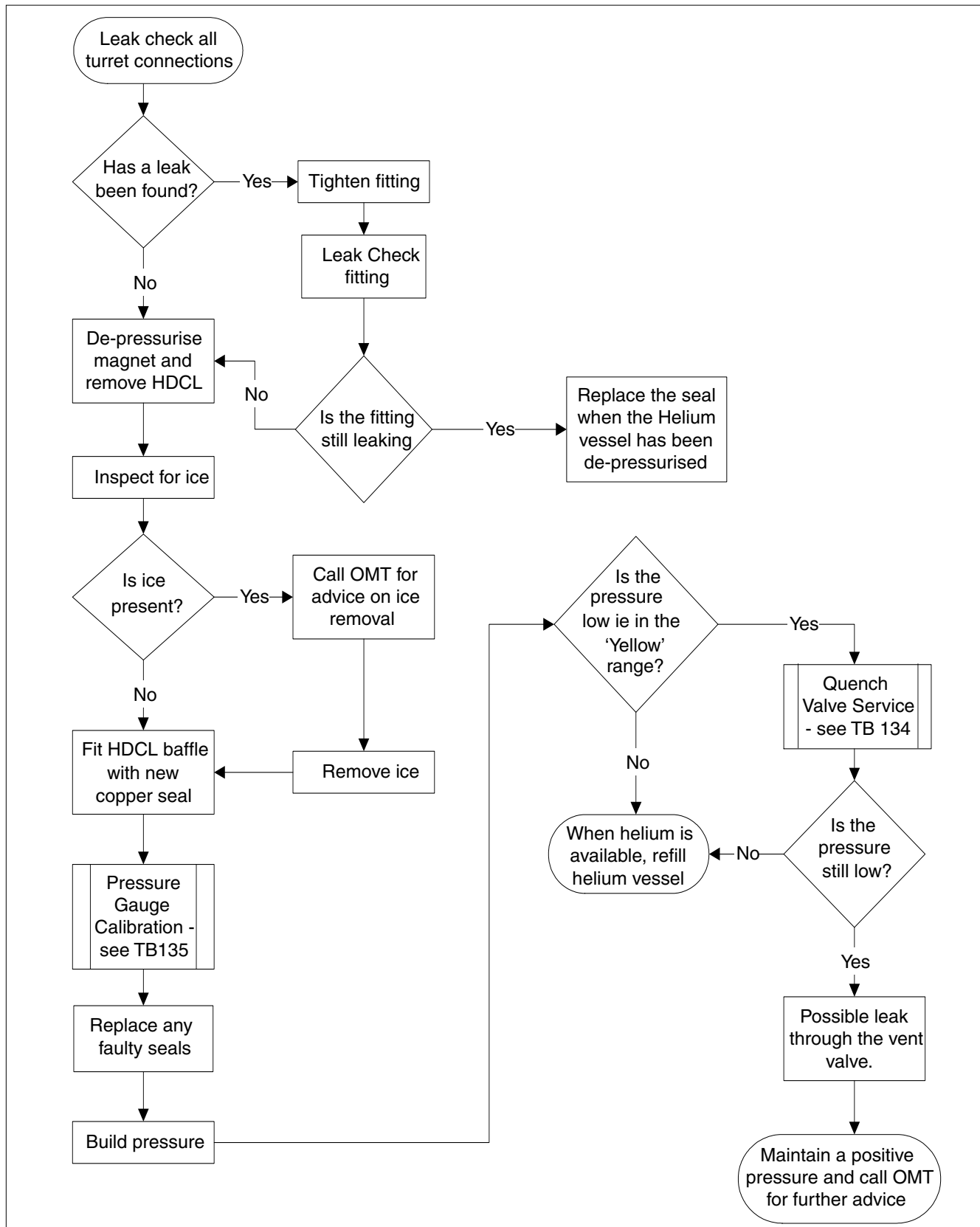
7.8.7.12 Flowchart for raising Pressure after Attended Quench

Fig. 185: Liquid helium available



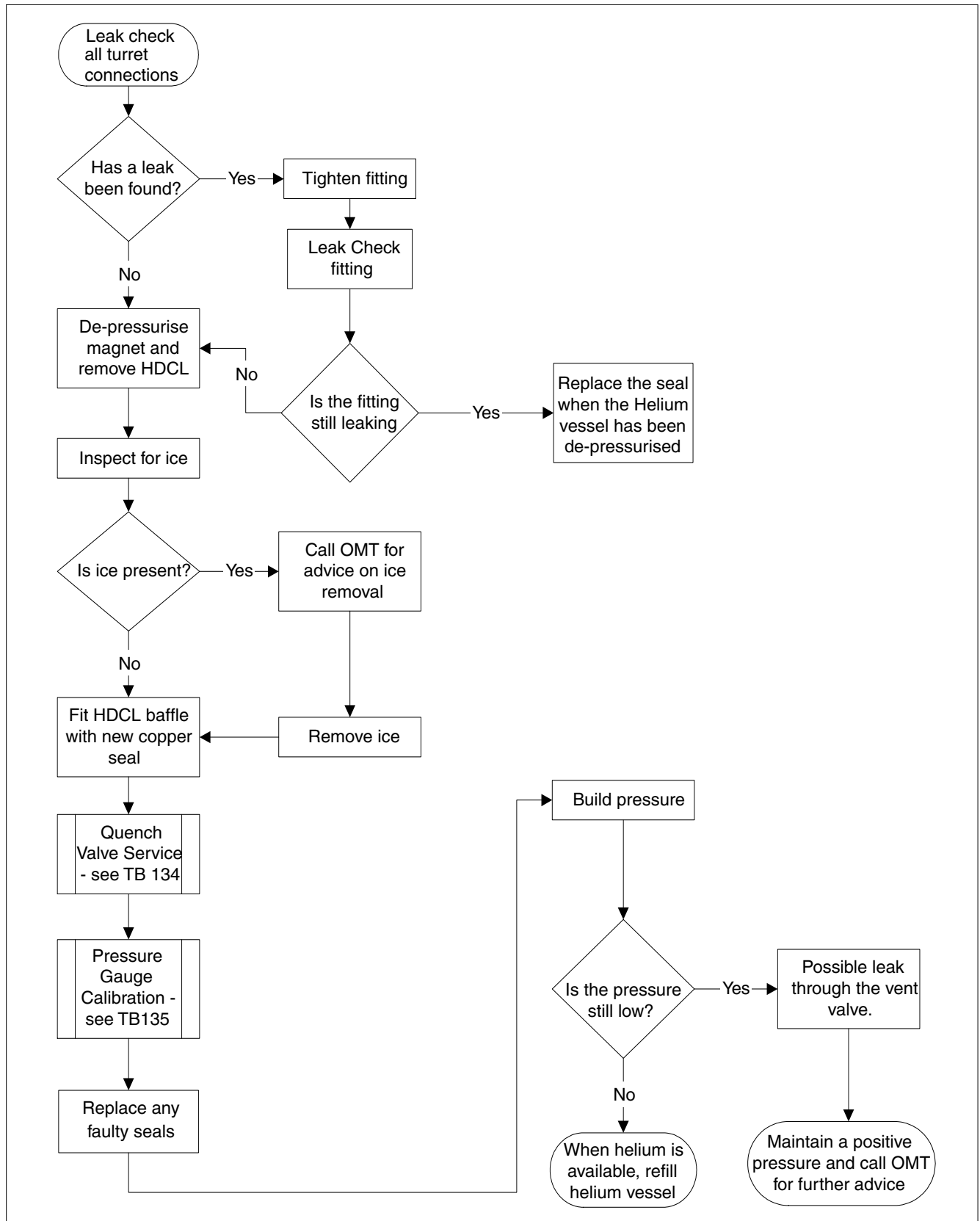
7.8.7.13 Flowchart for raising Pressure after Unattended Quench

Fig. 186: Pressure in the green band



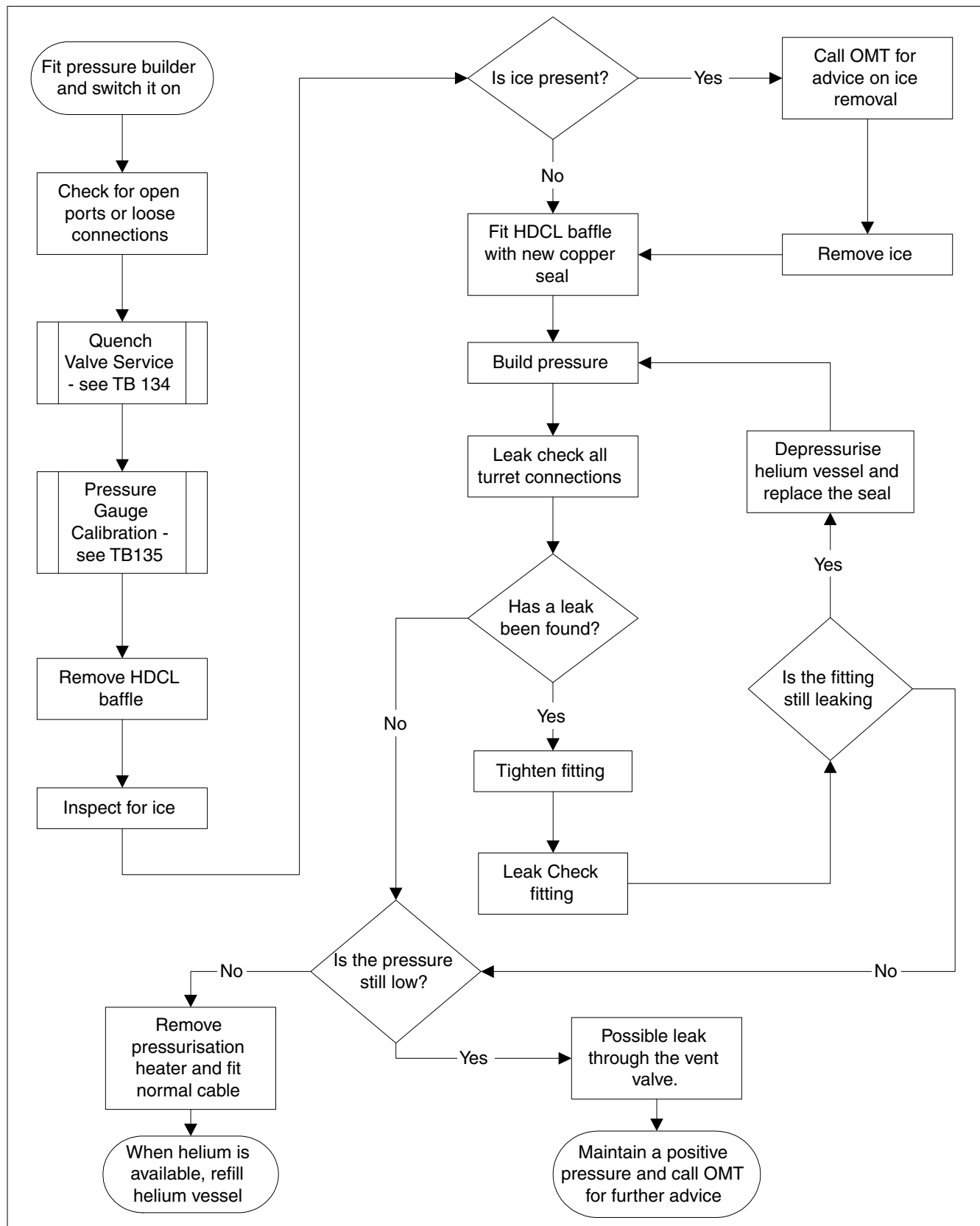
7.8.7.14 Flowchart for raising Pressure after Unattended Quench

Fig. 187: Pressure in the yellow band



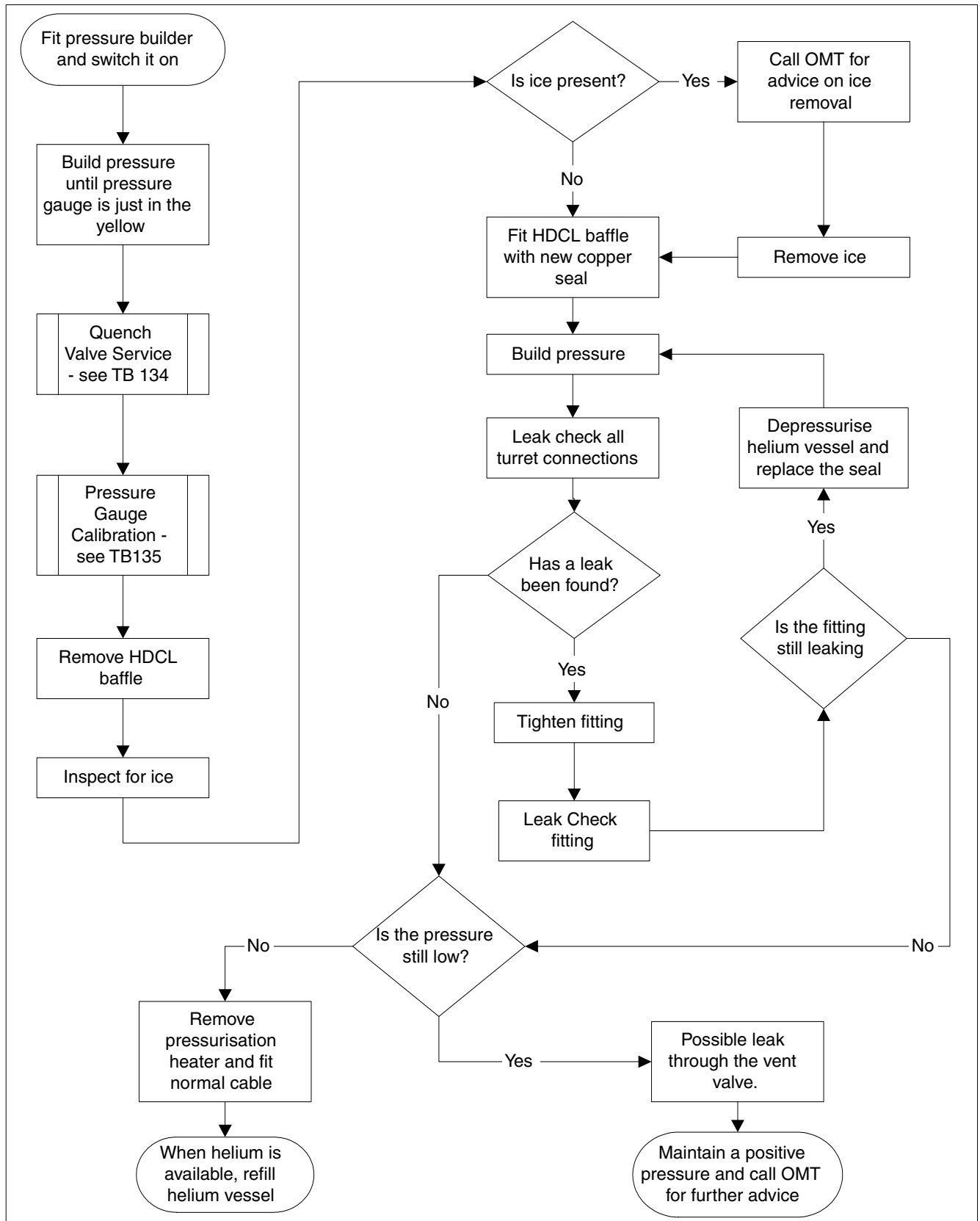
7.8.7.15 Flowchart for raising Pressure after Unattended Quench

Fig. 188: Pressure gauge reading ZERO pressure



7.8.7.16 Flowchart for raising Pressure after Unattended Quench

Fig. 189: Pressure gauge reading in the red band



8.1 General



Valid for Germany and certain other EU countries.

If the monitor of an MR system is used for medical reports, then a special acceptance test is required in accordance with DIN V6868-57. The monitor has to be labeled as such by the installation personnel.

In accordance with the documentation listed below:

1. Acceptance testing for image display devices (→ Acceptance testing for image display devices / MR00-000.820.01).
2. Acceptance test protocol for image display devices. (→ Acceptance test protocol for image display devices / MR00-000.820.02)



The monitor only needs to be checked (and adjusted as necessary) if at least one of the parts listed below is replaced.

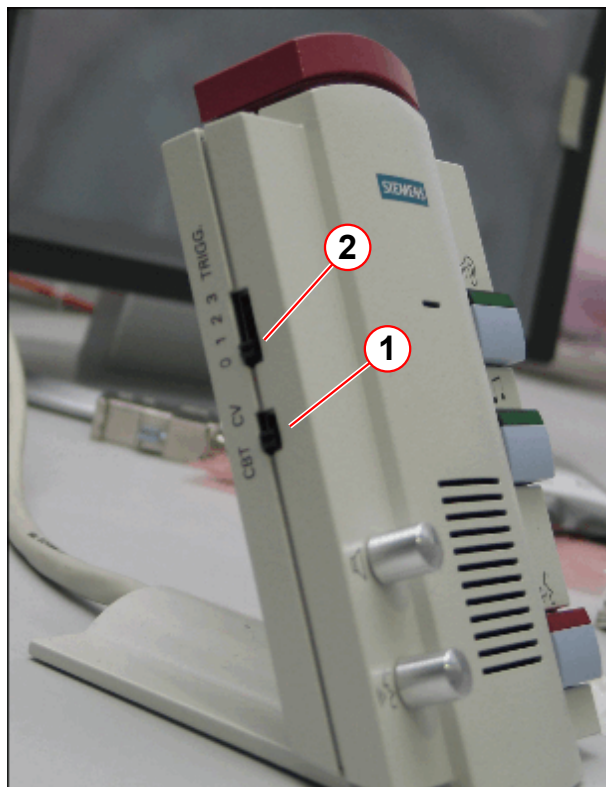
- Monitor
- Graphic board
- PC

Refer to (→ Host / Imager / M6-015.841.12)"Replacement of Parts"

9.1 Start-up

- Switch on and boot the system.
- Switch the operating unit (p.n. 4757261) to position "CBT". This switch is located on the left side of the operating unit.

Fig. 190: Intercom



- (1) CBT
- (2) TRIGG

- Close the patient application (Patient/ Close Patient) in case patient is active.
- Press <Ctrl> and <ESC> and select Programs/Accessories/Advanced User and enter the password: meduser1. Click <OK> to exit.
- Press <Ctrl> and <ESC> and select Programs/ Accessories/ Multimedia/ Volume Control. Set all six regulators in the volume control window to maximum (upper position), as shown in (→ Fig. 191 Page 198). Deactivate the four controls (CD Audio Bal., Synt., AuxB, ILS) on the right side of the window by clicking mute. "Mute" does not have to be activated for the two controls on the left side (Vol. Contr., Wave Output). In case the volume control menu has a different design than the figure below, set the "Volume Control" and wave output to maximum, and set all other controls to "Mute". Select Options/ Properties in the same window and click <Recording>. Press <OK> and set the regulators for the "Recording Control" as shown in (→ Fig. 192 Page 198). Set the recording control volume to approx. 25% and the volume for the other four regulators to approx. 50%. Click <Select> for the microphone. In case the recording control menu has a different design than the figure below, set the volume to approx. 25% and all others to 50%. To select the microphone, click <Select> with all others deselected, as shown in the example below. Refer to online help, if one of the controls for the device is missing. Close the window via Options/Exit.

Fig. 191: Volume Control

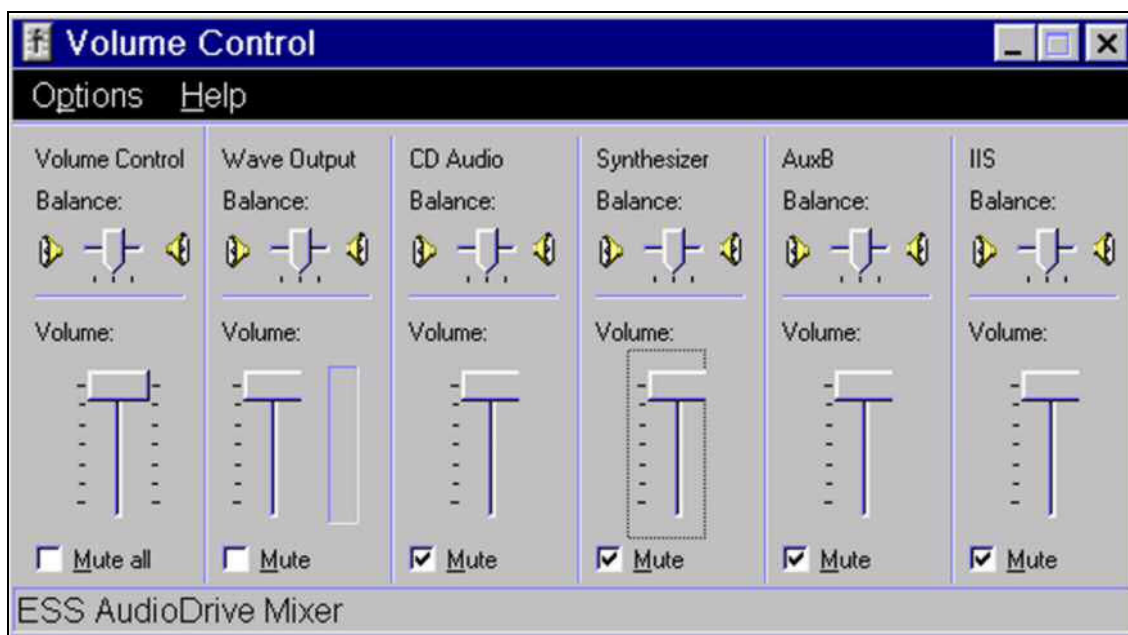
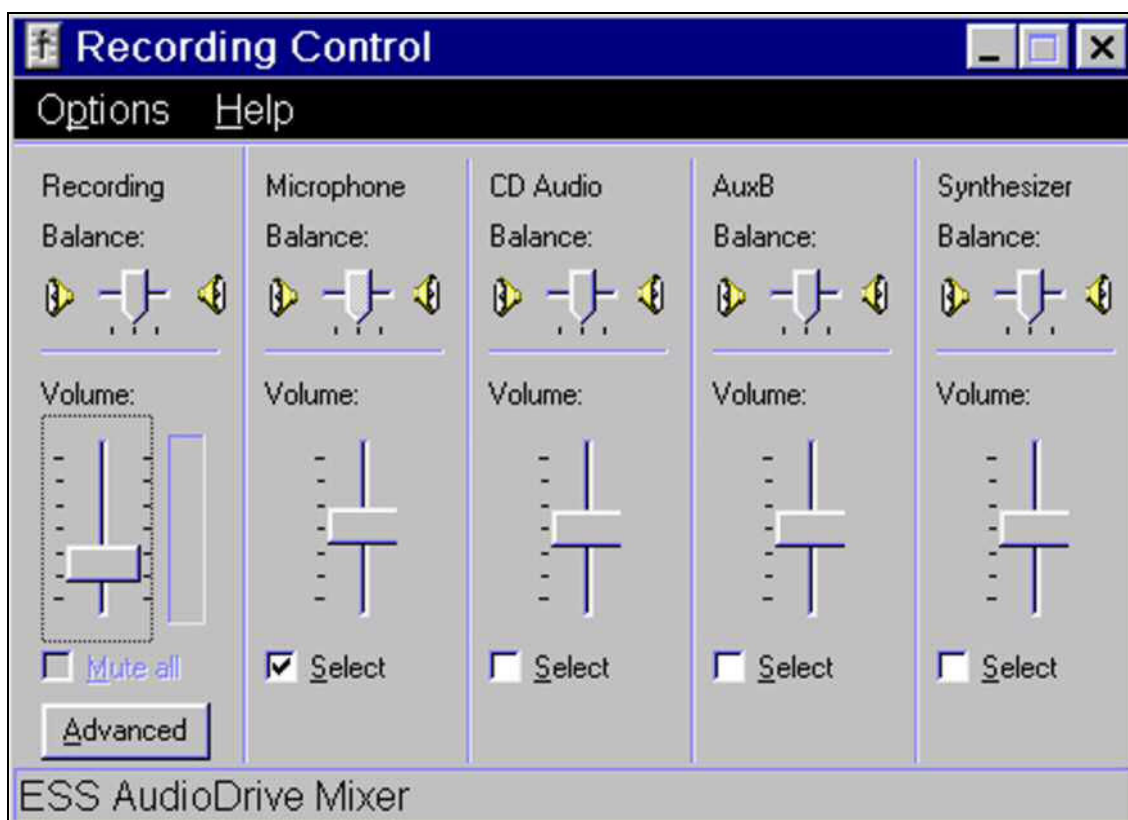


Fig. 192: Recording Control



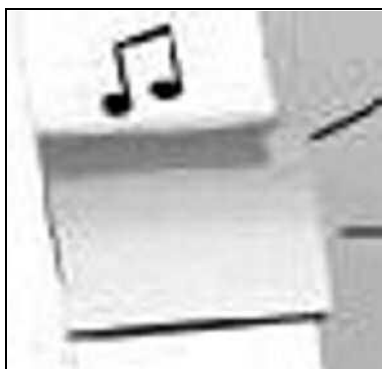
- Setting the computer voice (CV) volume: Select Options/ Configuration/ ExamUI and set the volume at approx. 70% of maximum. To save the settings and exit the windows, select <Apply> and <OK>.
- Setting the music volume, i.e. CD player: Turn both potentiometers on the left patient table control in clockwise direction to maximum. Switch on the music source, i.e. CD player, and adjust the output volume control on the music source to a volume just be-

low the level when the music starts to sound distorted (open the door to the RF room to listen). Keep this setting for the music source and adjust the two potentiometers on the left patient table control panel to a comfortable volume level (depending on the customer) for the speaker and the head phones.

9.2 Final Check

- Perform a final check for the voice output (new option as of VA21A). Close the patient application (Patient/ Close Patient) in case Patient is active. Select Queue/ Voice Output/ Organize... and click <New...>. Enter a name, e.g. "test", and select <OK> and click the ● button. Press the lower right-hand button on the intercom's operating unit and keep it pressed during this procedure. Click the ● button in the record voice window. Speak the command into the operating unit (keep a distance of approx. 40 cm). After finishing the text, click the ■ button in the record voice window and release the lower right-hand button on the operating unit. The voice recording is now finished. To check the recording, switch the operating unit from CBT to CV and click the >|| button. The recorded test command should be complete. A snapping sound at the beginning of the recording means that the procedure has been performed incorrectly. Compare the volume of your test recording with that of a default recording; both recordings should have just about the same sound level. If there is a difference in levels between the two recordings, readjust the volume of the recording control. Refer to the online help for more information regarding this new option. Close the voice output windows.
- The potentiometers for the operating unit (p.n. 4757261) affect the volume of the computer voice (CV) as well as the voice commands for the operator. The potentiometers on the operating unit do not require a basic volume adjustment, since these are set individually by the user. To select a starting value, turn the two potentiometers on the operating unit to approx. 50% of the volume. The user decides on an individual basis whether to use music or computer voice. The computer voice should not be used at the same time as music, because it may be difficult for the patient to understand the computer voice commands.
- If the customer prefers music: Switch the operating unit to "CBT" and press the play music button located in the center of the right side (music active = light switches on) as shown in (→ Fig. 193 Page 200).
- If the customer prefers the computer voice (CV): Switch the operating unit to "CV" and switch off the play music button in the center of the right side (music deactivated = switch light off). Notice: For computer voice recordings, the operating unit has to be switched back to "CBT". After the recording is complete, switch back to "CV".

Fig. 193: Music switch



9.3 Customer Information

Inform the customer about this new option by explaining how to record automatic voice output and what to keep in mind when using this new option.

10.1 General

10.1.1 Function Description of the VLM VESDA Device

The SIEMENS_VLM incorporates a very sensitive, laser-based smoke detection unit that can detect overheated connections in a very early phase. A little fan continuously draws air through a hose system that connects to the relevant connections that can overheat in an error scenario.

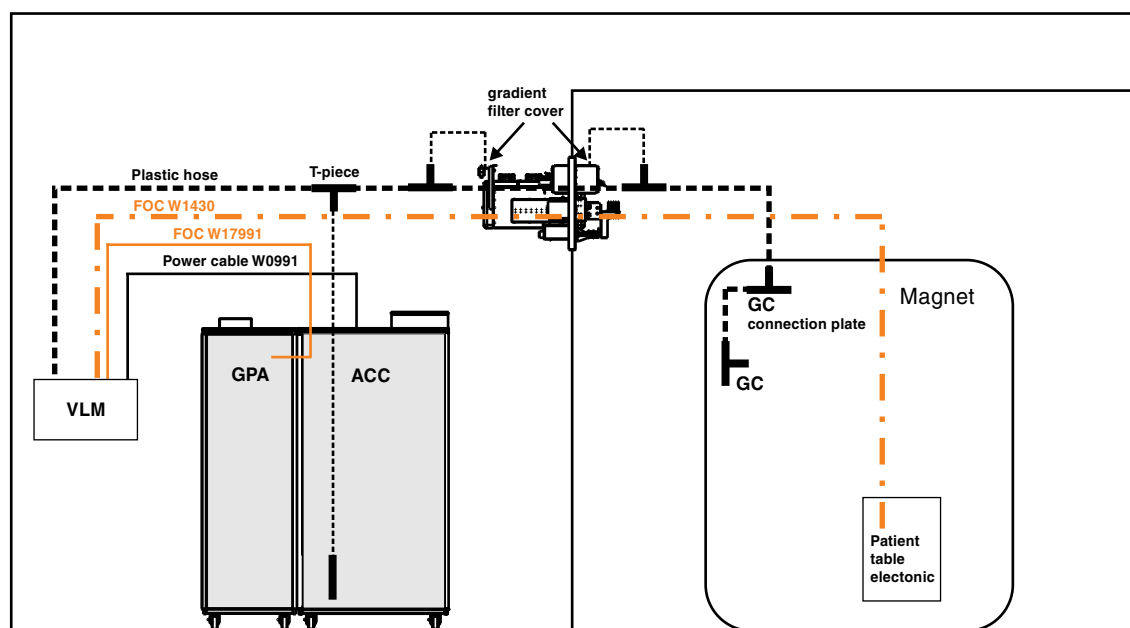
The aspirated air samples are fed through a filter into the laser detection chamber inside the unit.

Inside the calibrated detection chamber the air sample is exposed to a laser light source. If the sample contains small particles (e.g., emitted by overheated gradient connections), light is scattered within the detection chamber and is immediately recognized by the highly sensitive optical receiver system.



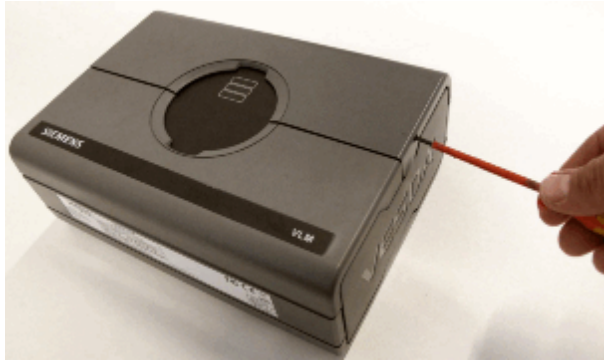
The VLM device is not designed to work as or to replace a fire warning system, but rather as an additional safety precaution to switch the MR system into “stand-by mode” in case of an overheated monitored connection.

Fig. 194: Gradient Connection Supervision (Tim systems, Biograph mMR)



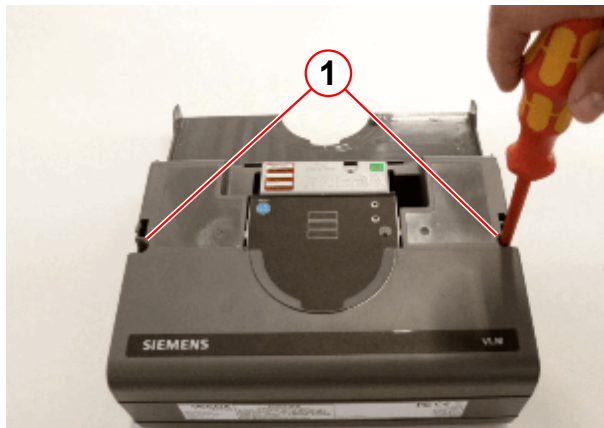
10.1.2 VLM Jumper Settings

Fig. 195: Opening the upper cover



- Open the upper cover of the device.

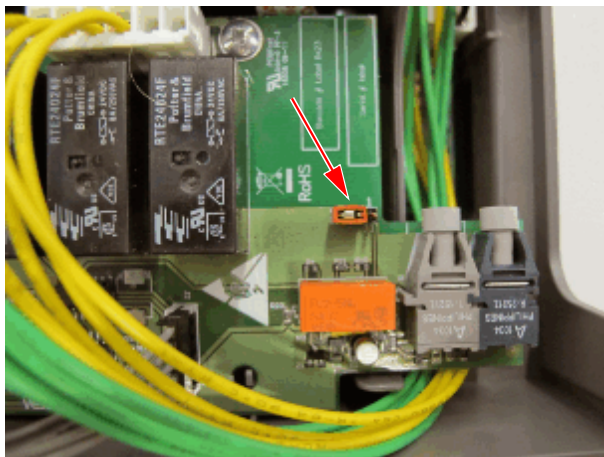
Fig. 196: Opening the lower housing



(1) 2x Phillips screws

- Remove the 2 Phillips screws (1) and swing out the housing.

Fig. 197: Jumper on 2-3 default setting - correct for Tim systems and Biograph mMR



- There is one jumper left from the fiber optic connectors. The default jumper setting is **2-3**, which is appropriate for all **Tim** systems.

10.2 Display

Fig. 198: VESDA device



- (1) OK (green)
- (2) FAULT (orange)
- (3) FIRE (red)
- (4) RESET

Legend	ON	FLASHING	Consequence
OK	Detector is powered and in normal operation		
FAULT	Indicates a major fault	Indicates a minor fault	» Normalize VLM device » Call Service
FIRE			» GPA-Modulator will be disabled » MR scanner will be switched off 1 minute later

10.3 Detector Control Button

There is only one button on front of the VLM device, which is the RESET button.

Icon	Button	Function
	RESET	■ Press and hold down this button to test the function of the LEDs on the unit. ■ Release this button to clear latched faults and alarms. » Alarm and fault lights will switch off, and if the system is still in alarm or fault mode, the lights will reappear after a certain delay. » The reset button does not operate during the delay fire notification interval. ■ Press and hold down this button for 10 sec. until the 3 LEDs start flashing to activate the set-mode for 100% air flow. » For details refer to (→ Start-Up Procedure of the VLM Device / Page 206).

10.4 Start-Up Procedure of the VLM Device

- Close the RF-cabin door.
- When F100 on top of the ACC cabinet is switched-on,
 - » The FAULT lamp is on for 5 sec.
 - » The VLM device is powered now and needs a 90 sec warm-up time.
- Open the front cover of the device and push the RESET button for 10 sec. until all lamps are flashing.
- Release the RESET button and wait until the lamps go off.
- When the lamps start flashing again, press the RESET button once more to confirm the actual setting.
 - » The current air flow through the device (which depends on the individual system configuration and hose length) is now set to 100%.
 - » Deviations from this nominal Air flow $100\% \pm 20\%$ will result in an internal error.
 - » A removed, broken, or blocked hose will result in a FAULT now.

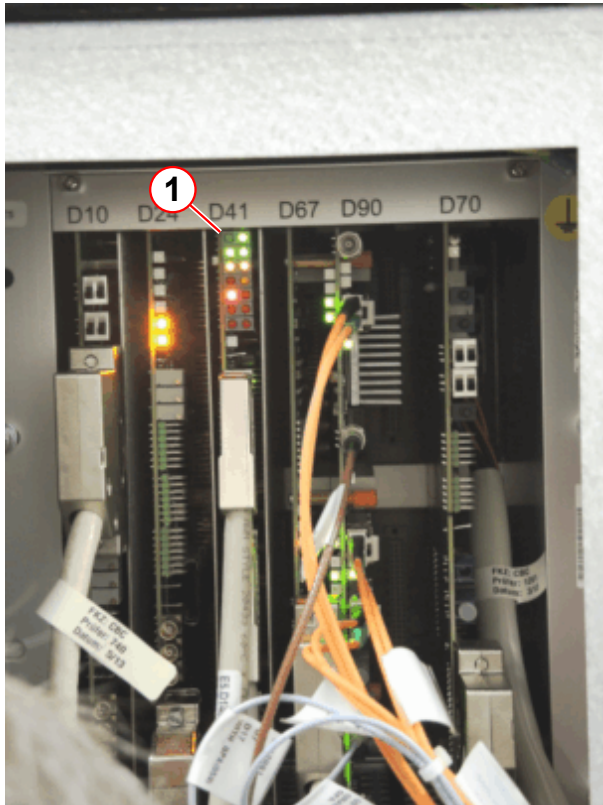


In case the normalization procedure described above does not work properly and the VLM device goes into failure mode, there could be a differential pressure between the RF-cabin and the equipment room, which exceeds the limits of the VLM device. In this case it is recommended to normalize the VLM at a mean pressure value between maximum cabin pressure (door closed) and a minimum cabin pressure (door open). This can be achieved by normalizing the VLM, when the cabin door is open for ~3 mm. When there is low pressure in the cabin, a pressure equalization tube must be fed from the exhaust of the VLM device back into the RF-cabin, according to (→ Pressure Equalization (optional) / Page 212).

10.5 Final Check

Modulator disable test

Fig. 199: Modulator LED in GPA cabinet off



(1) Modulator LED (V120)

- Remove the FOC (W1430) from U2
 - » The GPA modulator should be disabled (scanner signal at MRC is crossed through and modulator LED in GPA cabinet is off, refer to (→ Fig. 199 Page 207))

- » GPA error should be reported in the event log:

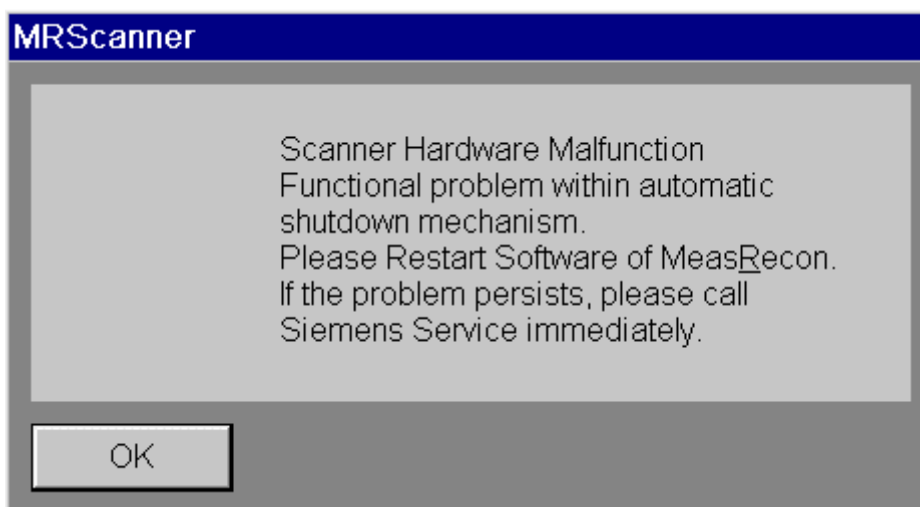
Date/Time	ID	Source	Seq.	Message Text
22-May-15 17:24:25	2016	MRI_GDV	0	Gradient Power Amplifier error: Hardware error. Please reboot

- Reattach the FOC W1430 to U2.
- Reassemble the VLM device.
- Perform "Reboot Scanner."

Air flow failure simulation

- Remove the hose from the VLM device.
 - » ~30 sec. later a fault should be displayed on front of the VLM device due to high air flow and a pop-up should be displayed at the console (→ Fig. 200 Page 208).

Fig. 200: Pop-up "FAULT" condition



- Refit the hose.
 - » The green OK lamp should go on.

Differential pressure test

- Open the RF-cabin door and wait for 5 minutes.
 - » The green OK lamp on the VLM device should remain on.



In case the normalization procedure described above does not work properly and the VLM device goes into failure mode, there could be a differential pressure between the RF-cabin and the equipment room, which exceeds the limits of the VLM device. In this case it is recommended to normalize the VLM at a mean pressure value between maximum cabin pressure (door closed) and a minimum cabin pressure (door open). This can be achieved by normalizing the VLM, when the cabin door is open for ~3 mm. When there is low pressure in the cabin, a pressure equalization tube must be fed from the exhaust of the VLM device back into the RF-cabin, according to (→ Pressure Equalization (optional) / Page 212).



In order to prove the proper function of the detector, a smoke test must be performed.



WARNING

Accumulation of oxygen during helium filling or magnet ramping.

The use of glowing smoke wicks during Helium filling or magnet ramping could result in an open fire in the area of the magnet service turret.

- » Do NOT perform a smoke test during helium filling or magnet ramping!



The smoke wicks, located in a box for the test below, are part of the Update-Kit. Each smoke wick will provide up to 30 minutes of continuous smoke, or 60 individual 30 second tests.

Smoke test

- Take one smoke wick out of the box (→ Fig. 201 Page 209).
- Hold wick at the end with black plastic handle mounted (→ Fig. 202 Page 209).
- Light the tip of the wick and allow it to burn for 10-15 seconds (→ Fig. 202 Page 209).

Fig. 201: Smoke wicks in a box



Fig. 202: Lighted smoke wick



- Extinguish the flame and hold the smoke wick behind the rear cover in the area of the lower sample plate for 5 seconds (→ Fig. 203 Page 210).
- Extinguish the wick by simply placing it in the empty box and close the lid carefully. This will extinguish the wick after approximately 15 seconds.
 - » With a delay of about 20-40 seconds (depending on the length of the hose) the "FIRE" lamp on the VLM device should light up (→ Fig. 204 Page 210).

Fig. 203: Smoke test at the magnet exhaust - SaTS

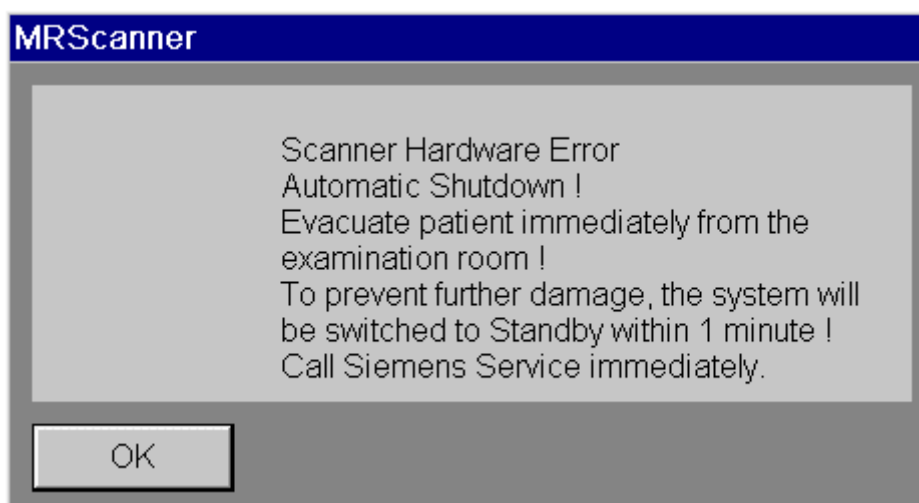


Fig. 204: Fire lamp ON



- » Subsequently, the modulator in the GPA cabinet should be disabled and the message (→ Fig. 205 Page 210) should appear on the screen of the host.
- » 1 minute later the scanner should be switched off (to stand-by).

Fig. 205: Pop-up "Automatic Shutdown"



Once the FIRE lamp has been activated, it must be reset with the RESET button (→ 4/ Fig. 198 Page 204). It may take several minutes until the background level of smoke particles has returned to a normal level and a reset is possible. After a successful reset, the system can be switched on again.

- To reuse a smoke wick, simply take it out of the box and follow the steps as described above.

- The smoke wicks have a long shelf-life and should be stored in a cool and dry place.



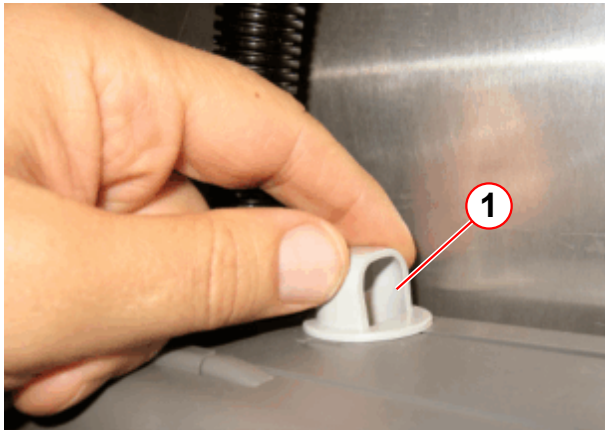
Use only as directed, do not eat and please keep away from children.

10.6 Pressure Equalization (optional)



This procedure is only valid if the differential pressure test fails.

Fig. 206: VLM device exhaust cap removal

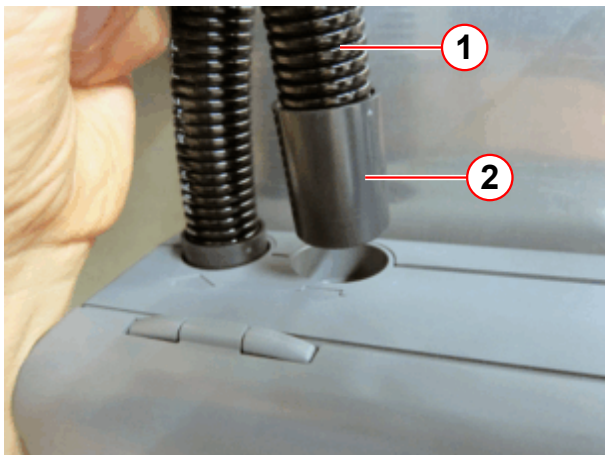


(1) Exhaust cap

Removal of the exhaust cap at the VLM device

- Remove the exhaust cap (1) on top of the VLM device

Fig. 207: VLM exhaust hose connection

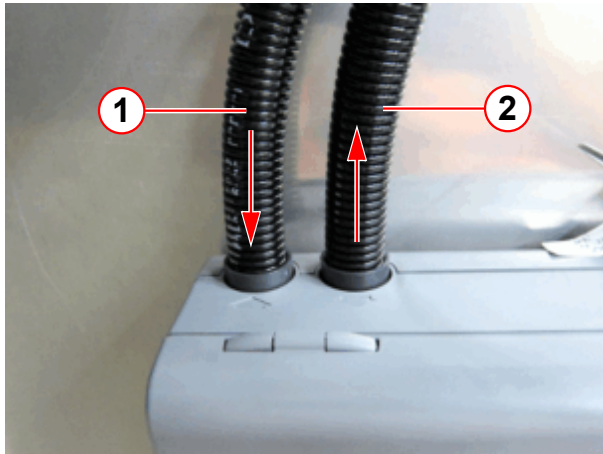


(1) Exhaust hose
(2) Bushing

Mounting of a pressure equalization hose

- Push-fit the bushing (2) onto one end of the remaining 22 mm hose roll.
- Push the bushing with the hose into the exhaust on top of the VLM device (→ Fig. 207 Page 212).

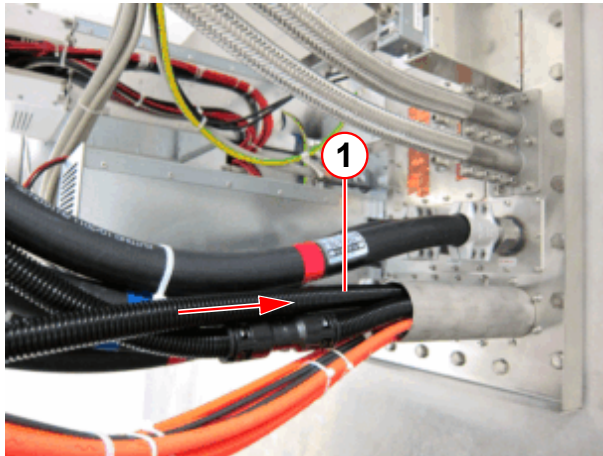
Fig. 208: VLM device with pressure equalization hose



- (1) Inlet
- (2) Exhaust

The air flow is now via hose (1) into the VLM device inlet and into the outlet hose (2).

Fig. 209: Hose fed from the equipment room

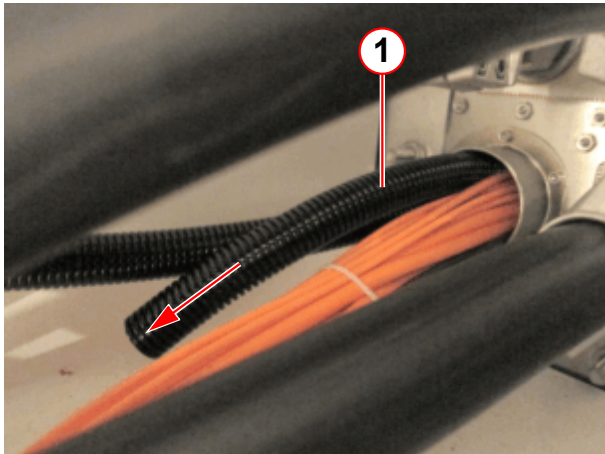


- (1) Pressure equalization hose

Feeding the pressure equalization hose through the filter plate

- Lay the pressure equalization hose (1) through the wave duct into the magnet room.

Fig. 210: Hose fed into the examination room



(1) Pressure equalization hose

Cutting the pressure equalization hose

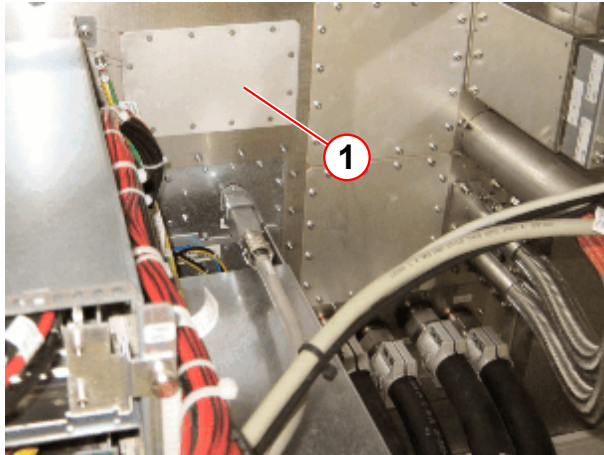
- Cut the pressure equalization hose (1) inside the magnet room at ~20-30 cm behind the wave duct.

10.7 Mounting of an additional Wave Duct (optional)



This procedure is only valid for systems where the original wave duct has not enough space for keeping the hose (-s). In this case, additional hardware is required and must be ordered from CS ML. Part-no.: 10684487

Fig. 211: RF filter plate (equipment room)



(1) Blind plate

Fig. 212: RF feed-through (optional)



Mounting the additional wave duct

- Remove the blind plate (1) from the filter plate.
- Replace it with the new wave duct (which can also be used with the MNO-option)
 - » Now the additional wave duct can be used for the hoses.

11.1 Prerequisites for Tune-Up/QA

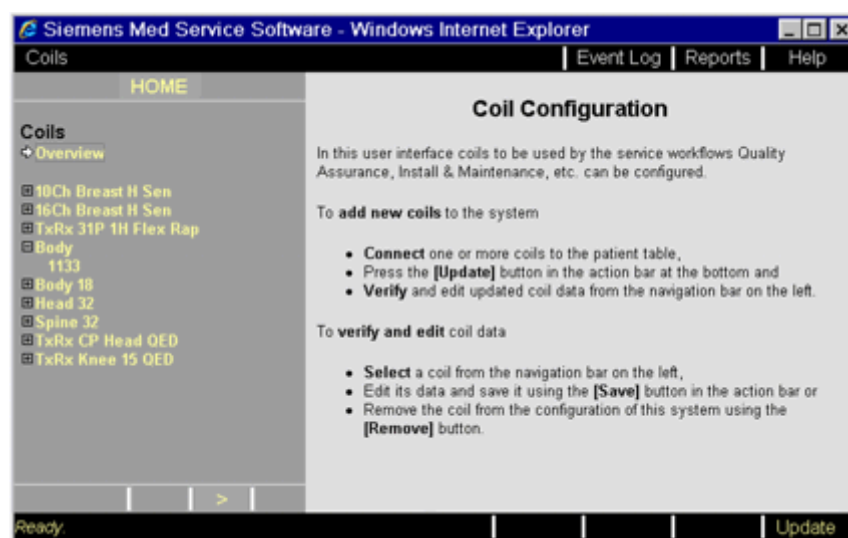
11.1.1 Configuration checks (Num4_VD13)



The status of the Coil QA will be set to "ToDo" after the coil configuration. This is necessary because of the new coil asset mechanism.

- Select **Coils** in Local Service

Fig. 213: Coil Configuration



- Check the correct configuration of the Body coil and the surface coils.
- Add missing coil configuration.

11.2 Tune-Up



If the system is not to be used with the option “Tim Whole Body Suite”, or there is only limited patient table travel, the PAMatrix Coil may only be selected in the configuration as F_PAMatrix (feet first).

This only indicates the coil in the service software QA and installation protocol.

The H_PAMatrix may not be used for quality assurance without the “Tim Whole Body Suite”.

The reason is that the patient table travel distance is too short to measure all of the coil elements.

This would result in an incorrect value in the installation protocol.



Tune-Up information is not published as hardcopy documentation. For more information, refer to the Online Help.

Performing the Tune-Up of the system (Options/ Service/ Local Service/ Tune-Up). Most Tune-Up steps are performed completely automatically. Follow the instructions shown on-screen. In addition, pop-up windows will inform you of the actions required. Refer to the Online Help for detailed information.

11.3 Testing pick-up coils/Body Coil



This test is part of the Quality Assurance (QA) with software version N4 VB15A and N4 VD13A.



This test is valid for all systems with software version N4 VB13.
Except for the Espree Straight Bore system and the Tim system with 8 Receiver Channels configuration.

Auxiliary tools:

- Loader with 24 cm spherical phantom

Preliminary steps:

- Register a patient (any)
- Move the table to HOME position
- Coil and phantom selection:
 - Remove all local transmit coils
 - Remove all local receiver coils
 - Position the loader together with the 24 cm spherical phantom (centered in the loader) in the upper part of the table and move it under the light localizer.
- Switch the light localizer on and off.
- Move the table into the CENTER position.

11.3.1 Performing the measurement

- Start the measurement with one of the protocols (e.g. a localizer).
 - » Measurement can be stopped immediately after it begins.

11.3.2 Evaluation

- Open the file "RFSWDHistoryListNew.log"
(it is located in c:\MedCom\Log Directory)
- Scroll back from the last entry in the file to the "604 DiCoTest Tra Power[0]" column
- Once there, ensure that the entries for the following parameters reach a minimum of 200 V or maximum 2500 V
 - $2500 > \text{DiCoTest PickUp Probe Ampl. [0][0]} > 200 \text{ V}$
 - $2500 > \text{DiCoTest PickUp Probe Ampl. [0][1]} > 200 \text{ V}$

This concludes the test.

- » If the conditions are not met, check the connections from the output BC to the input receiver.

11.4 QA (Quality Assurance)



Information regarding QA is not published as hardcopy documentation. For more information, refer to the Online Help.



The service software supports only the surface coil that has been configured in the software configuration menu (SeSo/Configuration/Measurement/Avail.Coil).

Perform the QA (Quality Assurance) of the system and of all available surface coils (Options/ Service/ Local Service/ QA). Most QA steps are performed completely automatically. Follow the instructions shown on-screen. Pop-up windows will inform you of the actions required. Refer to the Online Help for detailed information.

11.4.1 TestTools



Before handing the system over to the customer, please ensure that all tests of the TestTools have been performed and are within specifications. Please refer to the chapter "TestTools" in "Software Configuration".

11.4.2 Safety-related tests

Perform the "Safety-related tests".



Information regarding Safety-Related Tests is not published as a hardcopy. Refer to the SeSo/Install & Maintenance/Safety-Related Tests

12.1 Final Work Steps

12.1.1 Installation&Maintenance Protocol

Complete the software-based "Installation & Maintenance Protocol."

12.1.2 Quench Line Acceptance Protocol

Check that the Quench Line Acceptance Protocol is signed off by the Quench Line Manufacturer if present.

12.1.3 Dynamical Backup

Perform "Backup of dynamic data." For more information, refer to CB Doc, Software Installation.

12.1.4 Sign for bruising hazards



The bruising hazard sign must be posted in back of the patient table if the distance between patient table and back wall is less than 500 mm.

Fig. 214: Service end area



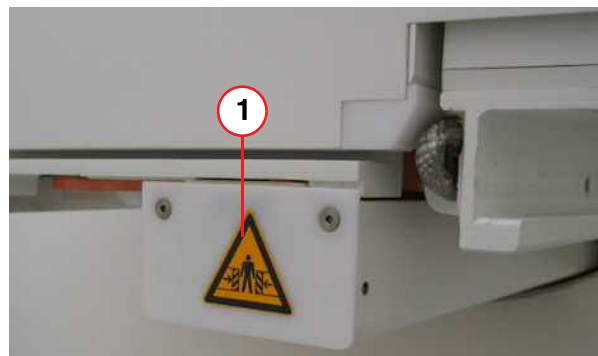
- (1) Patient table
- (2) Wall of RF room
- (3) Distance < 500mm

- Affix the label

Fig. 215: Bruising hazard sign



Fig. 216: Bruising hazard

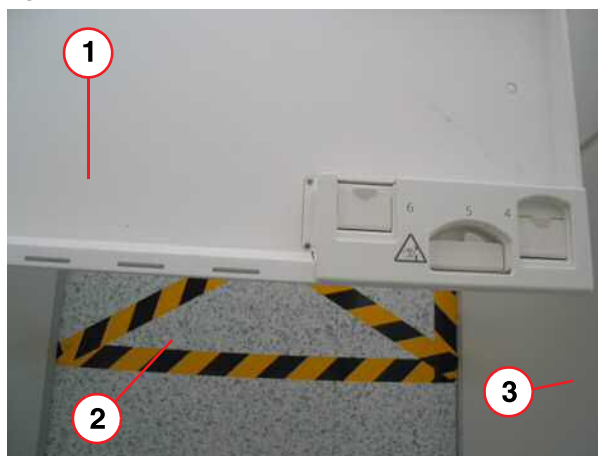


(1) Label

The controlled area is the area where the patient table is moving to the service end of the magnet.

- Mark the controlled access area with the tape.

Fig. 217: Service end area



- (1) Patient table
- (2) Marking on the floor
- (3) Wall of the RF room

Fig. 218: Service area end



- (1) Marking on the floor
- (2) Magnet cover

12.1.5 Hearing Protection Sign

The hearing protection sign is shipped together with the warning labels.



The hearing protection sign must be posted at the main entrance door of the examination room. Mount the sign at eye level as shown in

Fig. 219: Hearing protection sign



Fig. 220: Warning Signs



(1) Hearing protection sign

12.1.6 Cleaning up



This completes the start-up of your MR system. If no other options are being installed in your system, please complete the following final steps. If you are installing other options, complete these final steps after you have installed the options. Please bear in mind that your customer has purchased an excellent, high-quality MR system with a modern design. The visual impression of the system is equally important to obtain your customer's satisfaction.

- Assemble the surface coil cart according to the installation instructions.

- Close all magnet covers.
- Close all the doors of the electronics cabinet.
- Touch up paint chips with matching color paint.
- Clean fingerprints off the monitor screens.
- Clean fingerprints off the table and PMU displays.
- Clean fingerprints off the system with a mild cleaner.
- Clean the patient table and the surface coils with a mild cleaner.
- Clean the area of the system, dispose of empty cartons or other packing materials.
- Make sure that the MR system rooms are cleaned by the cleaning crew.

12.1.7 Documentation for Review

Send back the following mandatory documents
to Siemens Healthcare GmbH MR QT department.

immrqpininstallations.healthcare@siemens.com

- Safety-related Test Report
- Start-up Report
- Quench Line Acceptance Protocol incl. pictures of the quench line (provided by the project manager),
if not done by other participants like project manager or mechanical vendor.
- Pictures of the magnet room, shortly before ramping.

Version 02

- Chapter 02:
 - Hazard note added
 - Adjusting the primary flow rate
- Chapter 03
 - Check the MeasSettings added.
- Chapter 07:
 - (→ Spike Check / Page 175) added
 - (→ Ramping the magnet / Page 101) updated, transport test added
 - (→ Tim Whole Body Suite (optional) / Page 174) added

Version 03

- Chapter 08
 - TFT Color Monitor updated.

Version 04

- Chapter 0
 - RF cable 2 m BNC (→ Service Tools / Page 8)
- Chapter 02
 - (→ T-class (optional) / Page 41) added.
- Chapter 07
 - Limiting the patient table travel removed.

Version 05

- Chapter 01
 - Note for the safety check of the service platform added.
- Chapter 02
 - (→ Partial drive unit (optional) / Page 36) added.
- Chapter 07
 - (→ Checking the reference switch S10 of the patient table / Page 157) added.
- Chapter 08
 - Transferring monitor information to the REP.
- Chapter 11
 - Hearing protection sign added.

Version 06

- Chapter 02
 - Warning signs updated.
- Chapter 07
 - Safety notes before ramping.
- Chapter 11
 - Update of Installation&Maintenance Protocol.

Version 07

- Chapter 07
 - Information about shimming of reinstalled systems.

Version 08

- Chapter 02
 - (→ Checking the Gradient Connections / Page 12) was added.
- Chapter 07
 - Note for the Quench line installation was added.

Version 09

- Chapter 12
 - Company name changed.
- Chapter 10
 - Gradient Supervision added.

Version 10

- Chapter 07
 - Update of Ramping the Magnet with the ECL.
- Chapter 12
 - Quench Line Acceptance Protocol added.

Version 11

- All minor editorial changes.
- Chapter 02
 - Gradient cable tolerances at the GPA 35 ± 1 Nm changed to 35 ± 2 Nm (→ Gradient cable outside HF room / Page 12)
- Update 07
 - Chapter gradient supervision reworked (Charm MR_00458861) (→ Gradient Supervision / Page 202)
- Chapter 12
 - Changed Siemens AG to Siemens Healthcare GmbH.(→ Installation&Maintenance Protocol / Page 221)
 - Quench line acceptance protocol, deleted the sign-off by the project manager. (→ Quench Line Acceptance Protocol / Page 221)

Version 12

- Chapter 12
 - Minor editorial changes.
 - New section, documentation for review (→ Documentation for Review / Page 224) added.

Version 13

- Chapter 07:
 - Information regarding switch setting on 25 m extension cable 617-371-25T was added (→ CAN OPEN lead longer than 5 m / Page 115).

There are no Hazard IDs in this document.

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